

FEDERAL HIGHWAY ADMINISTRATION

TURNER-FAIRBANK HIGHWAY RESEARCH CENTER REPORT

***I-90 Seaport Portal Tunnel Partial Ceiling Collapse Investigation:
Sustained Load Behavior of Powers Fasteners Power-Fast+ Adhesive Anchors***

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JULY 2007

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CHAPTER 1. INTRODUCTION

Late in the evening of July 10, 2006, ten two-ton concrete panels suddenly fell from above the roadway of the eastbound I-90 Seaport Access Tunnel in Boston, Massachusetts. These suspended panels had been supported by adhesive anchors installed in the concrete ceiling of the tunnel, and the anchors had been subjected to direct tension loading over the 7-year life of the suspended ceiling panel installation. The National Transportation Safety Board (NTSB) is leading the investigation to determine the cause of the failure. In support of this NTSB effort, the Federal Highway Administration (FHWA) deployed staff and contractors to Boston on July 15, 2006 to evaluate the performance of the remaining adhesive anchors installed in the tunnel. This initial work by FHWA was summarized in a report titled *“I-90 Seaport Portal Tunnel Partial Ceiling Collapse Investigation: Adhesive Anchor Load and Load-Displacement Testing Results.”*¹ The load-displacement testing on in-situ adhesive anchors reported in that document used an incrementing loading protocol where individual load steps were maintained for two minutes. During these load holding periods anchor displacement often did not stabilize indicating that the anchor system may be susceptible to creep under sustained loads. Therefore, a series of sustained load or creep tests were conducted in the westbound tunnel. This work was reported in *“I-90 Seaport Portal Tunnel Partial Ceiling Collapse Investigation: Adhesive Anchor Sustained Load Testing Results.”*² Despite producing valuable data, a more controlled investigation on the response of this adhesive anchor system to sustained loads was deemed necessary.

As a result, a series of creep experiments were designed and conducted in the FHWA’s Turner-Fairbank Highway Research Center (TFHRC) Structure’s Lab. Twelve anchors were installed in concrete specimens and monitored under various sustained loads for 82 days. The anchors were installed overhead in cored holes using Powers Fasteners Power-Fast+ Fast Set or Standard Set epoxy. This report summarizes the methodology and results from this sustained load testing.

CHAPTER 2. EXPERIMENTAL METHODS

This chapter will describe the overall design of the creep experiments including the specimen design, data collection, and experimental procedure.

SPECIMEN MATRIX

The testing matrix was compiled collaboratively between NTSB and FHWA personnel. Creep testing under laboratory conditions at TFHRC was initiated as a result of the anchor pullout observed and the sustained load testing conducted in the I-90 tunnel after the accident. While the anchors in the I-90 tunnel had unknown installation and load histories, the tests at TFHRC were performed on anchors that were newly installed overhead with best practices.

The creep tests were conducted in the I-90 tunnel between July and November of 2006. Those tests used 1000, 2000 and 3000 pound concrete weights to load in-situ adhesive anchors, which were monitored for up to 63 days or until the anchor pulled out. Further details of those tests can be found in *I-90 Seaport Portal Tunnel Partial Ceiling Collapse Investigation: Adhesive Anchor Sustained Load Testing Results*.² The laboratory creep tests at TFHRC used loads of 1000, 2000, 3000 and 4000 pounds, with two replicate anchors tested at each load level.

The anchors used in all experiments at TFHRC were identical to the anchors installed in the I-90 tunnel. They are fabricated from AISI 316 stainless steel, have a diameter of 5/8 inch and 11 threads per inch. The anchors are 8 inches in length and were installed to an embedment depth of approximately 5 inches.

The adhesive used to install the anchors supporting the ceiling panels in the I-90 tunnel was a Powers Fasteners Power Fast+ epoxy, which comes in two variants; Fast Set and Standard Set. Eight anchors were installed with the Fast Set epoxy and tested at the four load levels indicated above. The test matrix also included four anchors installed with Power-Fast+ Standard Set epoxy, which were loaded at 2000 and 4000 pounds. Table 1 outlines the testing matrix used in the TFHRC creep tests.

LABORATORY SETUP

The TFHRC creep test anchors were installed overhead into holes cored into concrete blocks to mimic the installation procedure used in the I-90 tunnel. Once the sustained load was applied to each anchor, displacement was monitored for 82 days or until the anchor pulled out of the concrete block.

Figure 1 shows the detailing of the solid concrete blocks. The concrete specified for use was a 4000 psi compressive strength, #57 mix. Casting of 15 concrete blocks (which the anchors would be installed into), fifteen 4-inch x 8-inch concrete cylinders, and four 2000 pound concrete weights occurred in the late morning of 8 November 2006. Cylinders were made according to ASTM C192-05³ and match cured with the concrete blocks. Eight of the cylinders were

compression tested according to ASTM C39-05⁴ and the results are shown in Table 2. At the time that a sustained load was transferred to the anchors, the concrete had attained a 4400 psi compressive strength. The remaining seven cylinders were tested to attain an elastic modulus for the concrete in accordance with ASTM C469-02⁵. The modulus tests were conducted to provide data for potential analytical efforts.

The TFHRC creep tests were conducted in the basement of the structures lab. A schematic of the tests is shown in Figure 2. Figure 3 shows a plan view of the specimen arrangement. The dead load weights were fashioned from material available at the structures lab. The 1000 and 3000 pound weights were built-up from scrap steel plate. The 2000 pound weights were cast from the same batch of concrete that the concrete blocks were made from. The 4000 pound weights were made from two 1-ton lead weights stacked together. A picture of the dead load weights is shown in Figure 4. The exact weight of each dead load was measured with a load cell traceable to a NIST standard prior to and at the conclusion of testing. These two measurements and their average are reported in Table 1 for each dead load weight used. The average weight from the two measurements was used for all calculations and comparisons.

The 0.75 inch diameter holes for the adhesive anchors were wet cored using a Hilti DD-100 drill to a 5.625 inch depth after the concrete blocks were mounted to the lab strong floor. The holes were allowed to dry overnight then cleaned out with a three step process of blasting with compressed air, brushing with a nylon brush and concluding with a second burst of compressed air. The epoxy was injected overhead and the anchor was installed with a 5.0 inch embedment. If air was heard escaping from the hole during insertion of the anchor (usually a sign that a void had been created), the anchor was removed and the hole was refilled with epoxy prior to re-inserting the anchor. Eight anchors were installed with Power-Fast+ Fast Set formula (Lot #062570I1LC) and four anchors were installed with the Standard Set formula (Lot #062700H1LC).

DATA COLLECTION SYSTEM

Linear variable differential transformers (LVDT) were used to monitor the displacement of the adhesive anchors for the duration of the creep testing. The LVDTs were mounted to an aluminum plate that was in turn double-nut mounted to the threaded anchor protrusion. A schematic of the aluminum plate is shown in Figure 5. The advantage of the aluminum plate was that four separate LVDTs could be monitored simultaneously, all equidistant from the anchor, and paired in orthogonal planes. This permitted a correction to be made to each anchor's displacement for any bending caused during the transfer of load.

Two separate data acquisition systems were used during laboratory testing. The short-term system was used to monitor individual anchors during the initial load transfer. The long-term system monitored all 12 anchors through the initial weight drop and throughout the duration of testing.

Short-Term System

This system was setup to collect the data from three LVDTs at a rate of 10 Hz. This system, shown in Figure 6, was packaged in a portable enclosure, which included a laptop computer, a data acquisition module, signal conditioning modules, power supplies, transducers, and a back-up power supply. If the primary AC power supply was compromised, the back-up battery power supply insured the integrity and continuity of the data. The data collection and real-time output were controlled with a custom program using HPVee software.

The short-term system contained the following components.

1. Data Acquisition Module
 - a. Manufacture: Data Translation
 - b. Type: USB Data Acquisition Function Module
 - c. Model: 9804
 - d. S/N: 00562154
2. Laptop Computer
 - a. Manufacture: Dell
 - b. Model: Latitude C810
 - c. Model: PP01X
 - d. Service Tag: GL7XC11
3. Power Supplies
 - a. Type: +/-15V linear power supply for displacement transducers
4. Sensors
 - a. Displacement
 - i. Type: LVDT Displacement Transducer
 - ii. Manufacture: RDP Electrosense
 - iii. Model: DCTH400AG
 1. LVDT1 S/N: 66295
 2. LVDT2 S/N: 67816
 3. LVDT3 S/N: 67820

Long-Term System

The long-term system was used to collect displacement and temperature data throughout the duration of testing. This system took measurements from 12 displacement sensors and four temperature sensors, shown in Figure 7. A remote workstation connected to the data acquisition system was used for data storage and observation. The remote workstation was located on the main floor of the Structures Laboratory and connected to the data collector in the basement via an Ethernet cable.

The long-term system contained the following components.

1. Data Logger
 - b. Manufacture: Campbell Scientific
 - c. Model: CR1000

- d. S/N: 2103
- 2. Battery
 - a. Manufacture: Genesis
 - b. Model: NP100-12B, 12V 100-AHr
- 3. Sensor Power Supply
 - a. Manufacture: Tri-Mag Power
 - b. Model: TDB4W-1212D
 - c. Quantity: 2
- 4. Sensors
 - a. Displacement
 - i. Type: LVDT Displacement Transducer
 - ii. Manufacture: RDP Electrosense
 - iii. Model: DCTH400AG
 - 1. LVDT1 S/N: 94172
 - 2. LVDT2 S/N: 94176
 - 3. LVDT3 S/N: 91303
 - 4. LVDT4 S/N: 94175
 - 5. LVDT5 S/N: 94167
 - 6. LVDT6 S/N: 90502
 - 7. LVDT7 S/N: 94173
 - 8. LVDT8 S/N: 90503
 - 9. LVDT9 S/N: 95691
 - 10. LVDT10 S/N: 95690
 - 11. LVDT11 S/N: 95692
 - 12. LVDT12 S/N: 95693
 - b. Temperature
 - i. Type: Semiconductor
 - ii. Manufacture: Analog Devices
 - iii. Model: AD22100AT
 - iv. Quantity: 4

LOAD TRANSFER

Load was transferred to each anchor on December 20, 2006 between 8am and 2pm. Immediately prior to transfer, the dead weights were measured with a 10,000 pound load cell (StrainSert FL10U(C)-3DPKT). A forklift was used to maneuver the dead weights into position and to transfer load to the anchors. The load transfer typically took about 10 seconds and was accomplished by slowly lowering the forks out from under individual dead weights.

The long-term data collection system recorded data from all 16 sensors (12 LVDTs and four temperature gauges) connected to it at a rate of 1.25 Hz. Prior to the transfer of load to an individual anchor, the three LVDTs from the short-term data collection system were mounted to the aluminum plate and data was collected at a rate of 10 Hz. The higher rate of collection on the short term system captured more detailed information on anchor behavior during load transfer and any subsequent oscillations until they damped out. The short-term data collection continued to acquire data for 30 minutes after load transfer.

The four LVDTs used during load transfer were always orientated as shown in Figure 8. The sensors shown are all separated by an angle of 90-degrees. LVDT1, LVDT2 and LVDT3 were used with the short-term system. LVDT4 was used with the long-term system.

TEST DURATION

The last load transfer occurred at 2pm on December 20, 2006. The long-term system continued to collect and record data from each of the 12 LVDTs and four temperature gauges at 1.25 Hz until 3pm on December 20, 2006. From that time until the conclusion of testing the long-term system continued to scan the channels at 1.25Hz, however it continuously averaged the data over five minute intervals. Therefore one data point was produced every five minutes for the majority of the testing.

CONCLUSION OF TESTING

On March 12, 2007, the dead weights were removed from anchors still embedded in the concrete blocks to conclude the TFHRC creep tests. The weights were re-weighed using the same load cell used on December 20, 2006. These measurements make up the second set of weight readings reported in Table 1. Also shown in this table are the average weight values from before and after the tests. All differences recorded in the weight measurements are within the resolution of the load cell which is ± 10 pounds. The weight used for each block in any subsequent analysis was the average from the two weight measurements.

The long-term system continued to collect data for 24 hours after the dead weights were removed from the anchors in order to capture any recovery displacements in the epoxy. Finally, the adhesive anchors were removed from the concrete blocks via static load-displacement testing.

Table 1 Creep Testing Matrix

Specimen	Creep Weight 20 Dec. 2006 (lbs)	Creep Weight 12 Mar. 2007 (lbs)	Avg. Creep Weight (lbs)	Epoxy Type (Power-Fast+)
1	1050	1054	1052	Fast Set
2	1055	1058	1057	Fast Set
3	2003	2003	2003	Fast Set
4	1993	1990	1992	Fast Set
5	1988	1989	1989	Standard Set
6	1990	1992	1991	Standard Set
7	4080	4070	4075	Standard Set
8	4081	4074	4078	Standard Set
9	4092	4091	4092	Fast Set
10	4014	4012	4013	Fast Set
11	3056	3060	3058	Fast Set
12	3145	3140	3143	Fast Set

Table 2 Concrete Block Material Compressive Strength

Specimen	Test Date	Length (inch)	Diameter (inch)	Peak Load (pounds)	Peak Stress (psi)
CA/T #21 Creep	22 Nov 2006	7.949	3.999	48760	3883
CA/T #22 Creep	22 Nov 2006	7.887	3.998	48300	3848
Average	-	-	-	-	3865
CA/T #23 Creep	20 Dec 2006	8.006	3.999	57140	4549
CA/T #24 Creep	20 Dec 2006	7.962	4.000	57240	4556
CA/T #25 Creep	20 Dec 2006	7.958	4.002	57260	4552
CA/T #26 Creep	20 Dec 2006	7.995	4.00	53520	4260
CA/T #27 Creep	20 Dec 2006	7.905	3.999	56500	4499
CA/T #28 Creep	20 Dec 2006	7.898	3.998	54300	4326
Average	-	-	-	-	4457

Table 3 Concrete Block Material Elastic Moduli

Specimen	Test Date	Length (inch)	Diameter (inch)	Elastic Modulus (ksi)
CA/T #29 Creep	5 April 2007	7.931	4.000	3043
CA/T #30 Creep	5 April 2007	7.986	4.000	2961
CA/T #31 Creep	5 April 2007	7.917	4.000	3024
CA/T #32 Creep	5 April 2007	7.854	3.996	3207
CA/T #33 Creep	5 April 2007	7.931	3.999	3091
CA/T #34 Creep	5 April 2007	7.950	3.999	3148
CA/T #35 Creep	5 April 2007	7.967	3.993	3360
Average	-	-	-	3119

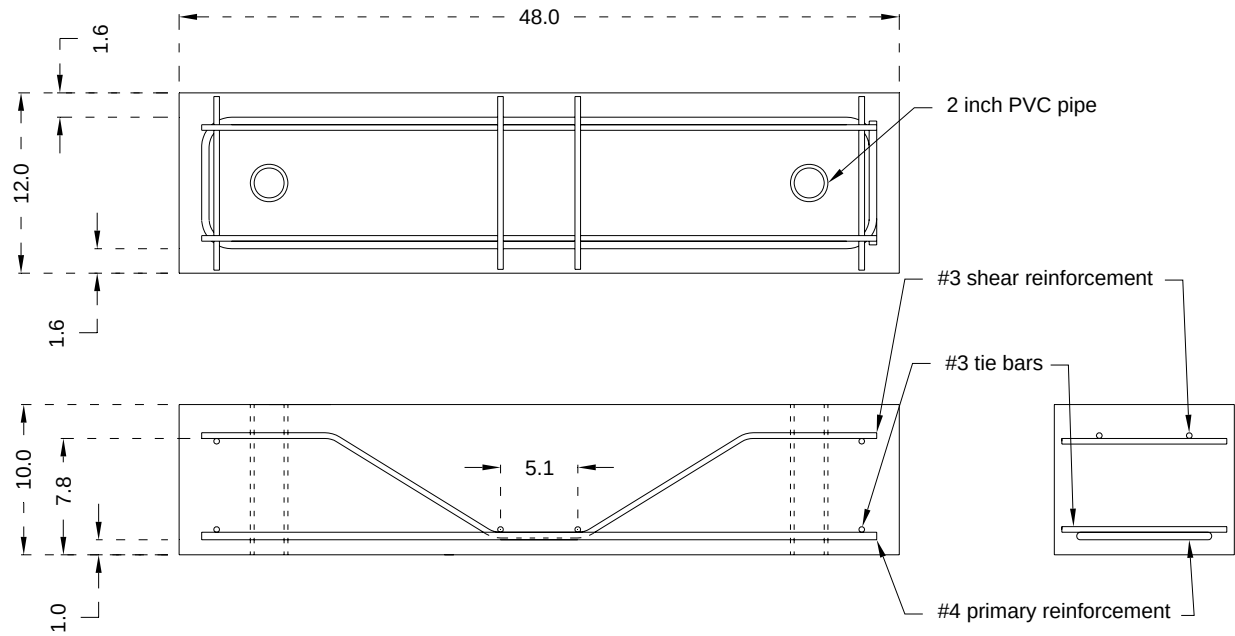


Figure 1. Schematic of concrete block dimensions and reinforcement detailing (units=inches).

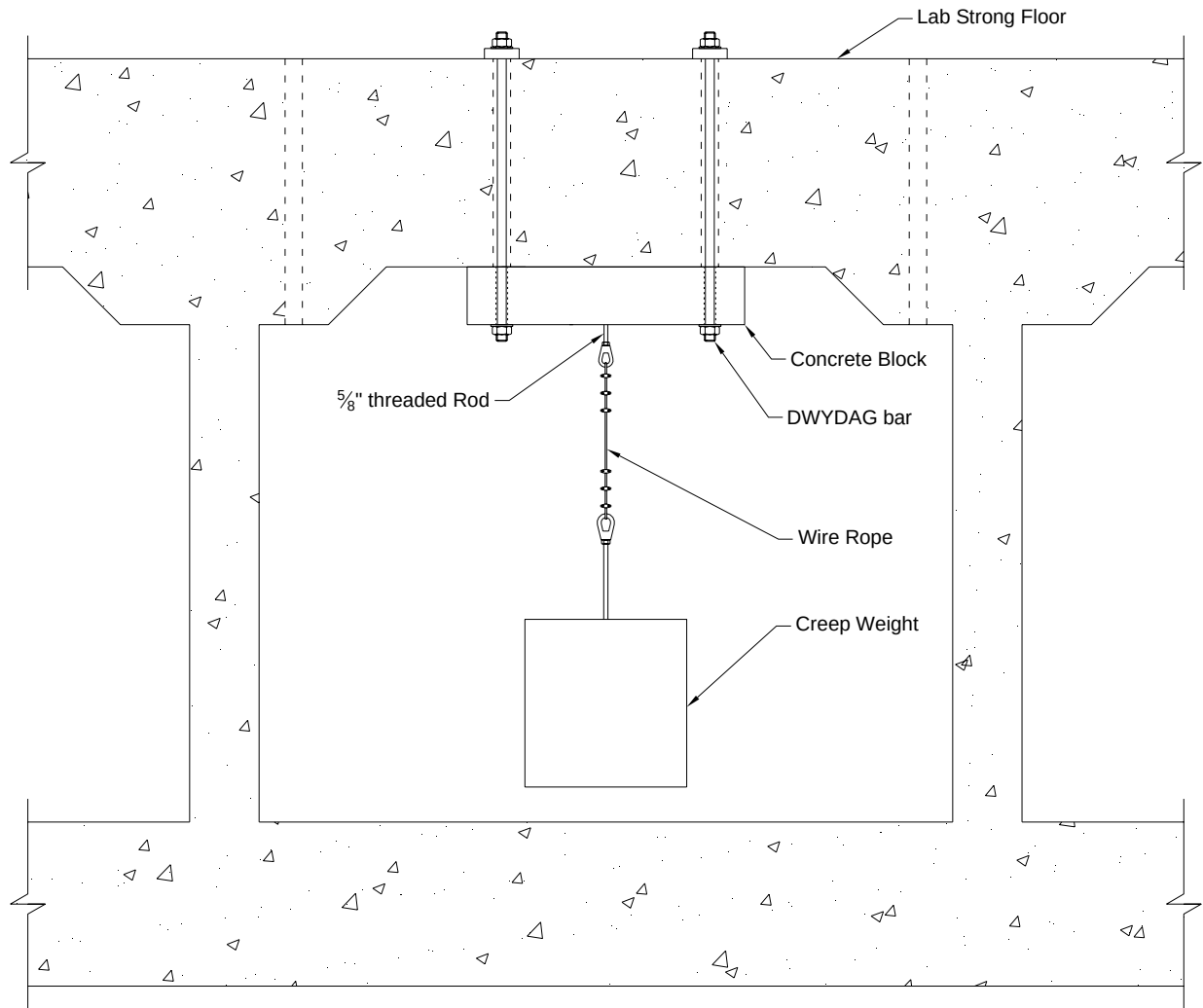


Figure 2. Cross-section (elevation) of basement tunnel showing layout of creep test.

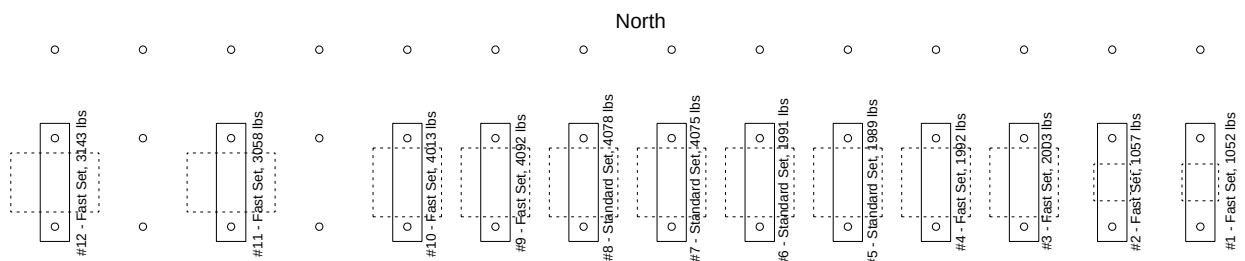


Figure 3. Plan view of basement tunnel showing layout of creep tests.

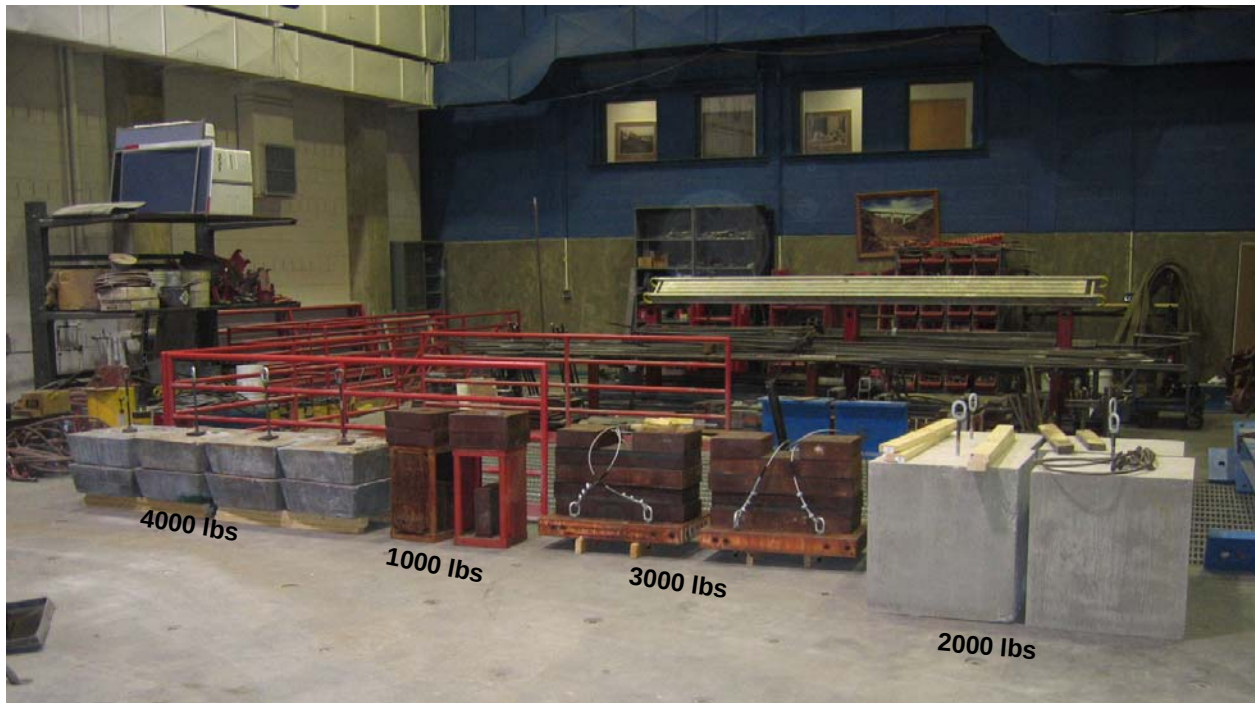


Figure 4. Creep weights. Left: 4000 lb from lead. Middle: 1000 and 3000 lb from steel plate. Right: 2000 lb concrete blocks.

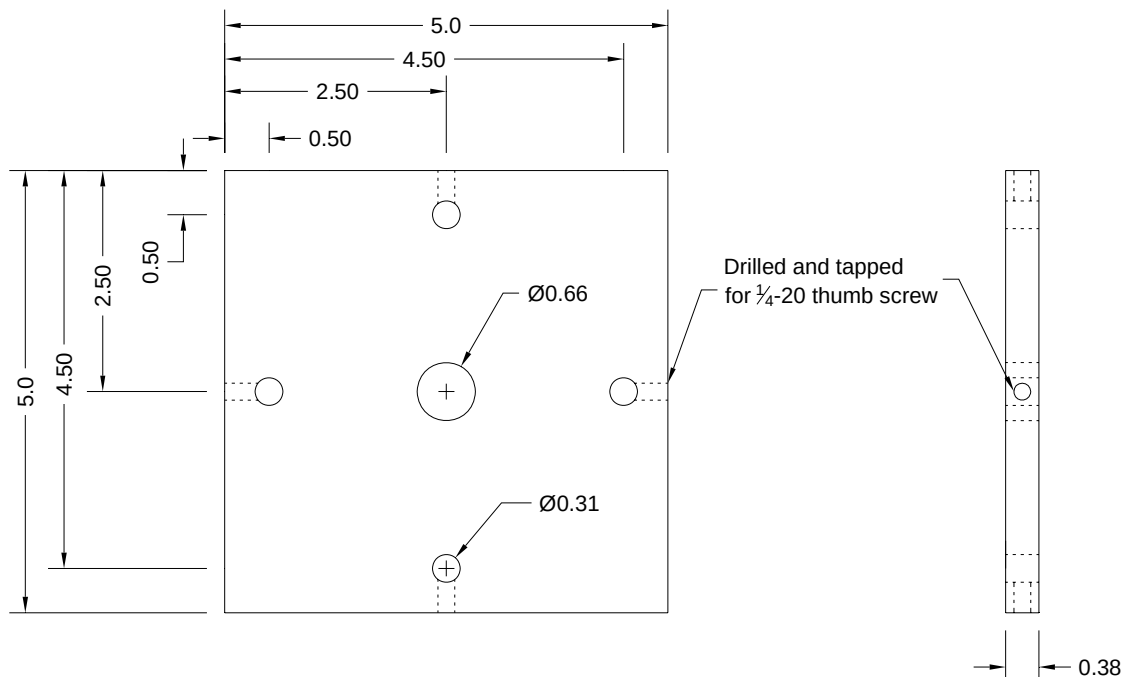
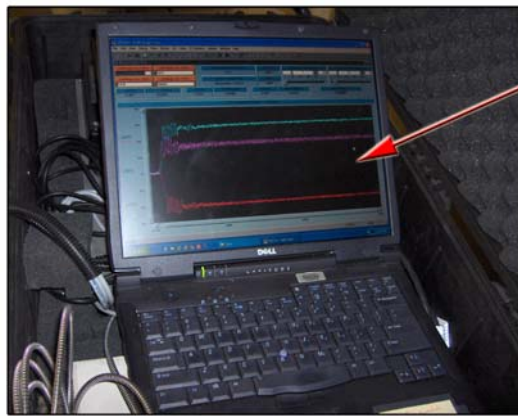
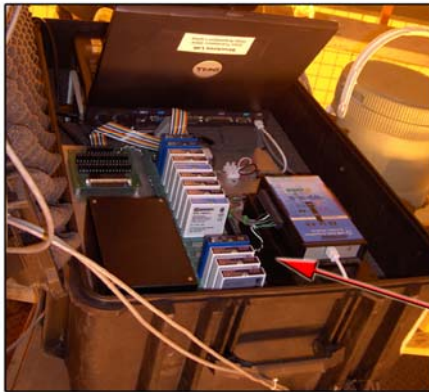


Figure 5. Aluminum LVDT bracket.



System Data Displayed During Testing

**System Collecting Data
During Testing**

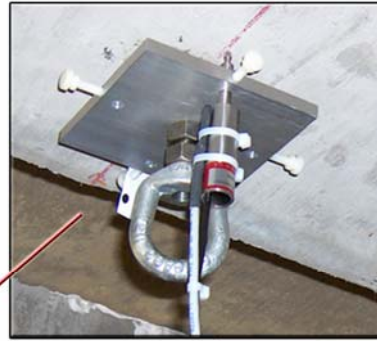


**Signal Conditionioing and
Data Acquisition System**

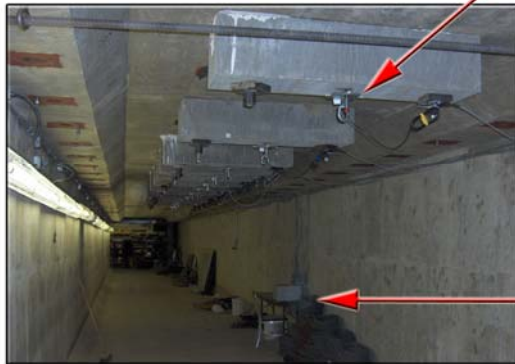
Figure 6. Short-Term system measuring three displacement sensors during weight drop.



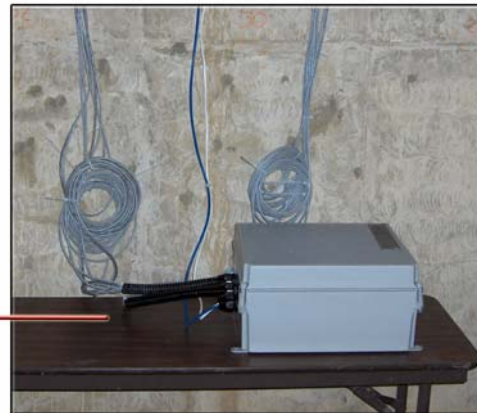
**Remote Workstation for
Data Storage / Display**



Displacement and Temperature Sensor



**System with 12 Displacement /
4 Temperature Sensors in Tunnel**



Data Acquisition System

Figure 7. Long-Term system measuring 12 displacement and 4 temperature sensors.

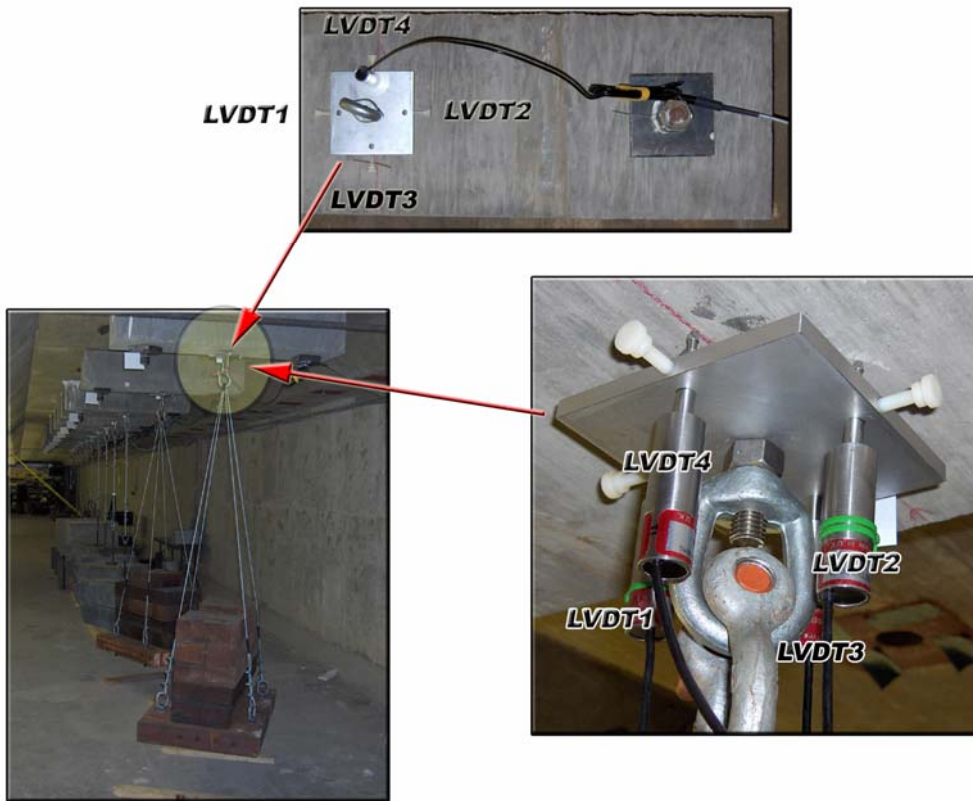


Figure 8. Sensor position and label information.

CHAPTER 3. CREEP TEST DATA

This chapter presents the time versus displacement data resulting from the sustained load testing for all twelve anchors in the TFHRC creep testing matrix. The anchors were monitored until they displaced completely from the concrete blocks or for a period of 82 days. Geometric curves fitted to this data to predict long-term anchor behavior to 600 days and 7 years are also described herein.

INDIVIDUAL ANCHOR DATA

Figures 9 through 20 plot the displacement behavior for each of the 12 anchors over the duration of testing. The data presented has been corrected for different sampling rates between the short and long-term data collection systems, bending in the anchor, and temperature variations in the lab. These corrections are described in detail in APPENDIX A. The scales used to present this data have been held consistent for 10 of the 12 plots. For the two anchors installed with Fast Set epoxy and supporting 4000 pound dead weights and which failed prior to the 82 day test duration, it was necessary to increase the displacement scale to display the full behavior history. A composite comparison of the anchor behaviors is made in the next section.

Unfortunately, over the Christmas holiday a loose connection in the long-term data collection system resulted in the loss of approximately 7.5 days of data for Anchors 5-12. This gap can be seen in the data plots for these anchors between 1 and 9 days after the transfer of load.

It should be recognized that the creep displacement of all the anchors tested were small, except for the two anchors that pulled completely out of the concrete blocks. In general, the displacements were so small that sensor drift, resolution, and temperature changes can have a significant effect on the measurements. These effects can be seen in the data collected from the anchors installed with Standard Set epoxy. This data looks noisy because the magnitude of the displacement measured is affected by the sensor resolution.

Daily variations in temperature were accounted for in the displacement records based on a manufacturer recommended factor. The temperature change of the TFHRC tunnel followed a downward trend for the first 50 days and a rising trend for the last 30 days. A graph of the TFHRC tunnel temperature can be found in APPENDIX A.

Within Figures 9-16 and 19-20 are four curves in addition to the test data which are empirical fits using different mathematical models. The four curves represent two different mathematical models fitted to two subsets of the test data. Each of these curves were fit by least-squares regression to a subset of the experimental data. The resulting empirical models allow for displacement predictions for much larger time periods than were tested. The first model (see Eqn. 1), herein referred to as the Logarithmic model, is a two parameter logarithmic law which is recommended by the ASTM E 1512 standard⁷ for creep data produced at 110°F and at 40% of the expected average ultimate load. The ASTM standard recommends a creep test be performed

a minimum of 42 days and creep results should be reported by extrapolating out to 600 days with the intent of using a model fitted to the last 20 days of data produced. The second model (see Eqn. 2), herein referred to as the Polynomial model, is a three-parameter polynomial. Each of these models were fitted to two subsets of the test data; the data from the last 20 days of testing and the data from day 10 to the end of testing. The constants derived through regression for each model and anchor data set are reflected in Table 4.

$$\Delta = A \ln(\text{time}) + B \quad \text{Eqn.1}$$

$$\Delta = C(\text{time}^D) + E(\text{time}) \quad \text{Eqn. 2}$$

where A - E are constants, and Δ is the displacement

The data sets were chosen in an effort to minimize the affects of the initial mechanical behavior of the adhesive anchor on the longer term creep displacement predictions that resulted from the modeling. For all anchors, considering the larger data set with both models resulted in lower creep displacement predictions than when using the smaller data set. The smaller data set (last 20 days of testing) was highly influenced by an increase in displacement rate evident in all anchor test data that is most likely due to the change in laboratory temperature trend that occurred during this same period of time (see Figure 51).

The empirical models were used to extrapolate the experimental data out to 600 days (as ASTM E 1512 recommends) and seven years. Seven years was chosen because this was the approximate age of the anchors in the I-90 Seaport Portal Tunnel when the collapse occurred. The results for all the anchors are presented in Tables 5 and 6 for the two respective time periods. Regardless of the data set considered, the Logarithmic model predicted much lower creep displacements at 600 days and 7 years than the Polynomial model as this logarithm based model assumes asymptotic behavior (a creep displacement rate approaching zero) over long periods of time.

Figure 21 shows the creep displacement projections to 600 days for Anchor #4 but is typical of the results for all projections made on the test data. Anchor #4 was chosen for this illustration because it was loaded at approximately the sustained load level of those anchors installed in the I-90 tunnel. The Polynomial model which is dominated by its linear term diverges from the Logarithmic model that relies solely on a logarithmic decay regardless of the data set considered. For this anchor at 600 days, the Logarithmic model predicts approximately a third of the creep displacement resulting from the Polynomial model.

Figures 17 and 18 show the creep data for Anchors #9 and #10 which both failed prior to the end of testing. The performance history of these anchors, both loaded with approximately 4000 pounds, indicates that at this load level and temperature, a critical displacement of approximately 0.2 inches exists after which the rate of displacement rapidly increases until failure occurs soon afterwards. Both of these two anchors were installed with Fast Set epoxy. Although the 0.2 inch displacement threshold is most certainly load level, temperature and adhesive dependent, it was

used conservatively with the empirical models to make comparisons of predicted time to failure for each of the anchors tested. The results of this analysis are presented in Table 7.

As shown in Table 7, anchors installed with Standard Set epoxy would be expected to fail between 13.1 and 55.3 years when using the Polynomial model and much greater than 100 years when using the Logarithmic model. These results are quite scattered because of the relatively shallow slope of the curves and the subtle differences in the equations extrapolated over very long periods of time. For the anchors installed with Fast Set epoxy, the assumed failure displacement of 0.2 inches could be reached in 0.2-6.1 years depending on the sustained load level using the Polynomial model and in most cases well over 100 years using the Logarithmic model (except Anchor 11 which is predicted to fail in 53 years using the Logarithmic model). These results are consistent with the actual observed behavior in the I-90 Tunnel.

Shown in Figure 22 is the time to displace 0.2 inches versus the bond stress induced by the sustained load, for Fast Set anchors using the Polynomial model. The bond stress is calculated by dividing the sustained load by the measured bond area which was evaluated after the load-displacement testing described in CHAPTER 4. This figure shows a decreasing linear trend with higher levels of stress corresponding to short periods of time to displace 0.2 inches. Not enough data was generated in this program to make similar conclusions for the Standard Set epoxy.

ANCHOR COMPARISONS

Figure 23 plots the time-displacement curves for all eight anchors that used Fast Set epoxy and Figure 24 plots the same curves for the Standard Set epoxy anchors. For the most part, the results are fairly repeatable among the duplicates.

Referring to Figure 23, there is a correlation between the magnitude of sustained load and the rate at which creep occurs. This trend appears to be linearly increasing for the data shown from the 1000, 2000, and 3000 pound sustained load tests. However, there is a drastic increase in rate of creep displacement at the 4000 pound sustained load level. In fact, the two anchors resisting the 4000 pound weights failed 63.2 and 75.8 days after load transfer. This indicates that there is a critical load between 3000 and 4000 pounds where the anchor adhesive as tested diverges from this linear behavior. Referring to the Standard Set epoxy behavior curves shown in Figure 24, a similar linear correlation is evident between load level and displacement rate based on the limited data shown. Creep displacements are twice as large for this material when the sustained load is doubled.

The dramatic difference in behavior between the Standard and Fast Set epoxies can be seen in Figures 25 and 26. Figure 25 plots the time-displacement behavior for the four anchors that sustained the 2000 pound loads during testing. The Standard Set epoxy anchors exhibited approximately 6 times less creep displacement than those anchors that used the Fast Set epoxy. Figure 26 makes this same comparison for the anchors loaded with 4000 pounds. The anchors using Fast Set epoxy completely failed after an average of 69 days but those using the Standard Set epoxy displaced no more than 0.01 inches during that same time period.

RECOVERY BEHAVIOR

After 82 days, the creep tests were concluded and the weights were removed from the remaining anchors. The long-term data collection system continued to collect data for ~22 hours on the unloaded anchors. This 22 hours of data represents the recovery of the displaced anchor. The data collected during recovery is separated from the remaining time-displacement curves so that comparisons between anchors could be easily illustrated.

Figure 27 plots the 10 recovery curves for the creep specimens that did not fail. The displacements are negative indicating the anchor retracted into the hole. The data for the anchors in Fast Set epoxy at the 1000 and 2000 pound load levels indicates that more recovery occurs in the adhesive that had sustained the heavier loads. One of the anchors loaded at 3000 pounds continues this trend, but the adhesive for the other anchor loaded at 3000 pounds, Anchor #11, actually recovered less than the two anchors loaded at 1000 pounds. Anchor #11 was not fully embedded to 5.0 inches and had large voids which could reasonably explain these differences in its behavior. The anchors that used Standard Set epoxy recovered very little, if at all, suggesting that the magnitude of the recovery is related to the magnitude of creep displacements previously incurred.

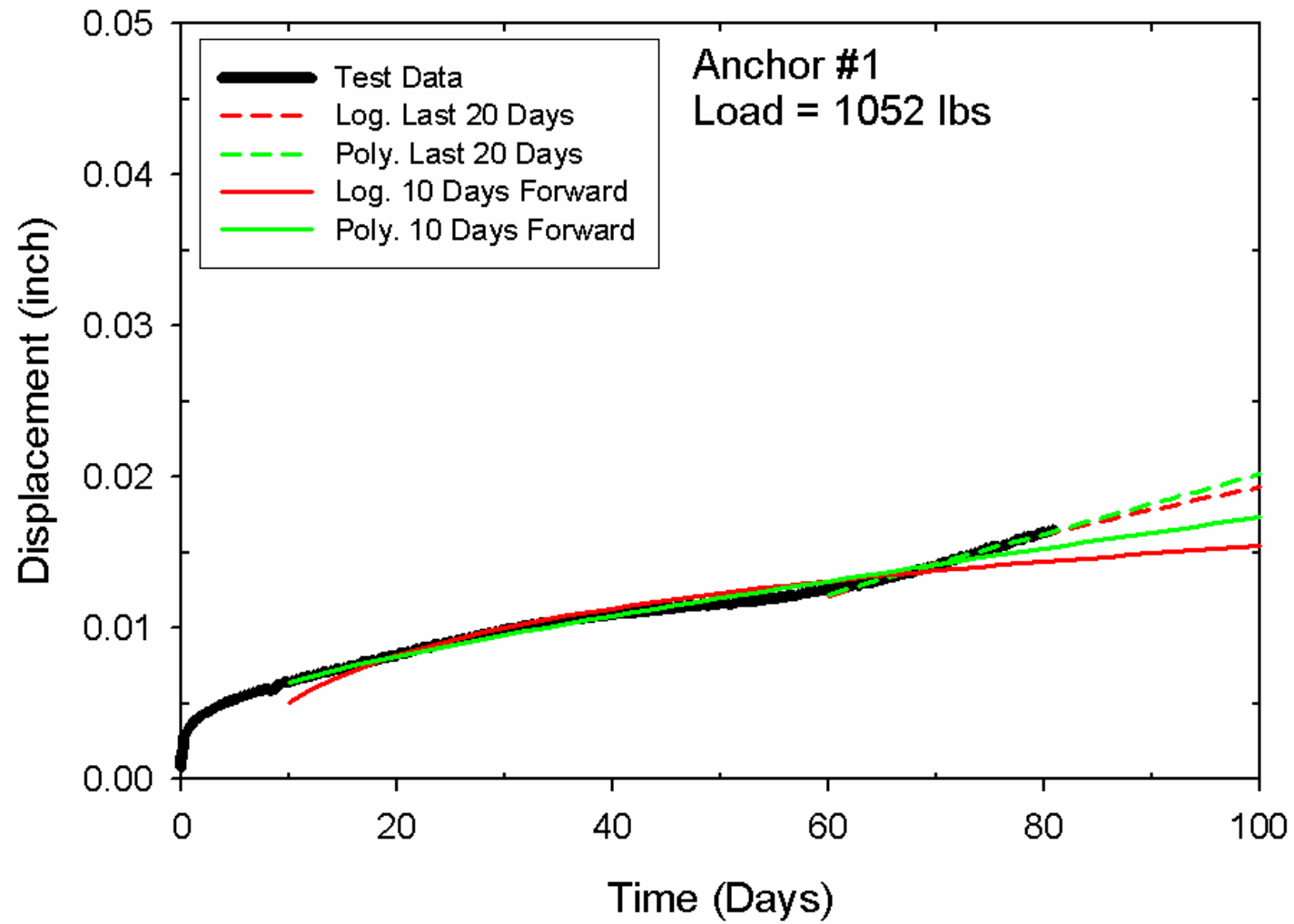


Figure 9. Full time/displacement history for Anchor 1.

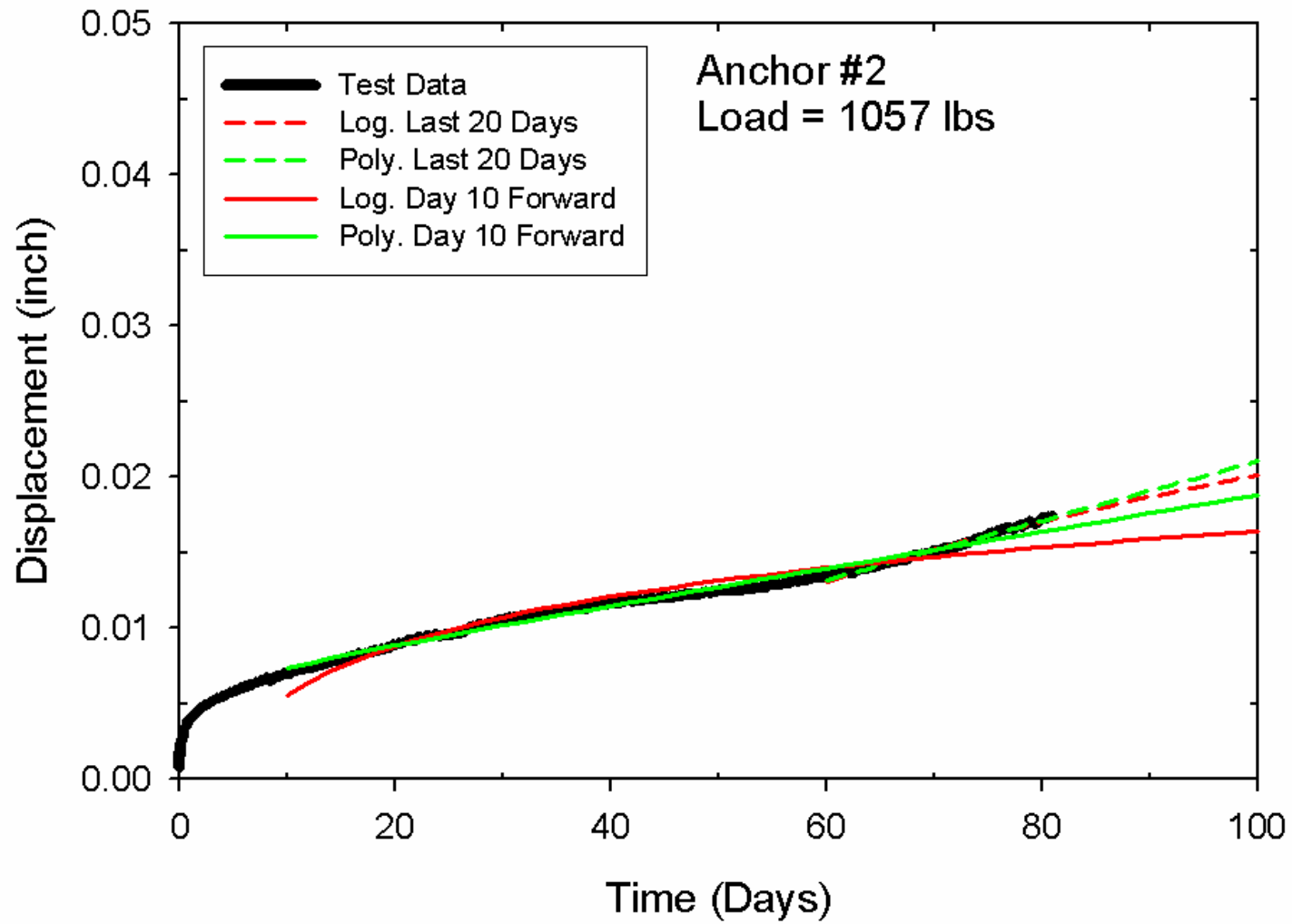


Figure 10. Full time/displacement history for Anchor 2.

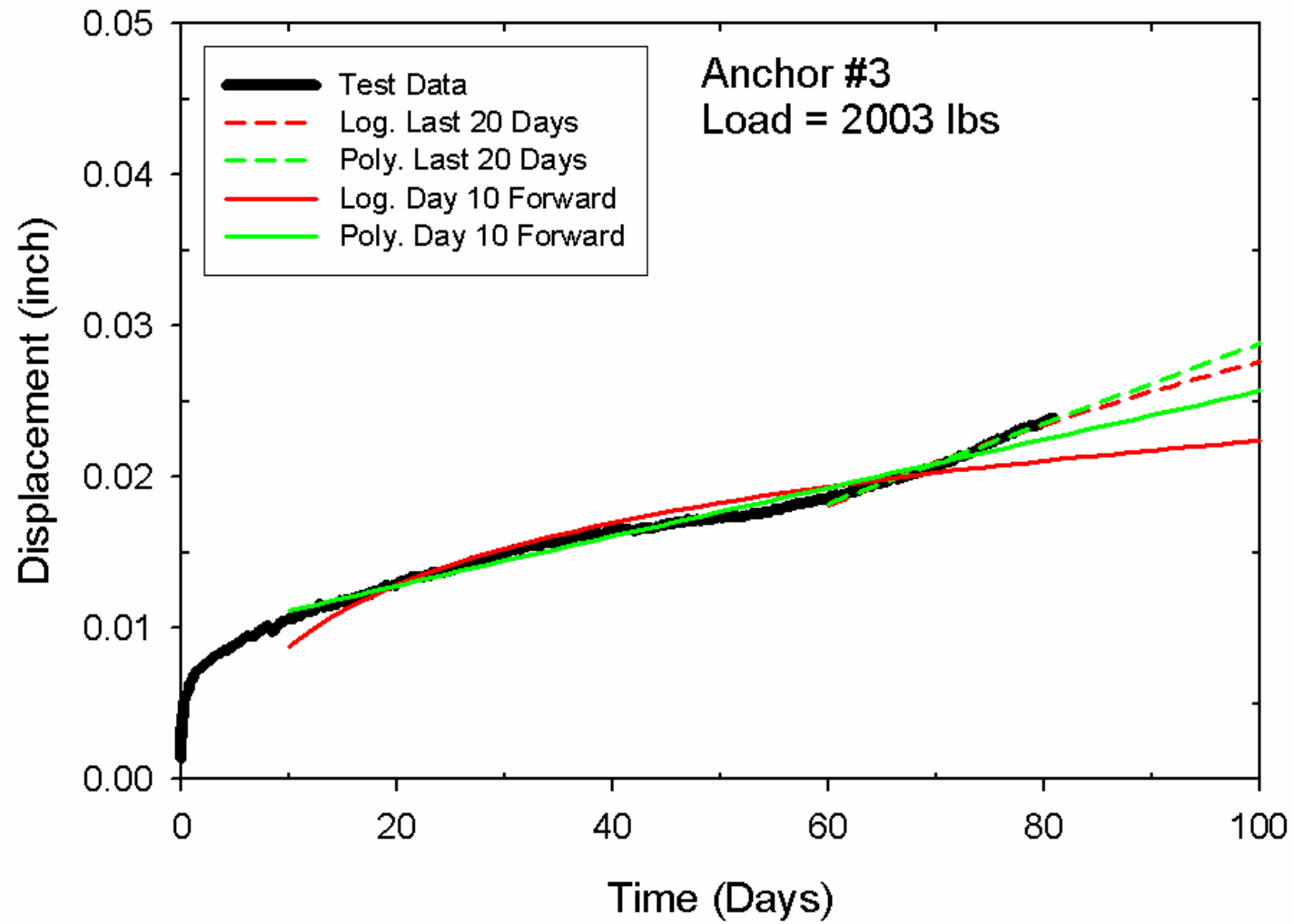


Figure 11. Full time/displacement history for Anchor 3.

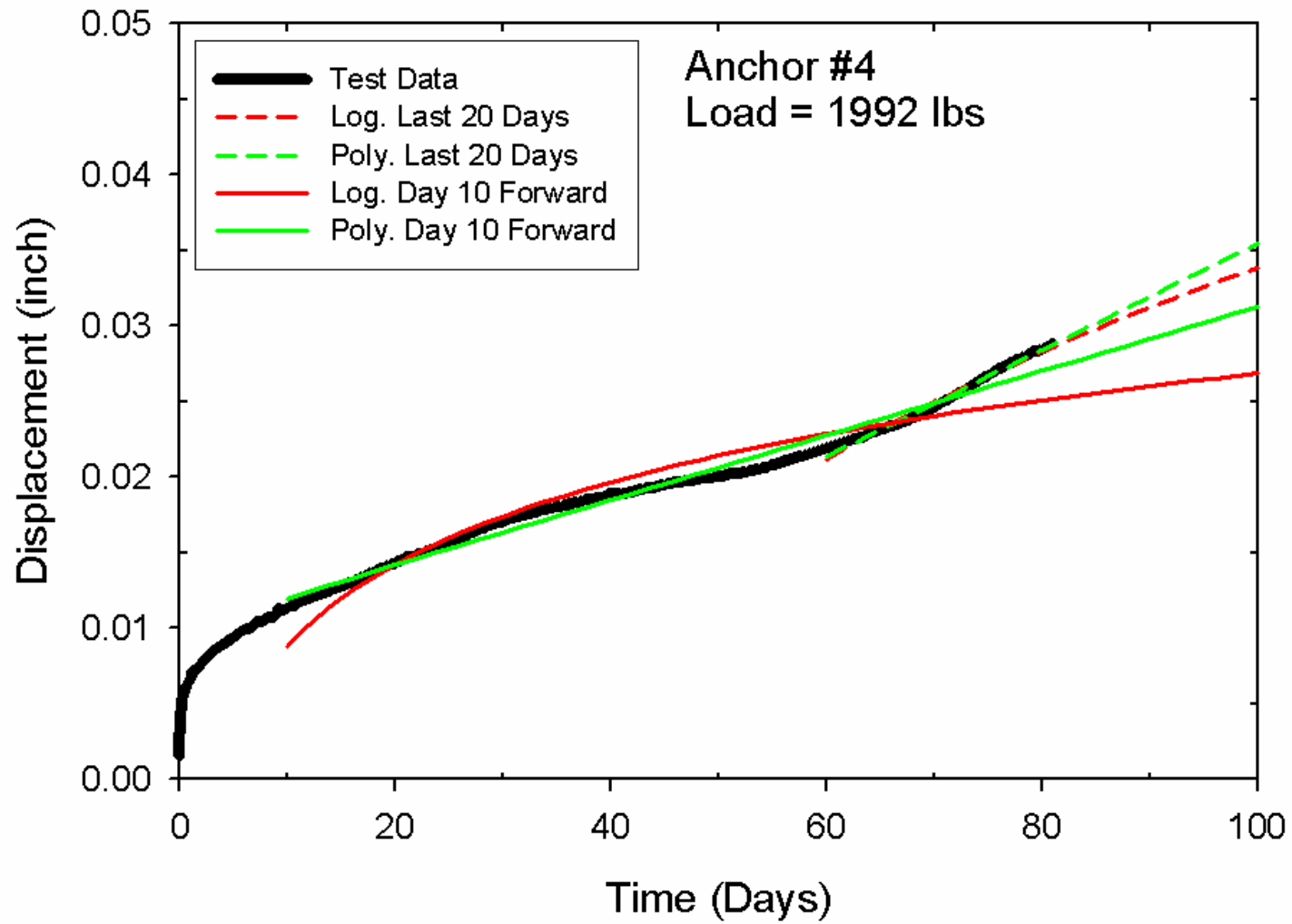


Figure 12. Full time/displacement history for Anchor 4.

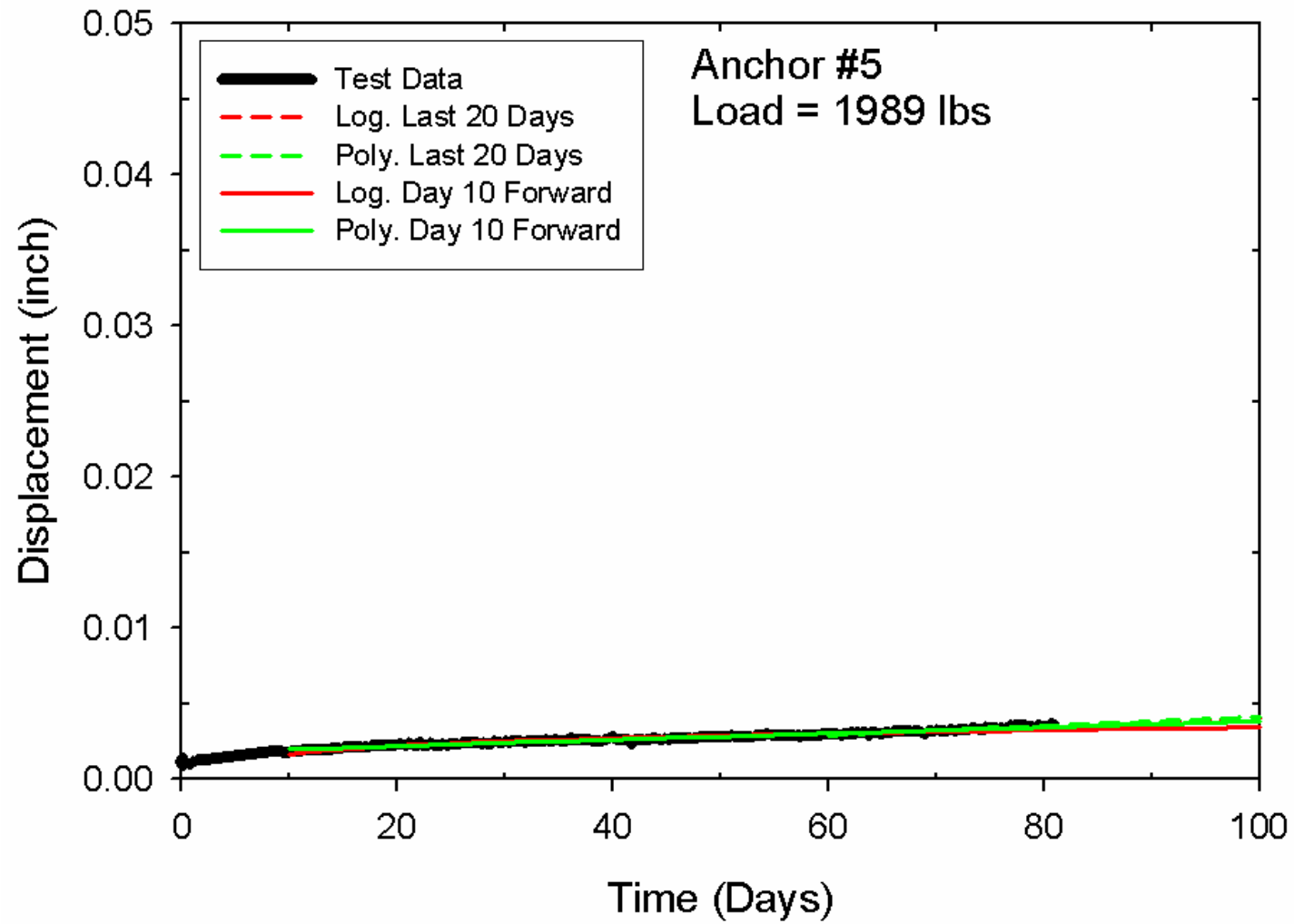


Figure 13. Full time/displacement history for Anchor 5.

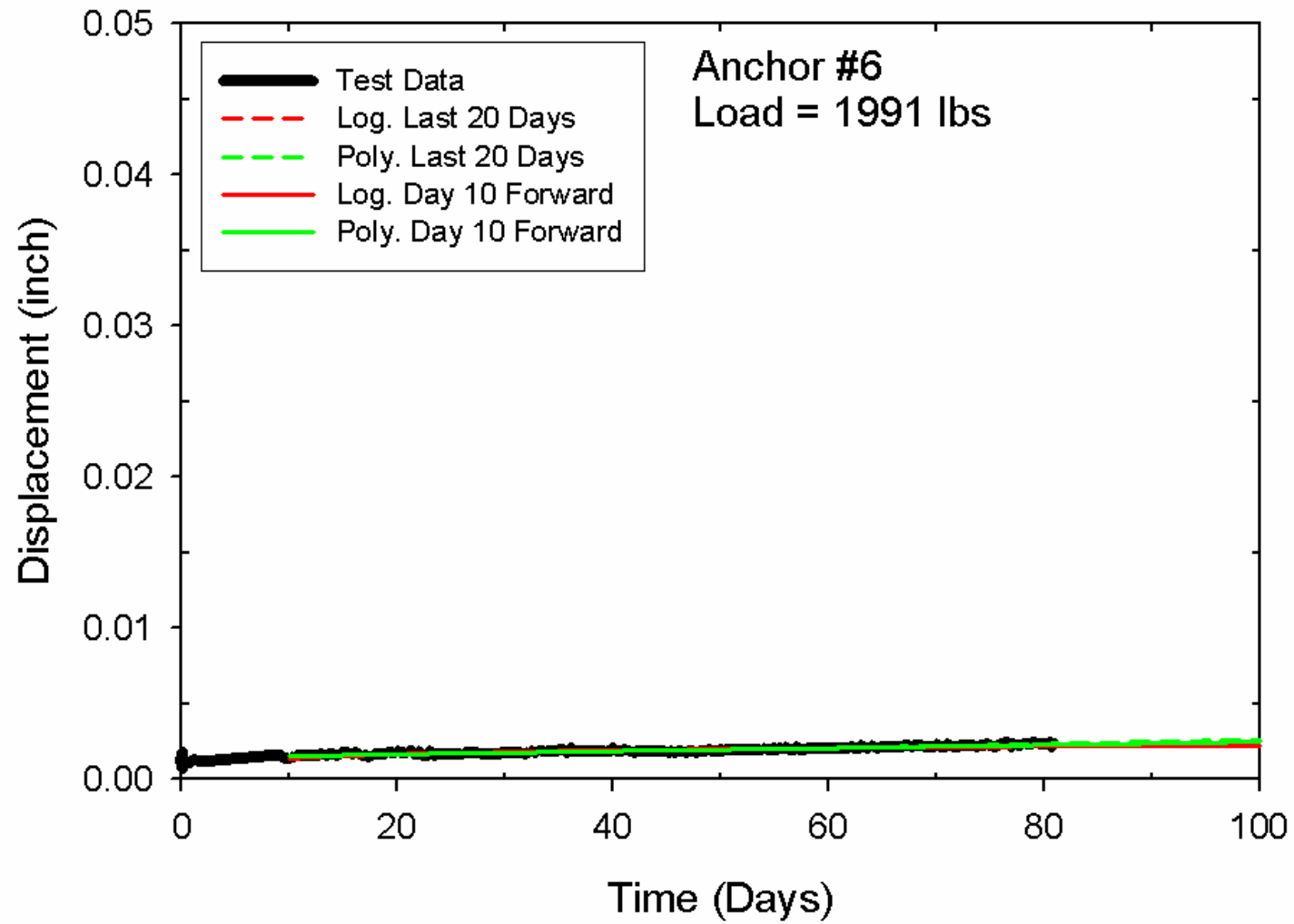


Figure 14. Full time/displacement history for Anchor 6.

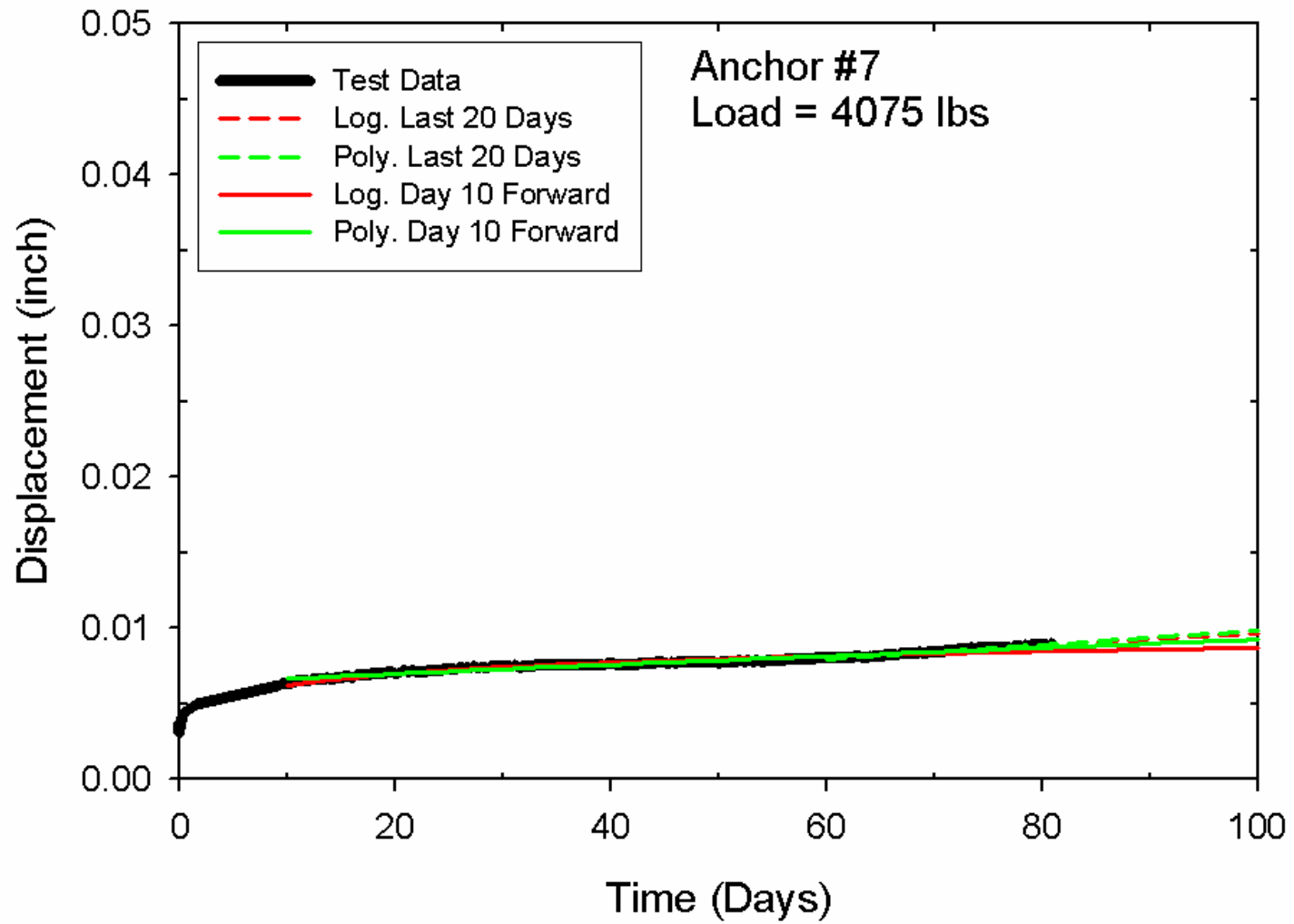


Figure 15. Full time/displacement history for Anchor 7.

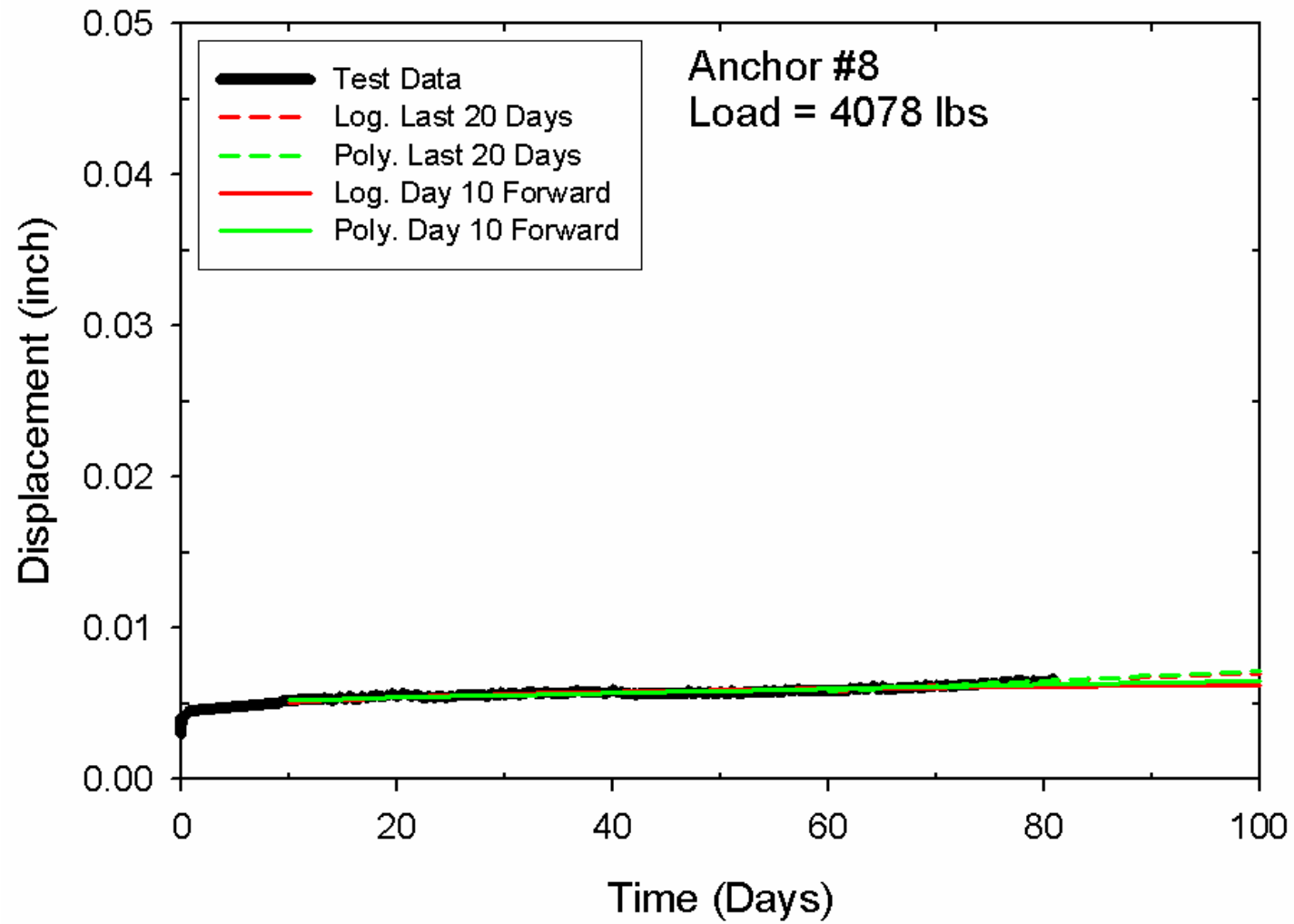


Figure 16. Full time/displacement history for Anchor 8.

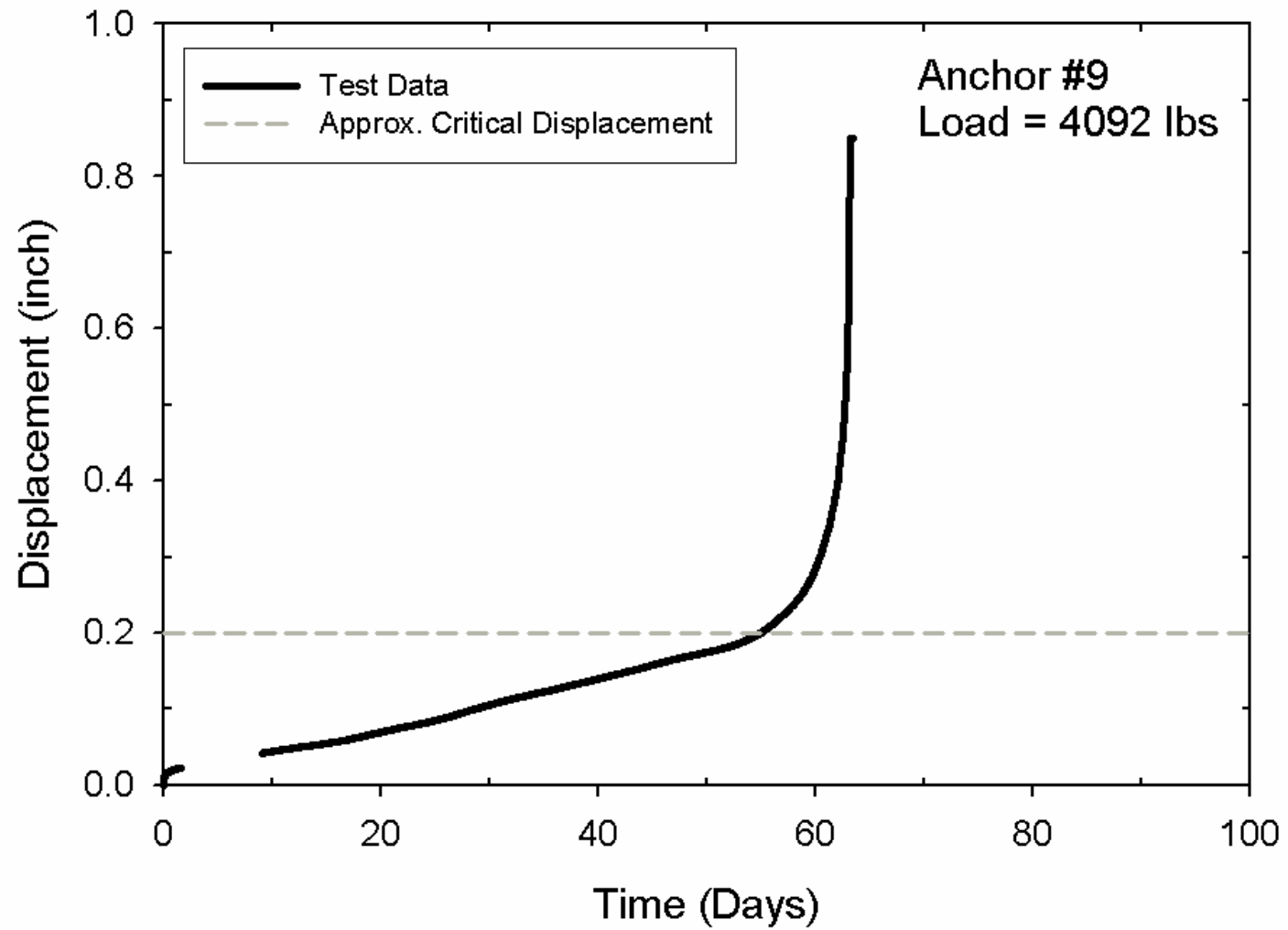


Figure 17. Full time/displacement history for Anchor 9.

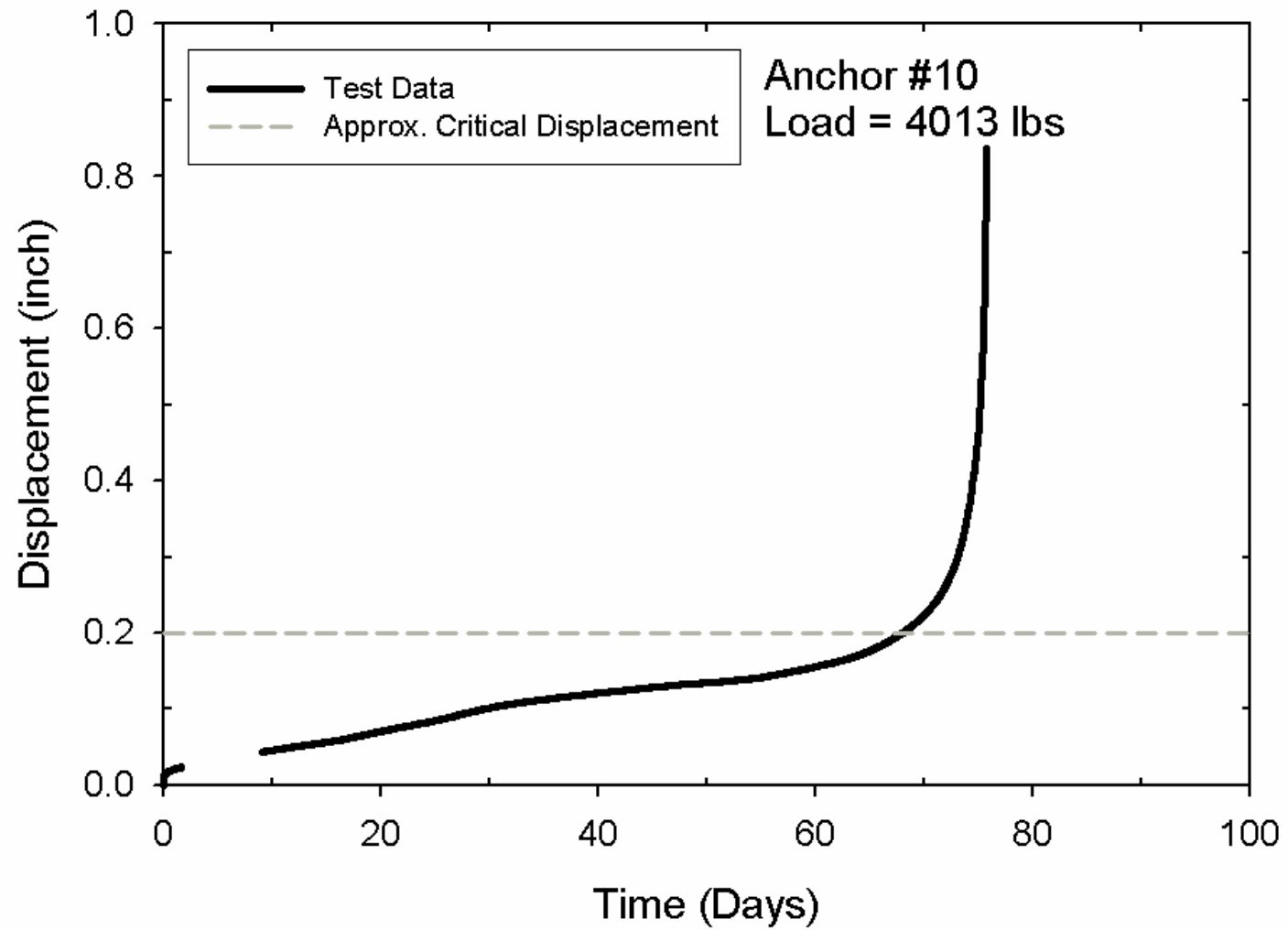


Figure 18. Full time/displacement history for Anchor 10.

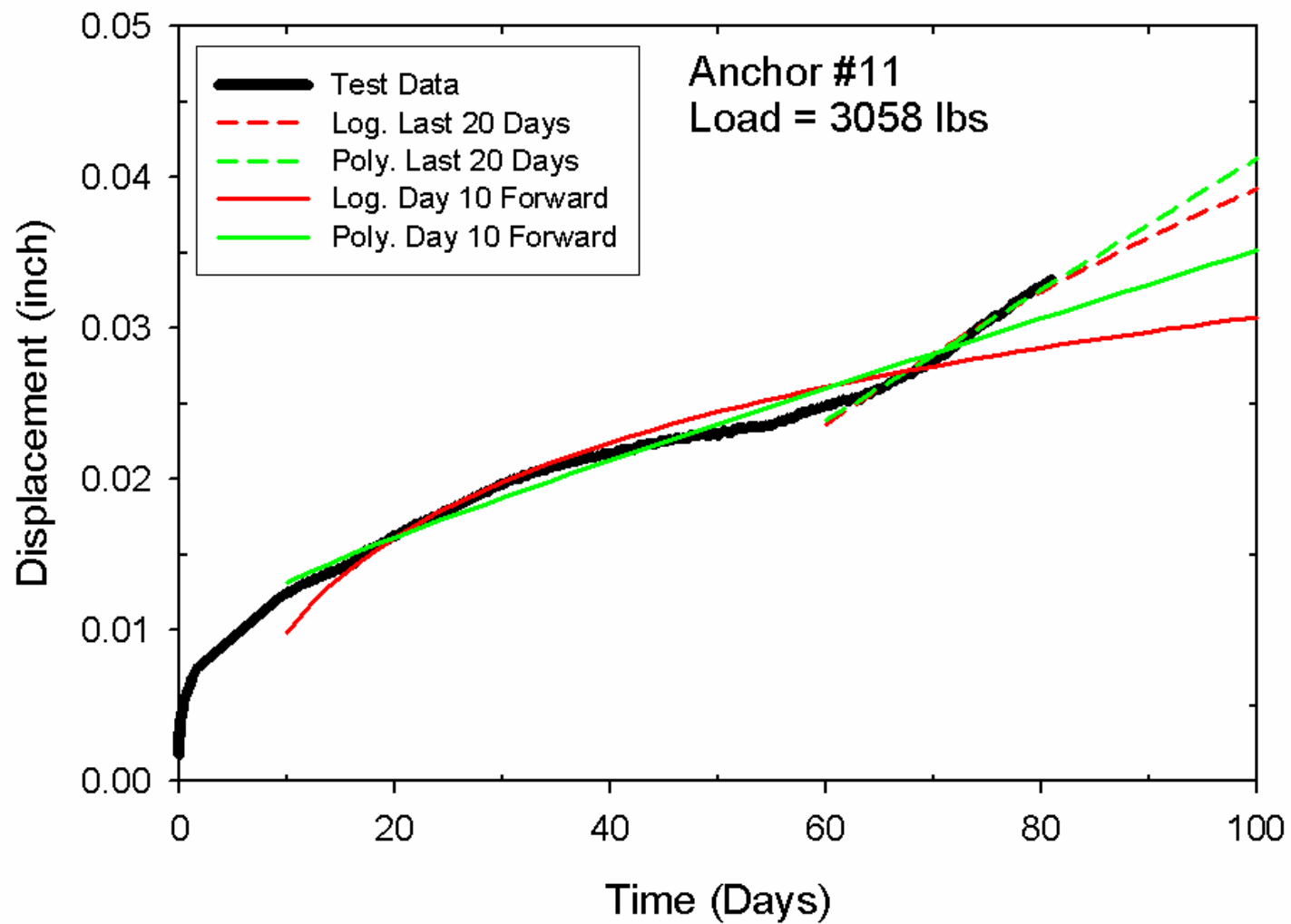


Figure 19. Full time/displacement history for Anchor 11.

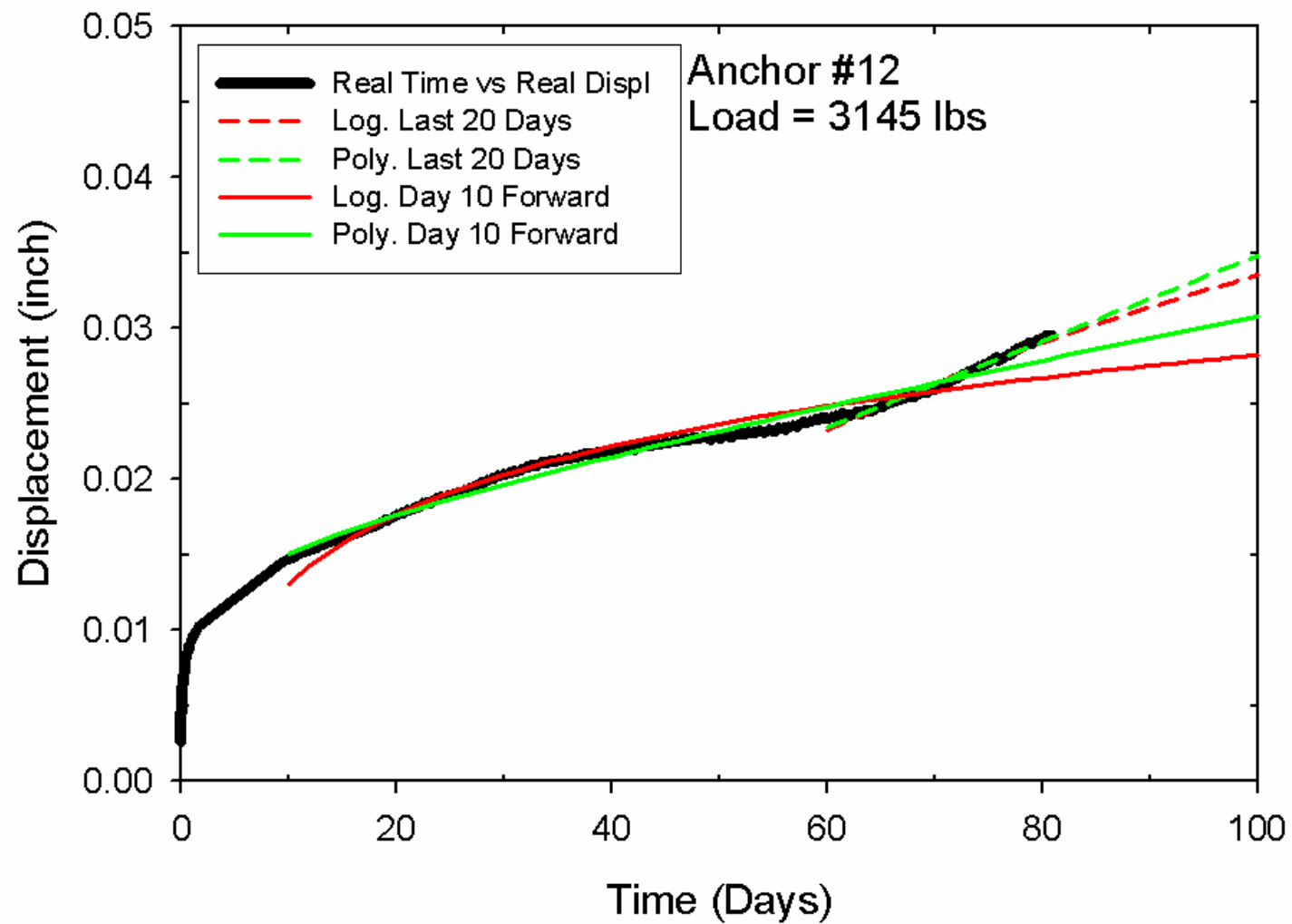


Figure 20. Full time/displacement history for Anchor 12.

Table 4 Constants for Empirical Models

	Logarithmic model		Polynomial Model	
Specimen	10 day forward $\Delta=A\ln(t)+B$	Last 20 days $\Delta=A\ln(t)+B$	10 day forward $\Delta=Ct^D+Et$	Last 20 days $\Delta=Ct^D+Et$
1 (1052 lbs)	A= 4.484×10^{-3} inch B= -5.247×10^{-3} inch	A= 1.413×10^{-2} inch B= -4.576×10^{-2} inch	C= 3.451×10^{-3} inch/day D= 2.097×10^{-1} E= 8.274×10^{-5} inch/day	C= 2.528×10^{-4} inch/day D= 9.794×10^{-2} E= 1.979×10^{-4} inch/day
2 (1057 lbs)	A= 4.703×10^{-3} inch B= -5.278×10^{-3} inch	A= 1.387×10^{-2} inch B= -4.374×10^{-2} inch	C= 5.250×10^{-3} inch/day D= 7.298×10^{-2} E= 1.144×10^{-4} inch/day	C= 9.995×10^{-4} inch/day D= 9.974×10^{-2} E= 1.947×10^{-4} inch/day
3 (2003 lbs)	A= 5.927×10^{-3} inch B= -4.904×10^{-3} inch	A= 1.866×10^{-2} inch B= -5.831×10^{-2} inch	C= 9.245×10^{-3} inch/day D= 1.442×10^{-2} E= 1.582×10^{-4} inch/day	C= 1.717×10^{-3} inch/day D= 9.794×10^{-2} E= 2.614×10^{-4} inch/day
4 (1992 lbs)	A= 7.858×10^{-3} inch B= -9.336×10^{-3} inch	A= 2.488×10^{-2} inch B= -8.074×10^{-2} inch	C= 9.522×10^{-3} inch/day D= 1.443×10^{-2} E= 2.111×10^{-4} inch/day	C= 1.171×10^{-4} inch/day D= 9.793×10^{-2} E= 3.529×10^{-4} inch/day
5 (1989 lbs)	A= 7.608×10^{-4} inch B= -1.021×10^{-4} inch	A= 2.107×10^{-3} inch B= -5.733×10^{-3} inch	C= 1.702×10^{-3} inch/day D= 1.430×10^{-2} E= 2.003×10^{-5} inch/day	C= 8.069×10^{-4} inch/day D= 9.794×10^{-2} E= 2.841×10^{-5} inch/day
6 (1991 lbs)	A= 3.727×10^{-4} inch B= 5.232×10^{-4} inch	A= 9.865×10^{-4} inch B= -2.011×10^{-3} inch	C= 1.370×10^{-3} inch/day D= 1.430×10^{-2} E= 9.959×10^{-6} inch/day	C= 8.698×10^{-4} inch/day D= 9.794×10^{-2} E= 1.229×10^{-5} inch/day
7 (4075 lbs)	A= 1.069×10^{-3} inch B= 3.763×10^{-3} inch	A= 3.394×10^{-3} inch B= -6.000×10^{-3} inch	C= 6.168×10^{-3} inch/day D= 1.431×10^{-2} E= 2.649×10^{-5} inch/day	C= 3.682×10^{-3} inch/day D= 9.794×10^{-2} E= 4.043×10^{-5} inch/day
8 (4078 lbs)	A= 5.153×10^{-4} inch B= 3.866×10^{-3} inch	A= 2.380×10^{-3} inch B= -3.955×10^{-3} inch	C= 4.925×10^{-3} inch/day D= 1.429×10^{-2} E= 1.263×10^{-5} inch/day	C= 2.762×10^{-3} inch/day D= 9.794×10^{-2} E= 2.805×10^{-5} inch/day
9 (4092 lbs)	Not applicable	Not applicable	Not applicable	Not applicable
10 (4013 lbs)	Not applicable	Not applicable	Not applicable	Not applicable
11 (3058 lbs)	A= 9.060×10^{-3} inch B= -1.099×10^{-2} inch	A= 3.057×10^{-2} inch B= -1.016×10^{-1} inch	C= 8.708×10^{-3} inch/day D= 1.049×10^{-1} E= 2.107×10^{-4} inch/day	C= -1.588×10^{-3} inch/day D= 9.790×10^{-2} E= 4.377×10^{-4} inch/day
12 (3143 lbs)	A= 6.565×10^{-3} inch B= -2.026×10^{-3} inch	A= 2.015×10^{-2} inch B= -5.928×10^{-2} inch	C= 1.006×10^{-2} inch/day D= 1.395×10^{-1} E= 1.168×10^{-4} inch/day	C= 4.553×10^{-3} inch/day D= 9.787×10^{-2} E= 2.768×10^{-4} inch/day

Table 5 Empirical Model Results at 600 Days For All Creep Specimens

		Predicted displacement at 600 days			
		Logarithmic model		Polynomial Model	
Specimen	Displacement after 82 days (inch)	10 days forward (inch)	Last 20 days (inch)	10 days forward (inch)	Last 20 days (inch)
1 (1052 lbs)	0.0165	0.0234	0.0447	0.0628	0.1192
2 (1057 lbs)	0.0174	0.0248	0.0450	0.0770	0.1187
3 (2003 lbs)	0.0239	0.0330	0.0610	0.1050	0.1600
4 (1992 lbs)	0.0288	0.0409	0.0784	0.1371	0.2120
5 (1989 lbs)	0.0036	0.0048	0.0077	0.0139	0.0186
6 (1991 lbs)	0.0023	0.0029	0.0043	0.0075	0.0090
7 (4075 lbs)	0.0089	0.0106	0.0157	0.0227	0.0311
8 (4078 lbs)	0.0066	0.0072	0.0113	0.0130	0.0220
9 (4092 lbs)	Failed in 63.2 days	Not Applicable	Not Applicable	Not Applicable	Not Applicable
10 (4013 lbs)	Failed in 75.8 days	Not Applicable	Not Applicable	Not Applicable	Not Applicable
11 (3058 lbs)	0.0332	0.0470	0.0940	0.1435	0.2596
12 (3143 lbs)	0.0295	0.0400	0.0696	0.0946	0.1746
Logarithmic model assumes the form; $\Delta = A \ln(t) + B$ Polynomial Model assumes the form; $\Delta = Ct^D + Et$ Where A-H are constants, Δ is the displacement , and t is time in days					

Table 6 Empirical Model Results at 7 Years For All Creep Specimens

		Predicted displacement at 7 years			
		Logarithmic model		Polynomial Model	
Specimen	Displacement after 82 days (inch)	10 days forward (inch)	Last 20 days (inch)	10 days forward (inch)	Last 20 days (inch)
1 (1052 lbs)	0.0165	0.0299	0.0651	0.2294	0.5065
2 (1057 lbs)	0.0174	0.0316	0.0651	0.3018	0.4998
3 (2003 lbs)	0.0239	0.0416	0.0881	0.4147	0.6717
4 (1992 lbs)	0.0288	0.0523	0.1145	0.5502	0.9023
5 (1989 lbs)	0.0036	0.0059	0.0108	0.0531	0.0743
6 (1991 lbs)	0.0023	0.0034	0.0057	0.0270	0.0333
7 (4075 lbs)	0.0089	0.0122	0.0206	0.0746	0.1113
8 (4078 lbs)	0.0066	0.0079	0.0147	0.0378	0.0777
9 (4092 lbs)	Failed in 63.2 days	Not Applicable	Not Applicable	Not Applicable	Not Applicable
10 (4013 lbs)	Failed in 75.8 days	Not Applicable	Not Applicable	Not Applicable	Not Applicable
11 (3058 lbs)	0.0332	0.0601	0.1383	0.5584	1.1153
12 (3143 lbs)	0.0295	0.0495	0.0988	0.3287	0.7172
Logarithmic model assumes the form; $\Delta = A \ln(t) + B$ Polynomial Model assumes the form; $\Delta = Ct^p + Et$ Where $A-H$ are constants, Δ is the displacement, and t is time in days					

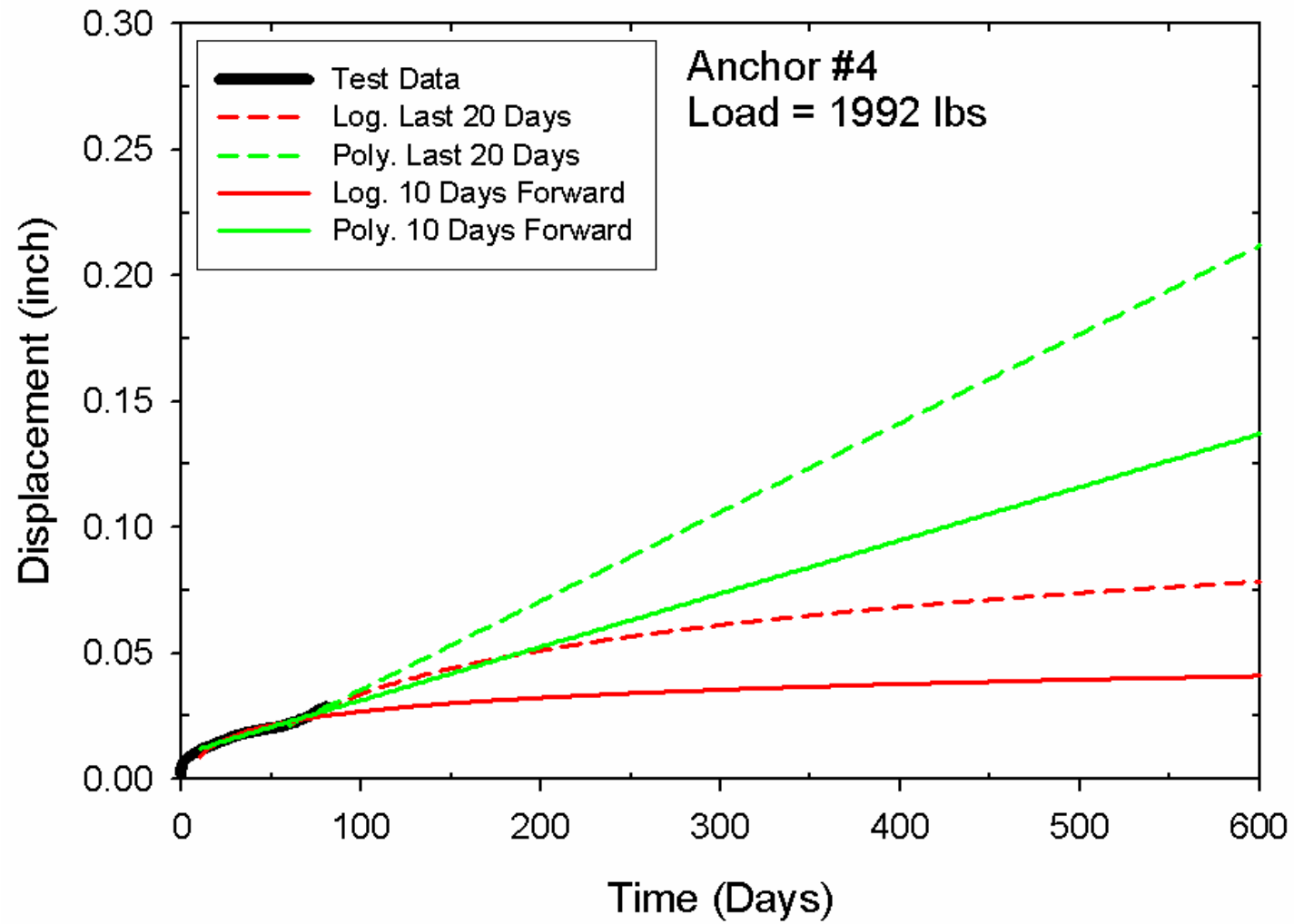


Figure 21. Creep Displacement Projections to 600 Days for Anchor 4.

Table 7 Time Predictions to Attain 0.2 Inches of Displacement

			Predicted time to creep 0.2 inches (years)			
			Logarithmic model		Polynomial Model	
Anchor	Bond Area ^a (inch ²)	Creep Stress ^b (psi)	10 days Forward	Last 20 days	10 days Forward	Last 20 days
1 (1052 lbs)	8.67	121	2.8×10^{27}	9.9×10^4	6.1	2.8
2 (1057 lbs)	9.13	116	2.3×10^{16}	1.2×10^5	4.6	2.8
3 (2003 lbs)	9.84	204	3.1×10^{12}	2909.9	3.3	2.1
4 (1992 lbs)	7.69	259	1.1×10^9	215.4	2.5	1.6
5 ^c (1989 lbs)	8.60	231	7.1×10^{37}	1.4×10^{31}	27.5	19.4
6 ^c (1991 lbs)	9.81	203	4.5×10^{66}	8.7×10^{59}	55.3	44.7
7 ^c (4075 lbs)	9.81	415	6.2×10^{76}	5.1×10^{23}	20.2	13.1
8 ^c (4078 lbs)	10.32	395	3.9×10^{105}	3.7×10^{34}	42.8	19.2
9 (4092 lbs)	8.16	501				
10 (4013 lbs)	8.46	474				
11 (3058 lbs)	6.65	460	4.0×10^7	53.3	2.4	1.3
12 (3143 lbs)	9.54	329	6.0×10^{10}	1044.5	4.1	1.9
^a - Bond area was calculated as $0.75\pi(8.0\text{-protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b – “Creep Stress” is the “Creep Weight” divided by the “Bond Area”. ^c – Anchor installed with Standard Set epoxy.						

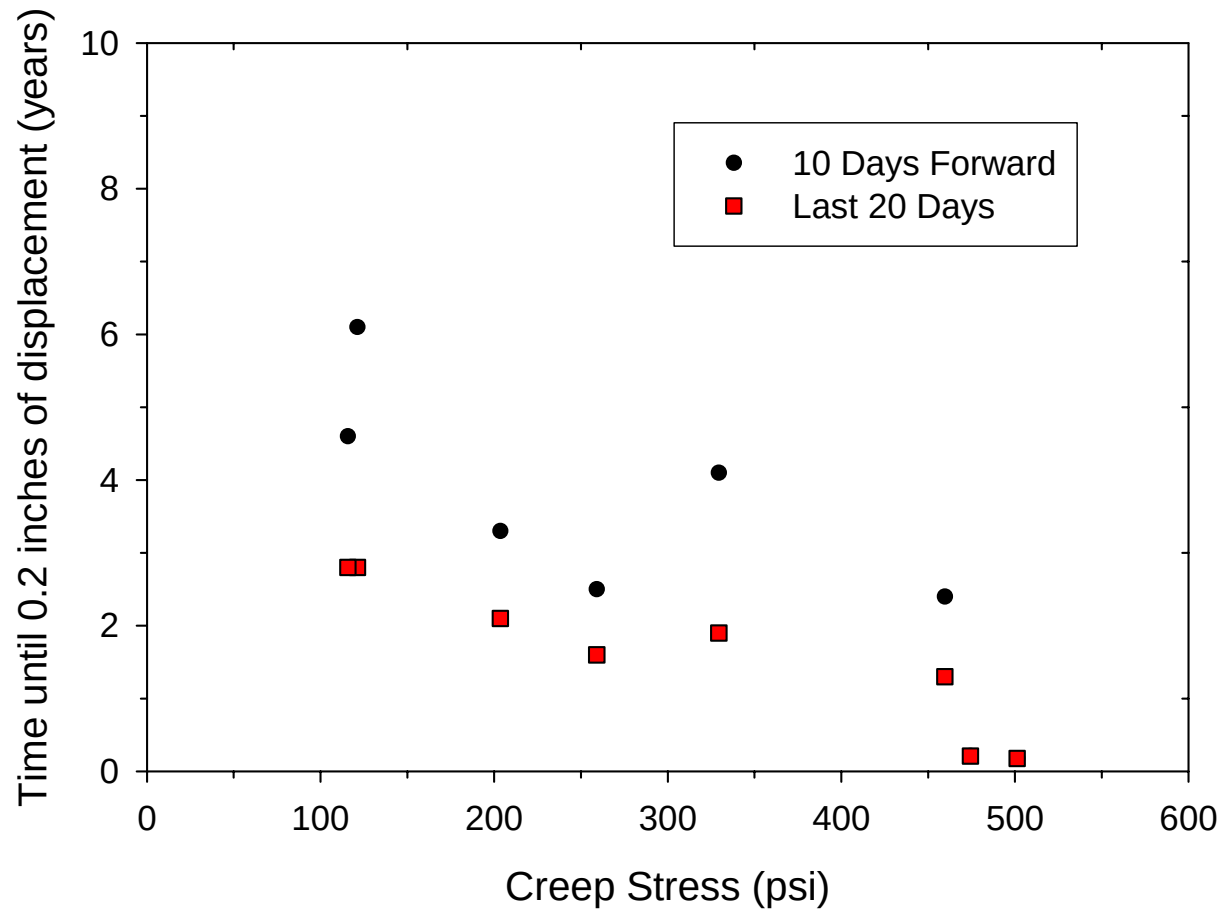


Figure 22. Graph of time to displace 0.2 inches versus creep stress for Fast Set anchors using polynomial model.

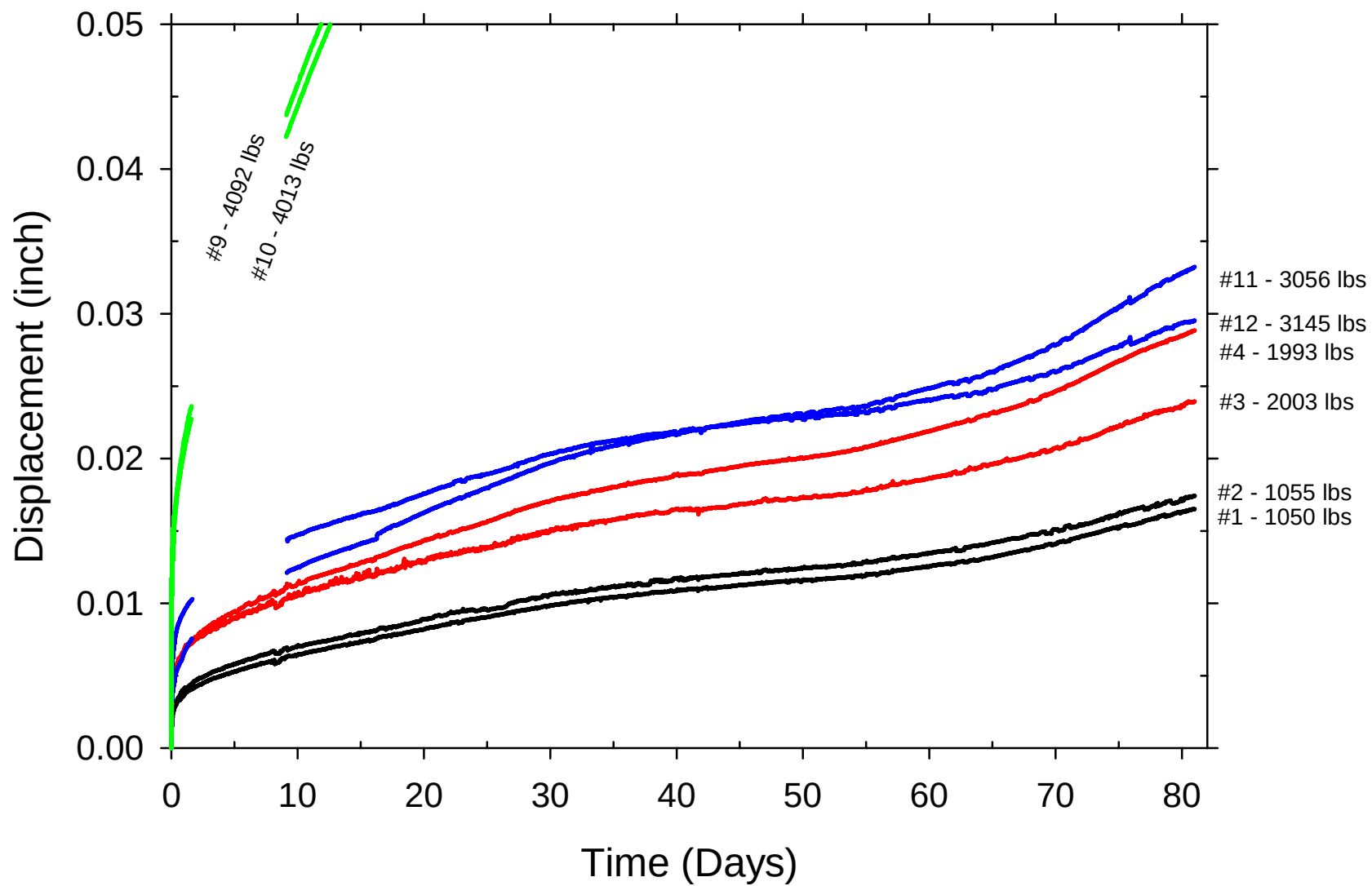


Figure 23. Comparison of Fast Set epoxy anchors.

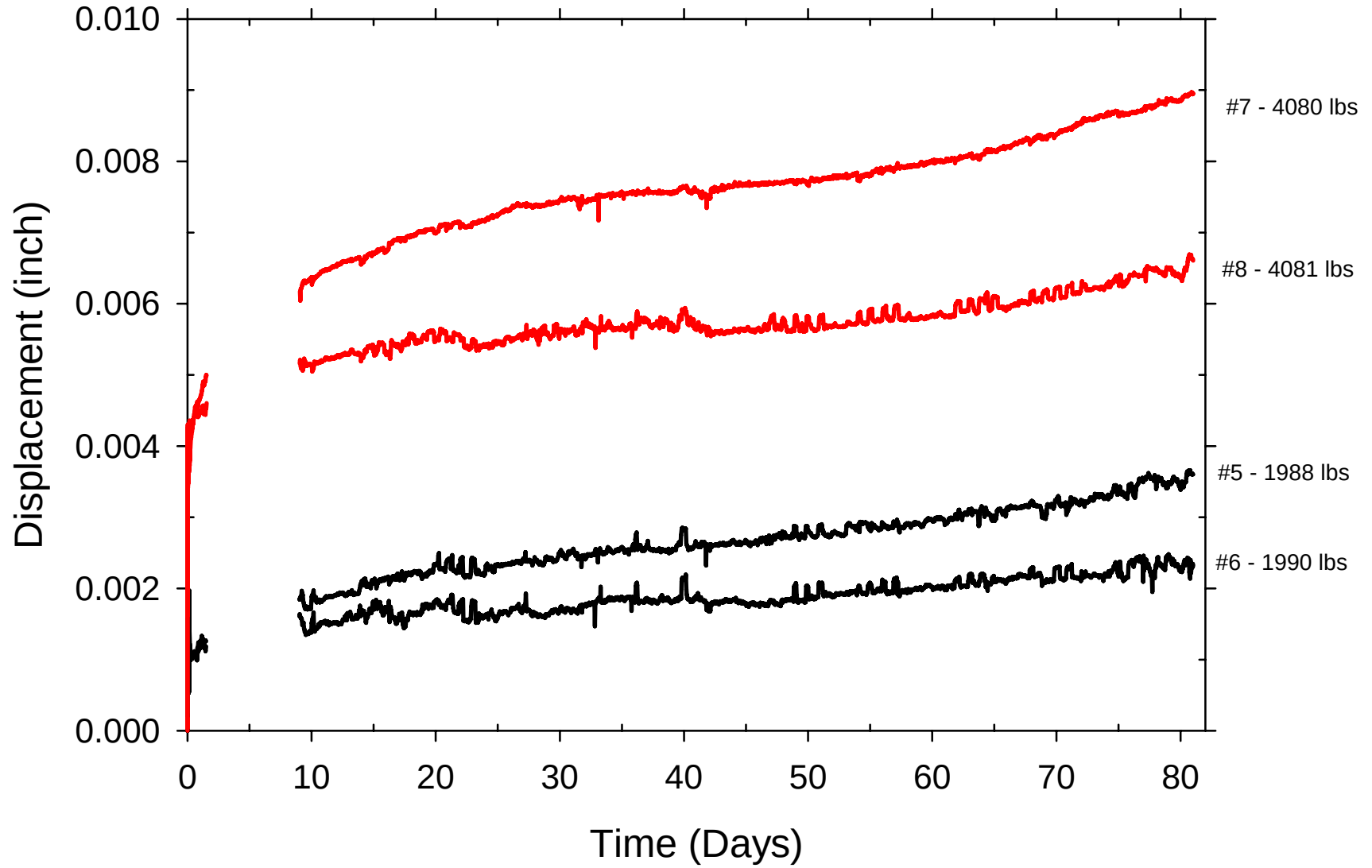


Figure 24. Comparison of Standard Set epoxy anchors.

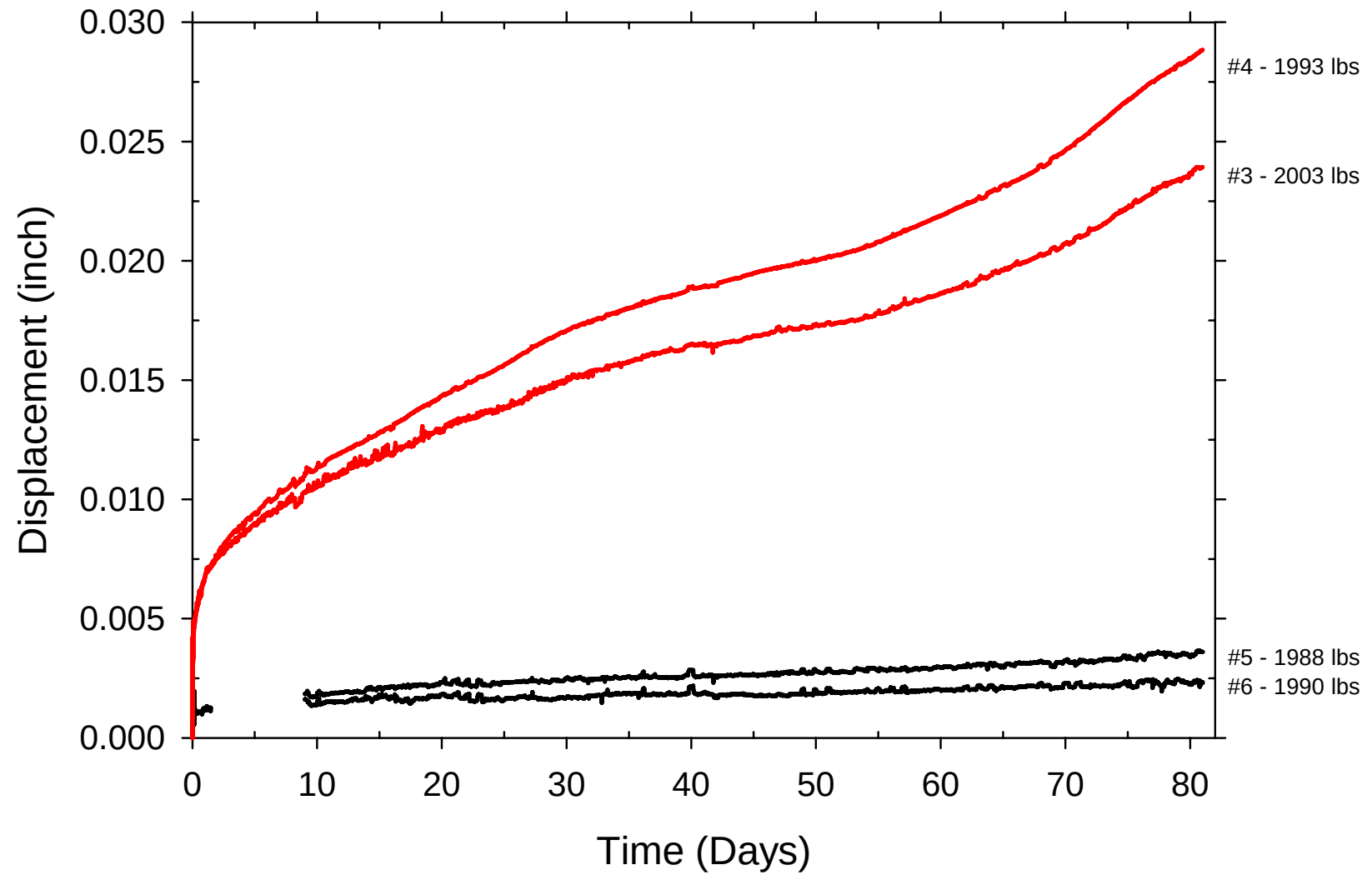


Figure 25. Comparison between fast and Standard Set epoxy at 2000lbs sustained load.

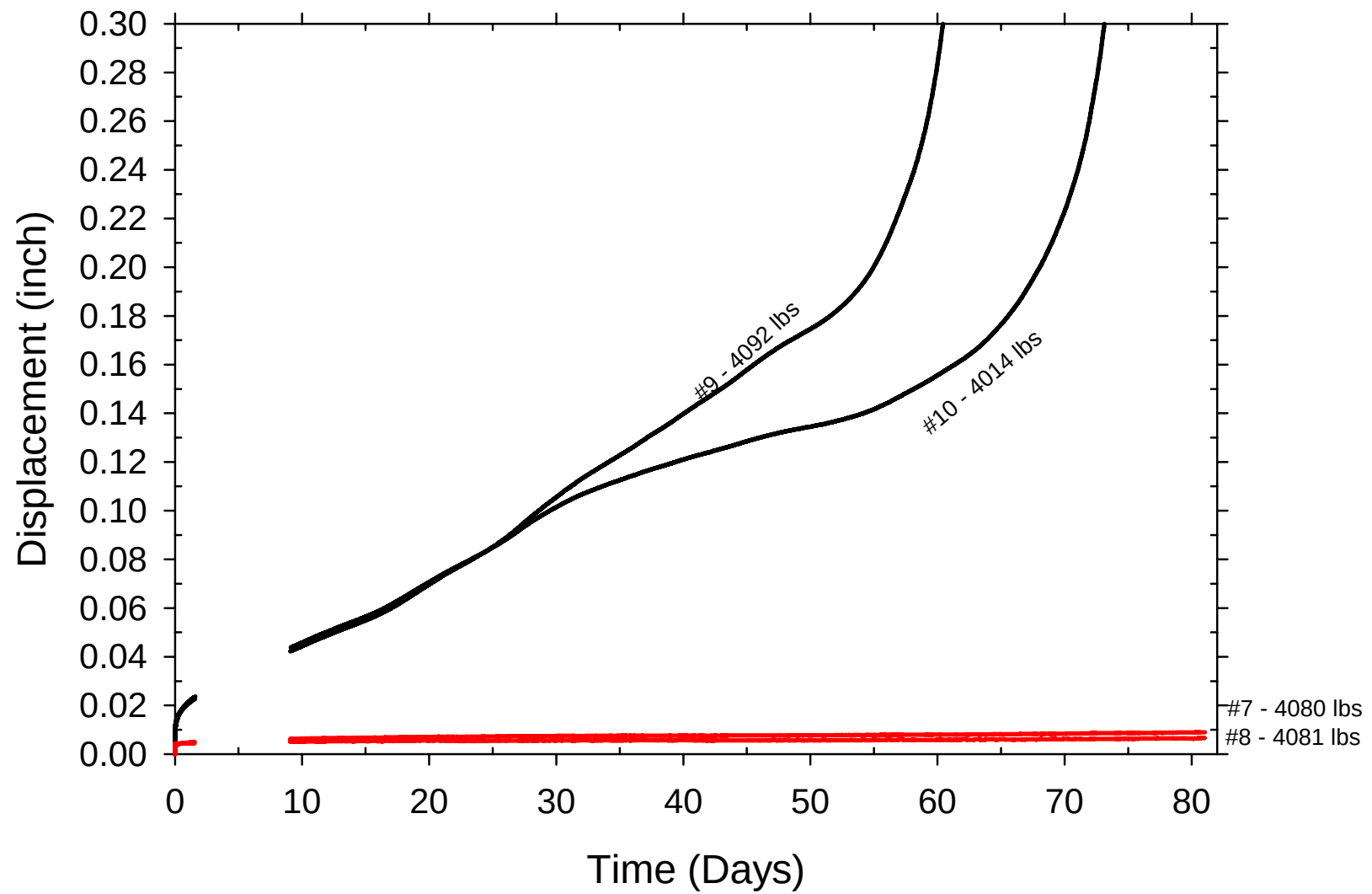


Figure 26. Comparison between fast and Standard Set epoxy at 4000lbs sustained load.

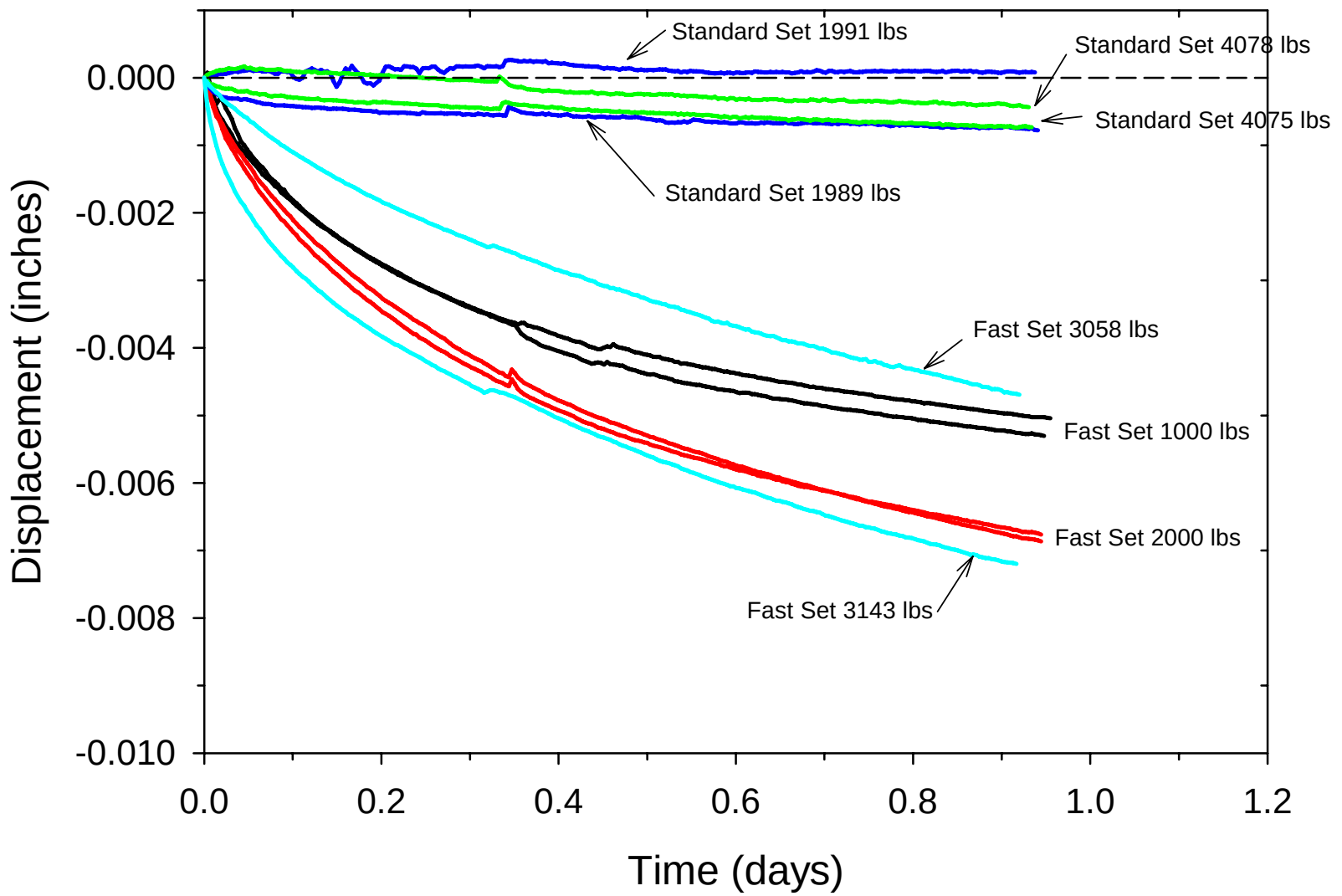


Figure 27. Relaxation curves for ten creep anchors.

CHAPTER 4. LOAD-DISPLACEMENT BEHAVIOR OF CREEP SPECIMEN ANCHORS

The sustained load or creep tests were concluded on 12 March 2007 for a creep test duration of 82 days. Two of the anchors failed during this period. Load-displacement testing was conducted on the remaining 10 anchors to determine how or if the sustained loading affected the static failure mode or capacity. In addition, once the anchors were out of the concrete blocks any void area in the epoxy could be evaluated. This chapter will summarize the load-displacement behavior of the anchors in addition to the condition of the epoxy bond.

LOAD FRAME AND DATA COLLECTION

The unconfined method of load-displacement testing was used allowing for concrete cone failures. This was accomplished using the portable load frame detailed in Figure 28. The frame consisted of two W8 steel sections placed transverse to the concrete block. A steel spreader beam (e.g. needle beam) built-up from channel sections was placed longitudinally atop the W8 sections. Atop the needle beam sat an Enerpac JDH-1508 double acting, hollow-core jack. Load was measured with a 100 kip Strainert model FL100U(C)-2DGKT (S/N 08905-16) flat load cell. The jack was run in load control by adapting a servo valve into the system via a custom made distribution manifold. Hydraulic pressure was provided by MTS pumps running at 3000 psi meaning the jack was limited to a capacity of 90 kips. Closed-loop control was attained with an MTS 458.20 controller using the Strainert load cell for feedback. The load cell was calibrated to the MTS 458.11DC conditioner using a calibrated load cell traceable to a NIST standard. The MTS controller also utilized a 458.91 MicroProfiler which allows for a precise load regime to be programmed into the controller which automated the testing.

A program was entered into MicroProfiler that produced a load/time history shown in Figure 29. The program applied load in 1500 pound load steps with two minute hold periods. The loads were incrementally applied in this manner until the anchor pulled out or fractured.

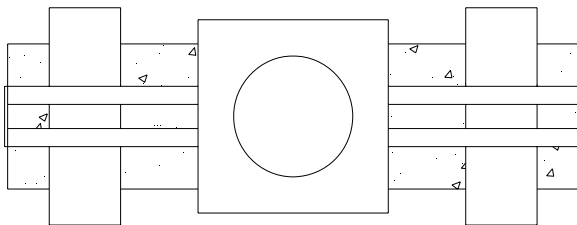
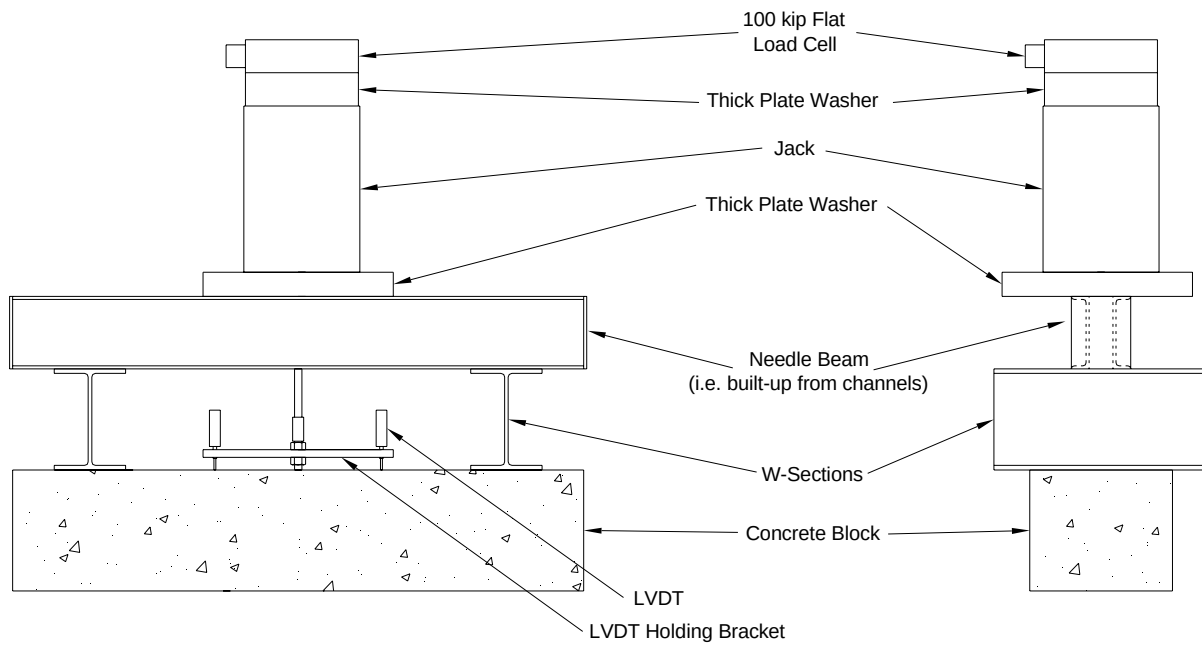
Data was collected at a rate of 2 Hz using the short-term data collection system with only two LVDTs. Both LVDTs were mounted 7 inches from the anchor into a steel mounting plate.

RESULTS

None of the 10 anchors tested slid out cleanly from the hole in the concrete block. Instead, all of the anchors pulled out a shallow cone of concrete. Out of the 230 anchors FHWA pulled from the ceiling in the I-90 tunnel, all but three slid out from the hole without removing concrete. Therefore, if creep due to the 7 years in service was a significant contributor to the anchor failure mode observed in the Tunnel, the 82 days of sustained loads experienced by these anchors was too little to affect performance.

Table 8 outlines the critical results from the load-displacement tests conducted on the creep anchors. The tables printed in APPENDIX B contain the pre- and post-pull pictures of the anchors along with a load-displacement plot for each anchor. The right column of Table 8 reports a strength ratio which is the peak stress for the individual creep anchor normalized by the average failure stress of optimally installed anchors produced in a separate study (See Reference 6). The strength ratio averaged 0.823 (coefficient of variation = 0.057) for the six anchors using Fast Set epoxy, and 0.801 (coefficient of variation = 0.058) for the four anchors with Standard Set epoxy. After experiencing 82 days of sustained load, these anchors exhibited an average strength reduction of 18.6% compared to newly installed anchor. This reduction appears to be independent of the creep stress and final creep displacement. The data is insufficient to determine whether additional creep would contribute to further reduction in anchor capacity demonstrated.

Elevation Views



Plan View

Figure 28. Static pull testing setup.

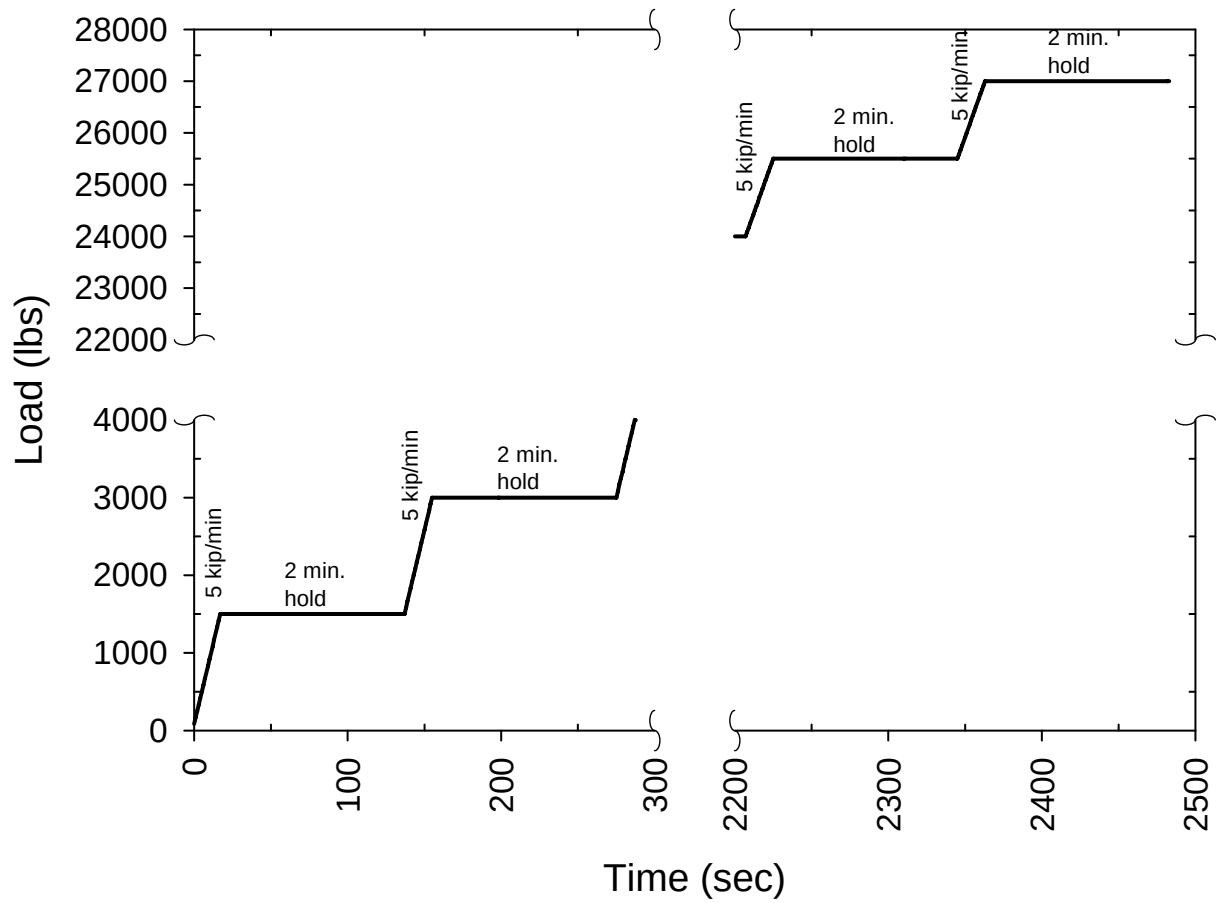


Figure 29. Static loading protocol for creep anchors.

Table 8 Static Pull-Out Data

Specimen	Bond Area to Concrete ^a (inch ²)	Creep Weight (pounds)	Peak Force ^b (pounds)	Creep Stress ^c (psi)	Peak Stress at Failure ^d (psi)	Pull-out Strength Ratio ^e
1 ^{D, V}	8.67	1052	15680	121	1808	0.818
2 ^{D, V}	9.13	1057	14985	116	1641	0.743
3 ^{D, V}	9.84	2003	17971	204	1827	0.827
4 ^{D, V}	7.69	1992	14983	259	1949	0.882
5 ^{D, V}	8.60	1989	18128	231	2108	0.832
6 ^D	9.81	1991	20994	203	2140	0.845
7 ^D	9.81	4075	19473	415	1985	0.783
8 ^D	10.32	4078	19478	395	1886	0.744
9 ^V	8.16	4092	not applicable	501	not applicable	not applicable
10 ^V	8.46	4013	not applicable	474	not applicable	not applicable
11 ^{D, V}	6.65	3058	11993	460	1803	0.816
12 ^{D, V}	9.54	3143	17955	329	1882	0.852
^a- Bond area was calculated as $0.75\pi(8.0-\text{protrusion}-0.68)$ minus the area of any voids. The 0.68 inches represents the average depth the red cap protrudes into the hole. Void area measured by Jim Wildey of NTSB. ^b- "Peak Force" was the maximum load imposed on the anchor during static testing. ^c - "Creep Stress" is the "Creep Weight" divided by the "Bond Area". ^d - "Peak Stress at Failure" is the "Peak Force" divided by the "Bond Area." ^e - "Pull-out Strength Ratio" is the "Peak Stress at Failure" divided by the average failure stress of a virgin anchor under monotonic loading. The average failure stress of a virgin anchor is reported in Reference 6 which is 2210 psi for Fast Set epoxy and 2534 psi for Standard Set epoxy. Failure Type: RF-rod fracture, D-cone failure, SO-slide out, V-void present, PF-partially filled w/epoxy						

Figure 30.

CHAPTER 5. CONCLUSIONS

This report summarized 82 day long creep tests conducted on 12 anchors, eight using Power-Fast+ Fast Set epoxy and four using the Standard Set formula. These tests were conducted in the Structures Lab of the FHWA Turner-Fairbank Highway Research Center.

Weights ranging from 1000-4000 pounds were hung from the eight anchors installed with Fast Set epoxy. For this adhesive, creep displacement rates were approximately proportional to the level of sustained load (or sustained shear stress in the epoxy) up to 3000 pounds. However, a load threshold or critical load exists at some level above 3000 pounds after which dramatically different adhesive behavior is created. The anchors sustaining 3000 pound loads displaced approximately 0.03 inches due to creep during the 82 day tests while the anchors loaded at 4000 pounds failed prior to the end of testing.

In stark contrast to the sustained load behavior of the Fast Set epoxy, the Standard Set epoxy specimens only displaced 0.004 inches sustaining 2000 pounds and 0.010 inches sustaining 4000 pounds for the 82 days of testing. These results indicate that there are inherent differences between the Fast and Standard Set epoxy formulas that significantly affect their creep behavior at the load levels and temperature exposure investigated.

Two empirical mathematical models were fit to subsets of the experimental data through least-squares regression. The first model, which is recommended by the ASTM E 1512⁷ standard using a two-parameter logarithmic form, produced lower creep level predictions. This model best describes creep which approaches asymptotic behavior. The remaining model used a three-parameter polynomial form which resulted in creep displacement predictions of approximately 3 times those that resulted from the Logarithmic model. Both models were used to predict displacements out to 600 days and seven years. When regressing the Logarithmic model over the last 20 days of data and predicting displacement out to seven years, the Fast Set epoxy anchors sustaining 1000, 2000, and 3000 pounds were predicted to displace 0.065, 0.101, and 0.119 inches respectively. Using the same criterion for anchors using the Standard Set formula sustaining 2000 and 4000 pounds is predicted to be only 0.008 and 0.018 inches respectively; essentially an order of magnitude less displacement than those using the Fast Set epoxy.

The Fast Set epoxy anchors sustaining 4000 pounds failed during testing shortly after their total displacement exceeded 0.20 inches. Although this displacement threshold is most likely load level, temperature and adhesive dependent, it was used to evaluate the performance of the remaining specimens in combination with the aforementioned predictive models. Applying this threshold using the Logarithmic model in all but one instance predicted lives much greater than 100 years. The Polynomial model indicated that all the Fast Set epoxy specimens tested would be expected to fail within seven years. Using the Polynomial model to evaluate the Standard Set epoxy specimens results in predicted failures in 13 to 56 years. However, this range is most likely a lower bound of behavior for this material.

In general, infrastructure projects, bridges and tunnels, are designed and detailed to provide 75 to 100 years of service with an appropriate level of maintenance. The results of this investigation indicate that even at the 1000 pound sustained load level, anchors installed with the Fast Set epoxy would be predicted to fail decades before nearing this service life. While the Standard Set epoxy specimens studied performed significantly better than the Fast Set epoxy specimens, the results also indicate that these anchors would fail prior to the expected service life. The behaviors observed certainly indicate a need for rigorous and frequent inspection of adhesive anchors already in use and similarly loaded elsewhere, and suggest that the continued use of adhesive anchors subject to sustained tension loads should be very limited if not eliminated for life safety applications.

CHAPTER 6. REFERENCES

- 1 Hartmann, J., Ocel, J., Shipp, J., Wright, W., and Fuchs, P. "I-90 Seaport Portal Tunnel Partial Ceiling Collapse Investigation: Adhesive Anchor Load and Load-Displacement Testing Results ," Federal Highway Administration, Turner-Fairbank Highway Research Center, McLean, VA, October 2006
- 2 Hartmann, J., Ocel, J., Wright, W., Fuchs, P., and Adams, M.; "I-90 Seaport Portal Tunnel Partial Ceiling Collapse Investigation: Adhesive Anchor Sustained Load Testing Results," FHWA TFHRC report, March 2007.
- 3 ASTM C192/C192M-05, "Standard Practice for Making and Curing Concrete Test Specimens in the Laboratory," American Society for Testing and Materials (ASTM) International, West Conshohocken, PA, 2005.
- 4 ASTM C39/C39M-05, "Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens," American Society for Testing and Materials (ASTM) International, West Conshohocken, PA, 2005.
- 5 ASTM C469-02, "Standard Test Method for Static Modulus of Elasticity and Poisson's Ratio of Concrete in Compression," American Society for Testing and Materials (ASTM) International, West Conshohocken, PA, 2005.
- 6 Ocel, J., and Hartmann, J.; "Mechanical Behavior of Powers Fasteners Power-Fast+ Adhesive Anchors under Various Installation Techniques," FHWA TFHRC report, April 2007.
- 7 ASTM E 1512-01, "Standard Test Methods for Testing Bond Performance of Bonded Anchors," American Society for Testing and Materials (ASTM) International, West Conshohocken, PA, 2001.

APPENDIX A

The final displacement data has been calculated using data from both the short-term and the long-term systems. The following information describes the types of data files used in processing and analysis and provides details for all steps of the data processing.

SUMMARY OF DATA PROCESSING STEPS

Data processing is broken into two steps. The first involves preprocessing of data to create new files offline. Here the short-term and long-term data are prepared and average displacement values are found. The long-term data is filtered to remove a section of bad data.

Prepare Short-Term System Data

Data are stored in the short-term system as ASCII data files with each displacement sensor data recorded as a voltage. In the raw data files the voltage is not converted to displacement. The conversion factor for each sensor is in the header of each ASCII file. The first preprocessing step is to convert the raw sensor voltages to displacement. This is done with a separate processing step that reads each data file, applies the proper conversion factor, and re-saves the data to a new ASCII file now in units of displacement. The conversion factor for each sensor is given in Table 9.

After conversion to displacement the short-term data needed further preprocessing prior to analysis for two reasons. First, this data was not sampled at standard intervals which created gaps in the data set. In addition, there were also duplicate time (X) values for some entries. For example, LVDT1 data for anchor 12 had 478 duplicate time values out of a total of 18005 points. In order to properly process this data the duplicate values must be removed and the signal must be re-sampled at a standard interval. These preprocessing steps are necessary in order to directly compare the short-term and long-term data.

Figure 31 shows short-term system data for LVDT1 on #12 for the period near the weight drop. This figure shows that the re-sampled waveform represents the original data. Figure 32 shows a close-up of this data with the original irregularly sampled data as blue dots with red circles. The re-sampled data is shown as black points. This figure shows duplicate time values in the original data. Figure 33 shows sections of the original data with missing time data. Here the re-sampling algorithm uses linear interpolation to calculate new sample points.

Prepare Long-Term Data

The long-term data also had to be preprocessed in order to be analyzed directly with the short-term data. The long-term system collected data at 1.25 Hz (0.8 seconds per sample) and the short-term system was re-sampled at 10 Hz (0.1 seconds per sample). The data logger data must be re-sampled eight (8) times the original rate ($1.25 \text{ Hz} * 8 = 10 \text{ Hz}$). A linear interpolation was used to re-sample the data.

Figure 34 shows the long-term data plotted as a function of sample points, shown in black. Here the original long-term data can be seen to contain many fewer points than the short-term data, shown in green, red, and blue. This same figure also shows the re-sampled long-term data, shown in cyan. All algorithms that work with both short-term and long-term data work on a point-by-point basis. The re-sampled long-term data will directly correspond to the short-term data to permit further processing. For further clarification, a close-up of the long-term data before and after re-sampling as compared to the short-term data is shown in Figure 35.

The long-term data before and after re-sampling are shown with the short-term data as a function of time (as opposed to sample points as above) in Figure 36. A zoomed in view of the re-sampled long-term data is shown in Figure 37 in order to illustrate that the re-sampling process does not alter the original data, rather it just interpolates between the existing data.

Align Time Values

The first step to process short-term and long-term data together is to line up the time (X) values. The two systems use a different clock and therefore must be adjusted to correspond with one another. This adjustment was done manually by plotting the data and finding when the long-term (LVDT4) data matched the short-term data for LVDT3. Figure 38 shows both short-term LVDT3 (red) data and long-term LVDT4 (cyan) data after the adjustment. Here the two data sets are equal and opposite of one another. An iterative process of selecting a time offset, creating a plot, and observing the LVDT3 and LVDT4 relative positions was done in order to find the proper time offset.

Find Average Values

After the previous steps have been completed the short-term and long-term datasets can be directly compared. The average of all four values represents the true axial displacement of the anchor independent of any bending during the initial loading. Figures 39-50 plot the data for LVDTs 1-4 for each of the anchors along with the averaged data. The averaged data looks much cleaner than any of the individual LVDT outputs because bending effects from the weight swinging are cancelled out.

Filter Long-term Data

After about three days in operation a problem occurred with the long-term system. Data from Anchors 5-12 were very noisy. Due to the holiday time period several days elapsed before the problem was identified. The system was examined and fixed on 12/30/06. A faulty ground and signal wire were identified on the system multiplexer. After correction of this problem the instrument has been working fine. The data from the period 12/22/06 00:15:00 to 12/29/06 11:15:00 was removed for the sensors LVDT 5-12. In order to make each data set the same size, removed values were replaced with a NAN (not-a-number) value. Filtered data was replaced with a NAN value solely to mark the filtered data point and keep each data set the same size. No

data was filled in for the filtered section. A total of 2151 points were removed representing data from a 7.47 day period (2151 points at 5 minutes/point = 7.47 days).

PROCESSING OF FINAL DATA

Temperature Correction

The displacement sensors need correction for temperature variations. The temperature in the tunnel varied over a range of about 5-7 degrees C during testing as shown in Figure 51.

Four temperature sensors, at sensors 1, 4, 7, and 10 were used to monitor changes in temperature during testing. This data was used to correct for temperature variation in the displacement sensor data.

$$D_{corrected} = D_{uncorrected} - (Temp \cdot 0.00014)$$

Where *Temp* is the temperature sensor measurement in degrees C, 0.00014 is the temperature scale factor in inches/degree C.

The temperature in the laboratory tunnel had a maximum of 31.9 degrees C and a minimum of 24.5 degrees C, or a total change of 7.4 degrees C. The temperature change would result in a maximum displacement error of about 0.0010 inches.

Figure 52 shows the temperature correction of data for Anchor 1. In order to show the scale of the temperature correction the data for Anchor 10 is shown in Figure 53. Here the total displacement is much greater than the temperature errors and the correction factor is less significant to the data set.

Long-Term Data Offset Adjustment

The average displacement during the weight drop represents the true downward displacement of the test anchor. This average data is only available for about 30 minutes after the weight drop. After this point the three short-term system sensors were removed and only the single long-term sensor remained. This assumes that any bending effects occur immediately and any displacement after 30 minutes from the weight drop are purely from creep. However, because of the bending effects, the single LVDT connected to the long-term system has an offset displacement relative to the averaged data. Therefore, an offset displacement needed to be evaluated for each single LVDT such that it can be cleanly spliced into the initial averaged data.

To determine the offset value, a portion of data at the end of the average data set was used. The mean of 300 data points (about 30 seconds of data) for both the average data and the LVDT4 data were used to find the offset. Figure 54 shows the LVDT before (blue) and after (yellow) offset to match the average (green) data. Table 10 shows a list of the offset value calculated for each anchor.

After the long-term data has been offset, it is spliced together with the average data. A sample is shown in Figure 55. The long-term data prior to the splice point is not used.

Calculate Data Statistics

The weight drop time was always 400 seconds after the average data zero time. The splice time was always 1950 seconds after the average data zero time.

Pre-drop and Post Drop Displacement Values

The displacement value before and after the weight drop were calculated for each anchor. Values were found for both the LVDT4 long-term data and the average data. The mean value over a 30 second window was used to find the pre-drop and post-drop displacement values. Figure 56 shows the window regions of the long-term LVDT4 data used for calculations. Figure 57 shows the window regions for the average data. The post-drop window was chosen just after the data began to be more stable after the weight drop.

Table 9 Short-term LVDT Calibration Constants

Sensor	Calibration Constant (in/V)
LVDT1	0.082110
LVDT2	0.081871
LVDT3	0.081691

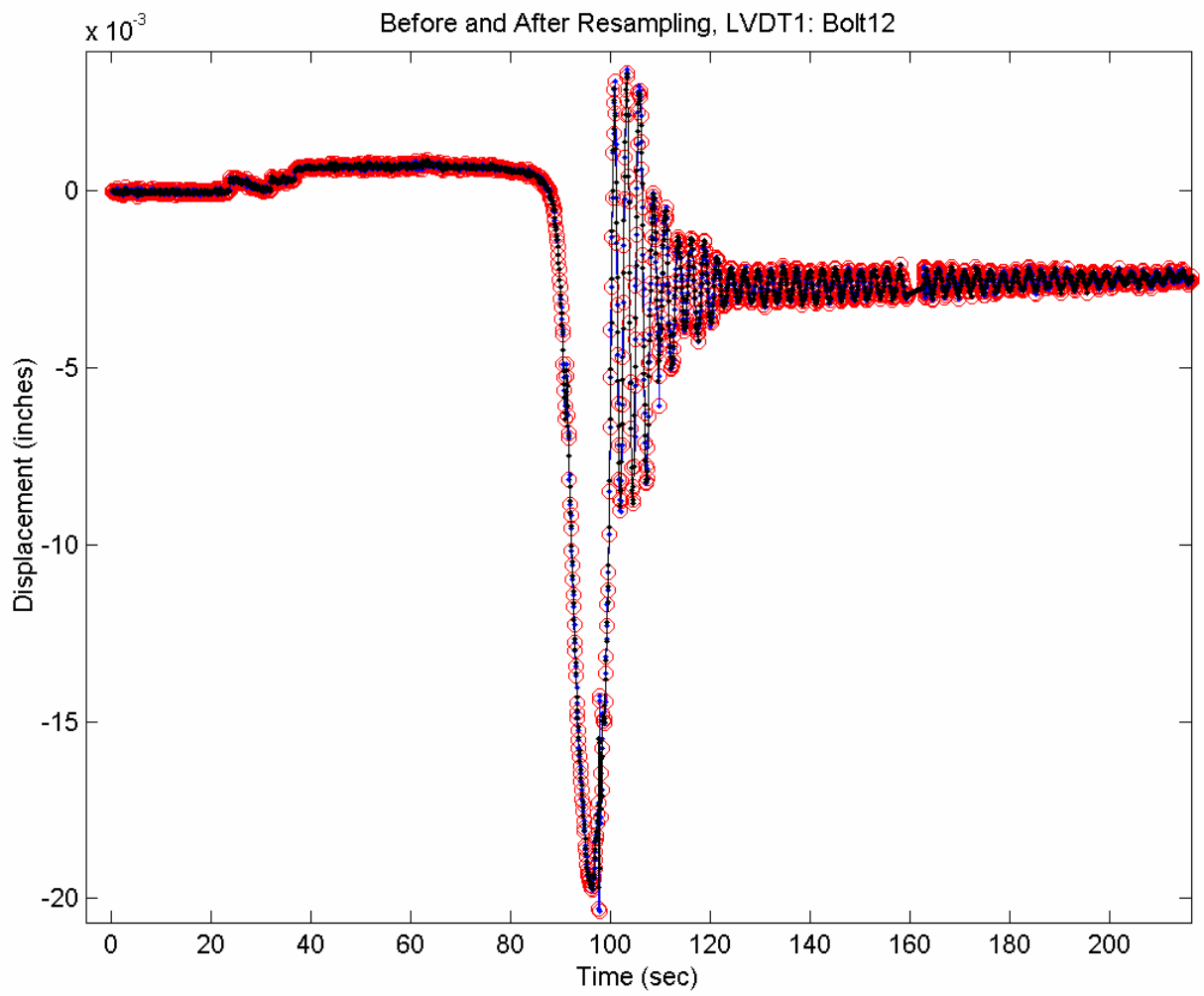


Figure 31. Short-term data before and after re-sampling showing a segment near the weight drop.

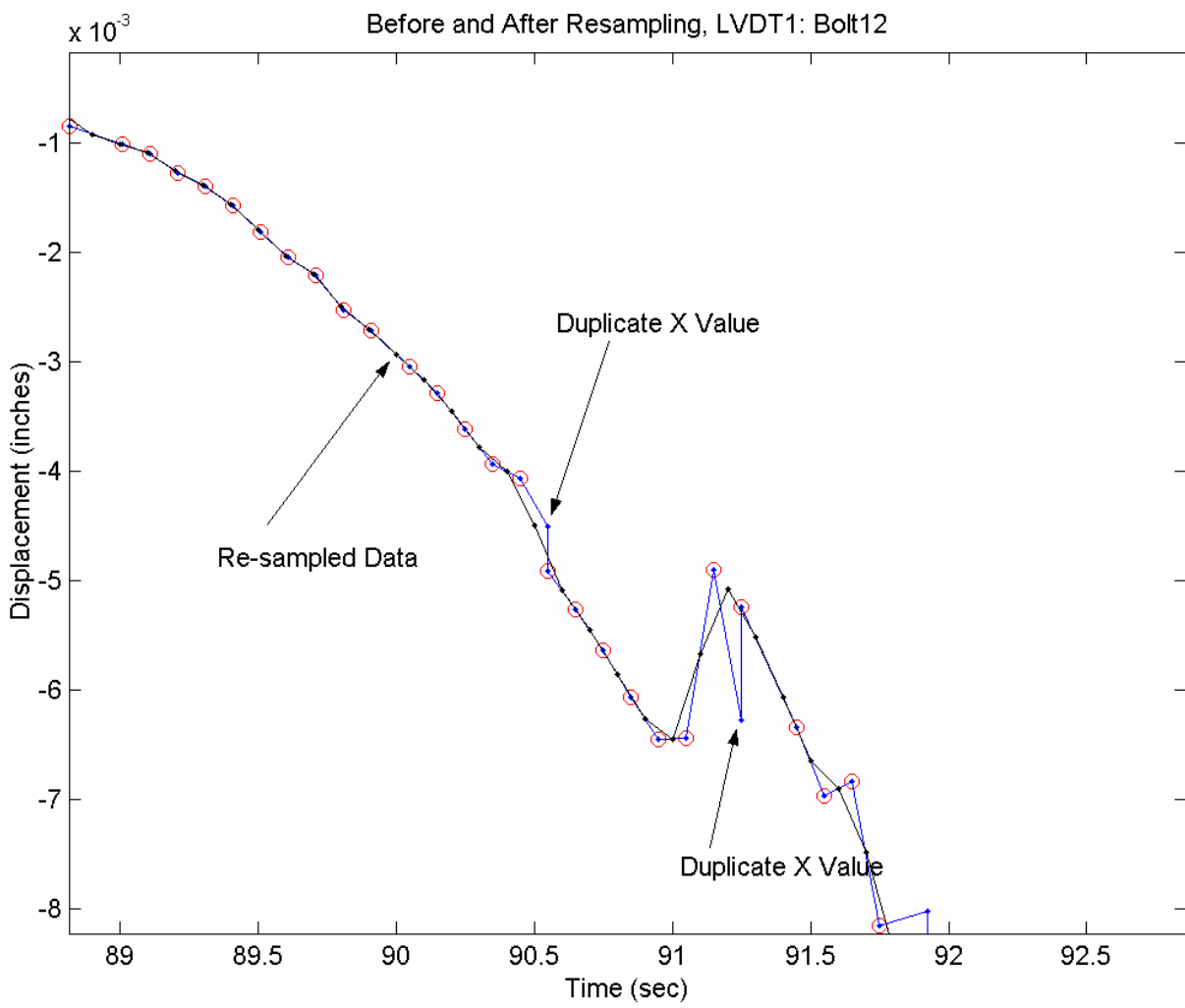


Figure 32. Close up showing re-sampled data and duplicate X values.

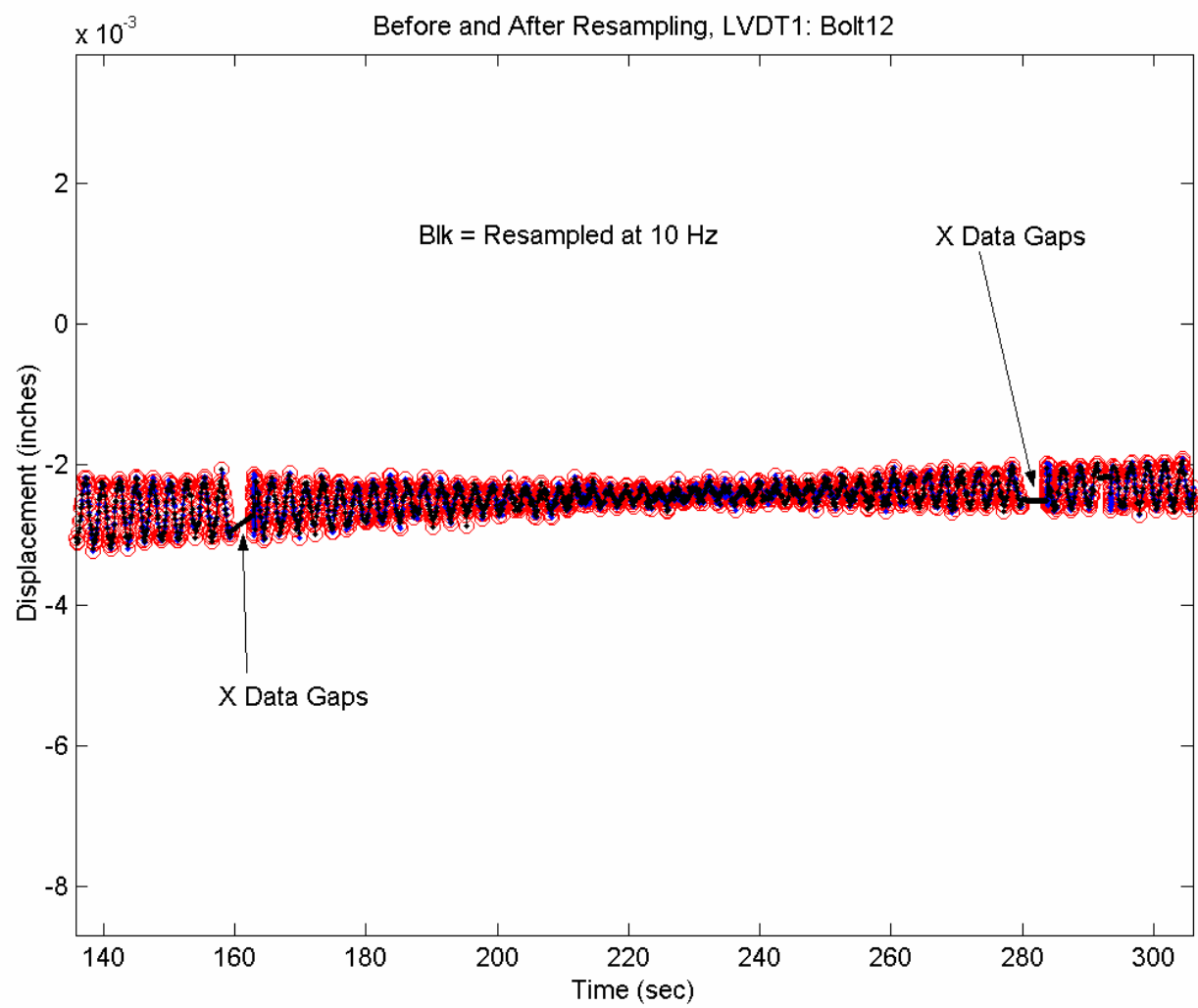


Figure 33. Re-sampled data showing data gaps in original dataset and re-sampled data.

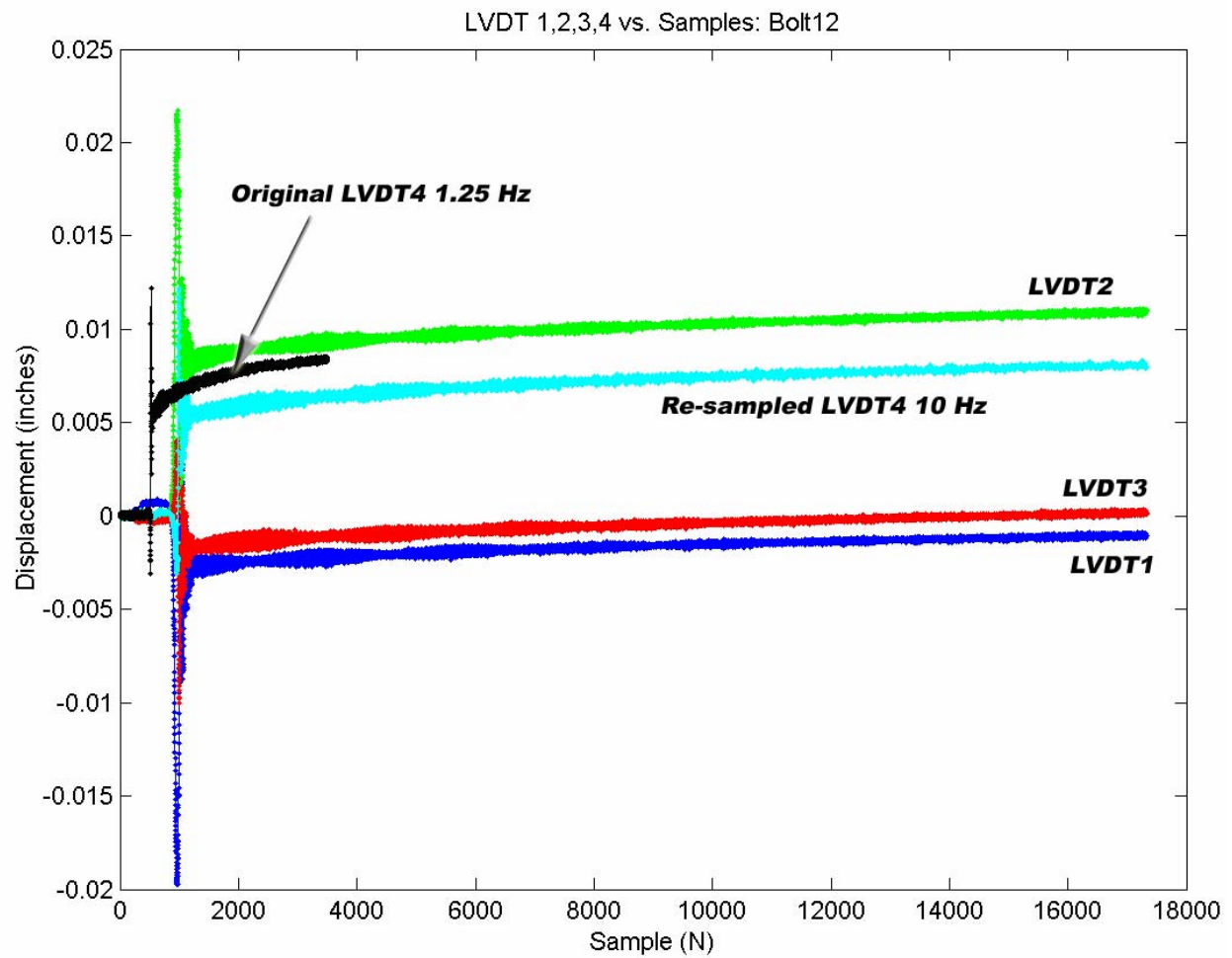


Figure 34. Long-term data before (LVDT4: black) and after (LVDT4: cyan) re-sampling plotted against sample points along with the short-term data (LVDT1,2,3: blue, red, green).

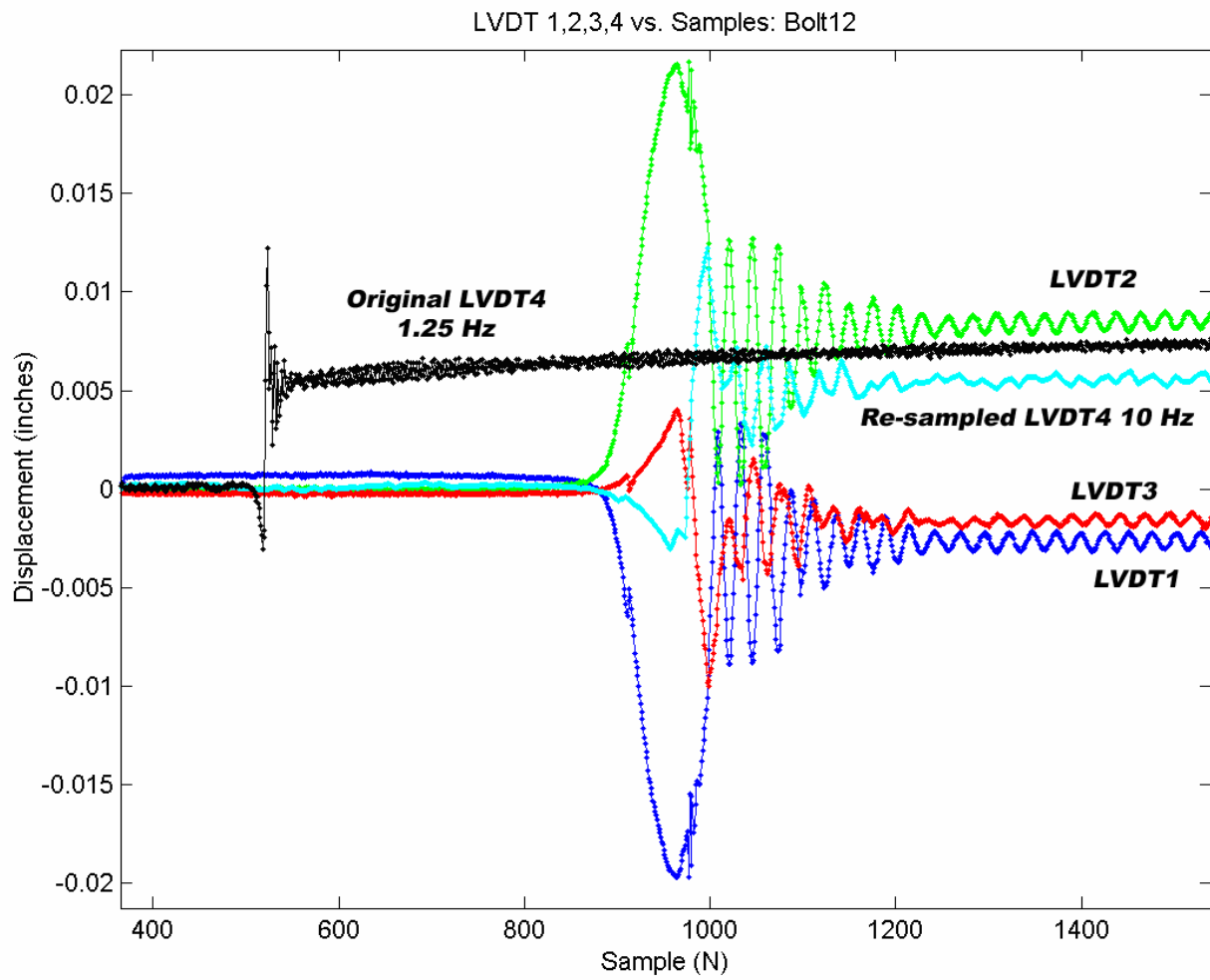


Figure 35. Close-up of long-term data before (black) and after (cyan) re-sampling.

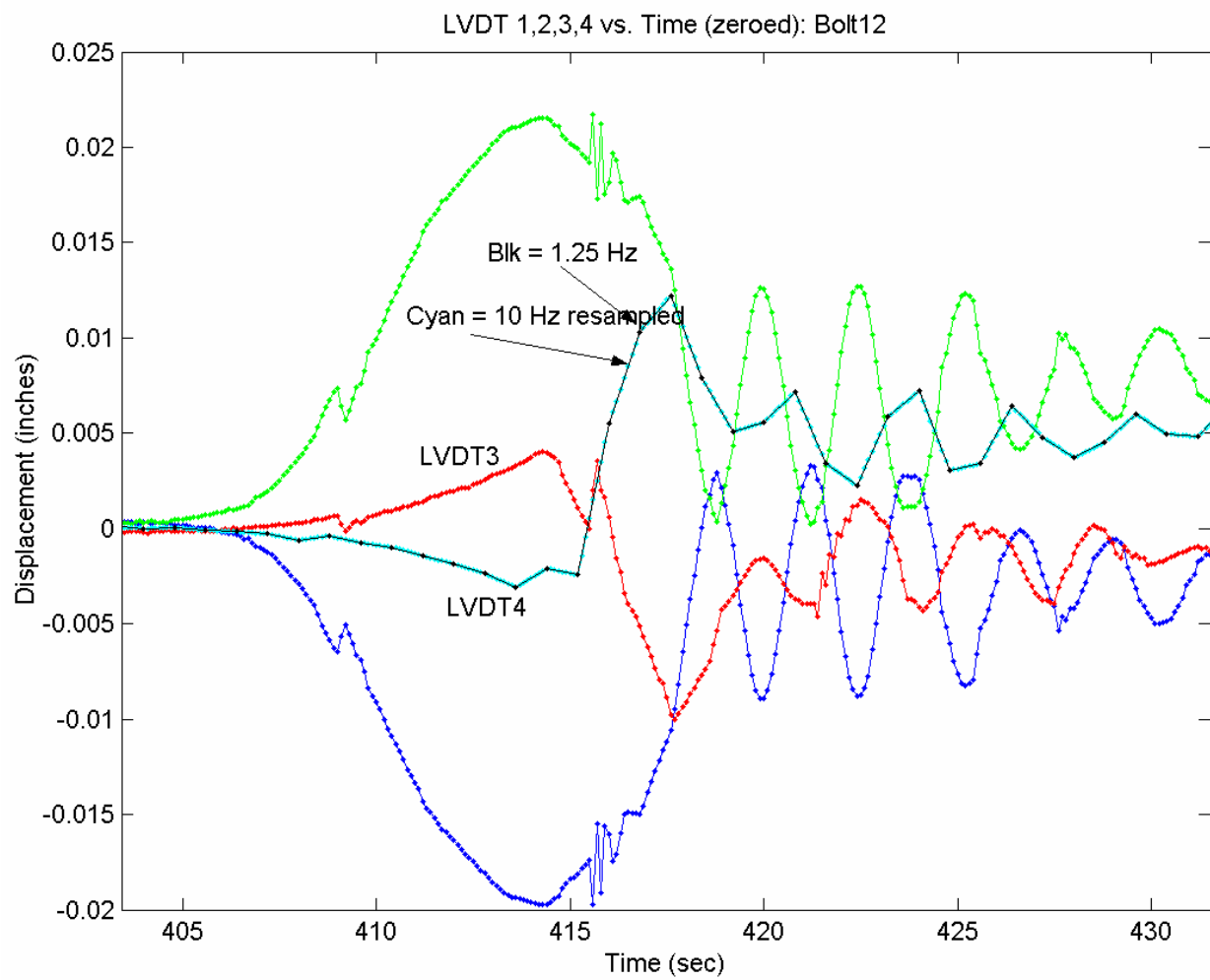


Figure 36. Long-term data before and after re-sampling plotted against time.

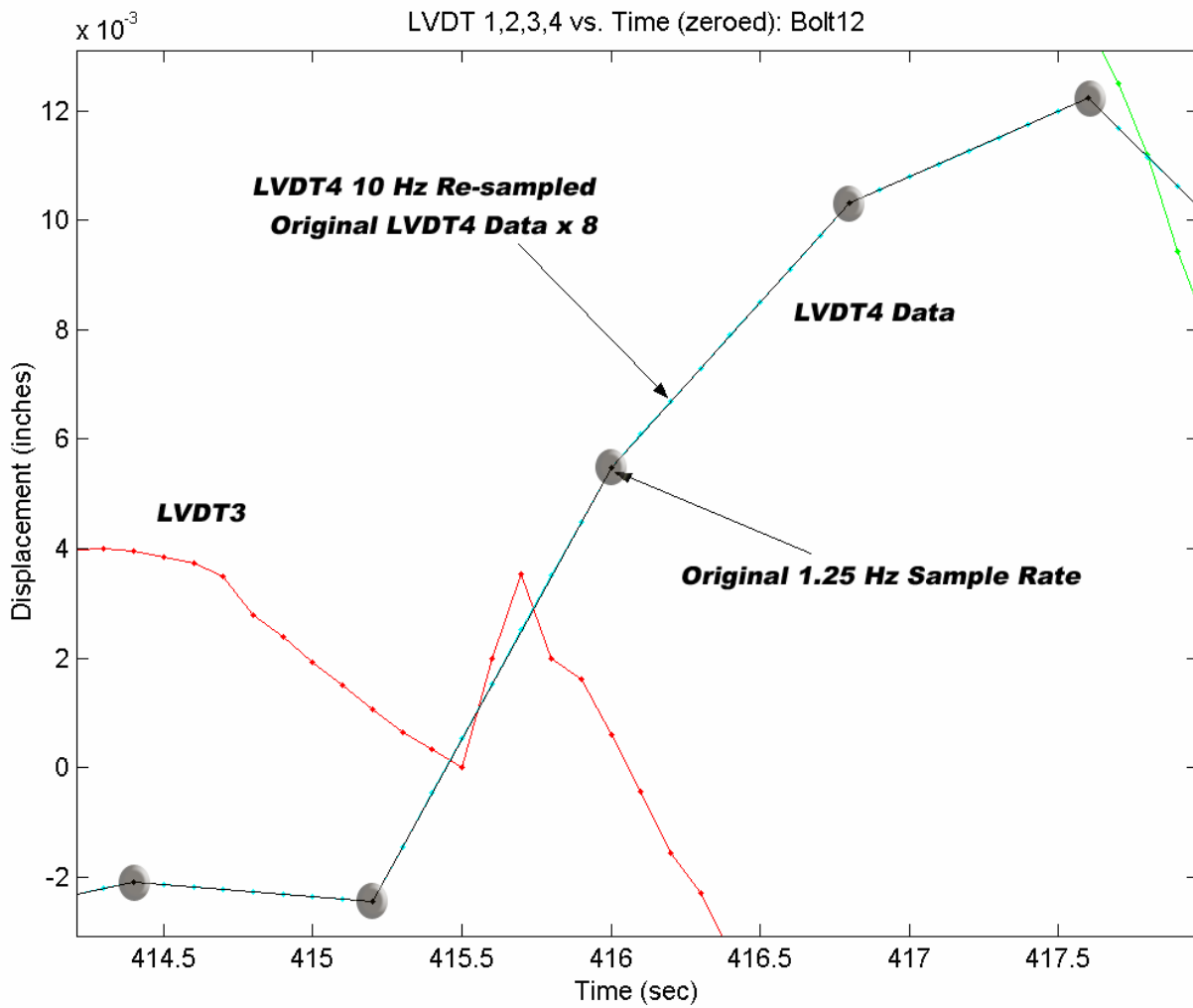


Figure 37. Close-up of long-term data before and after re-sampling plotted against time.

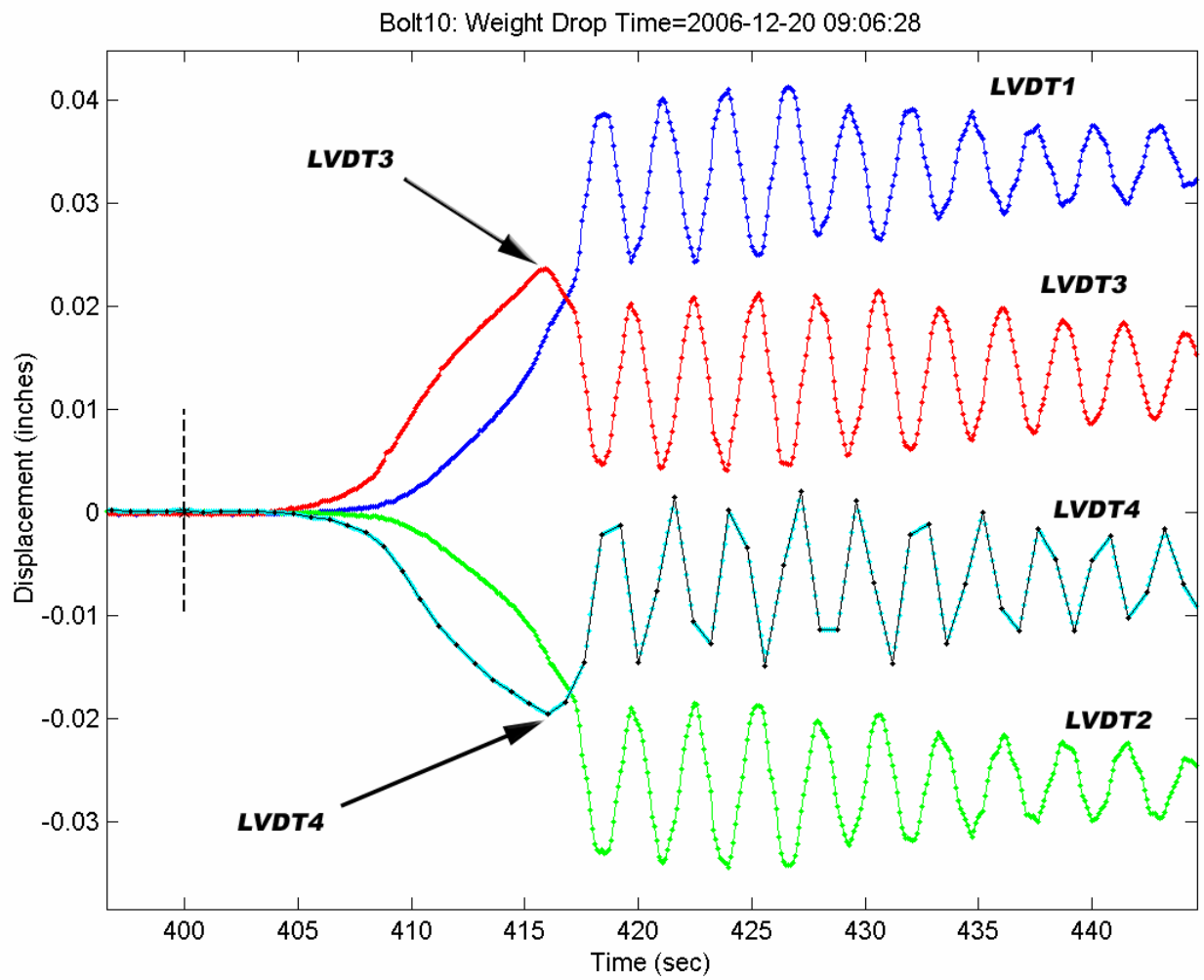


Figure 38. Short-term data manually offset in time to match long-term data.

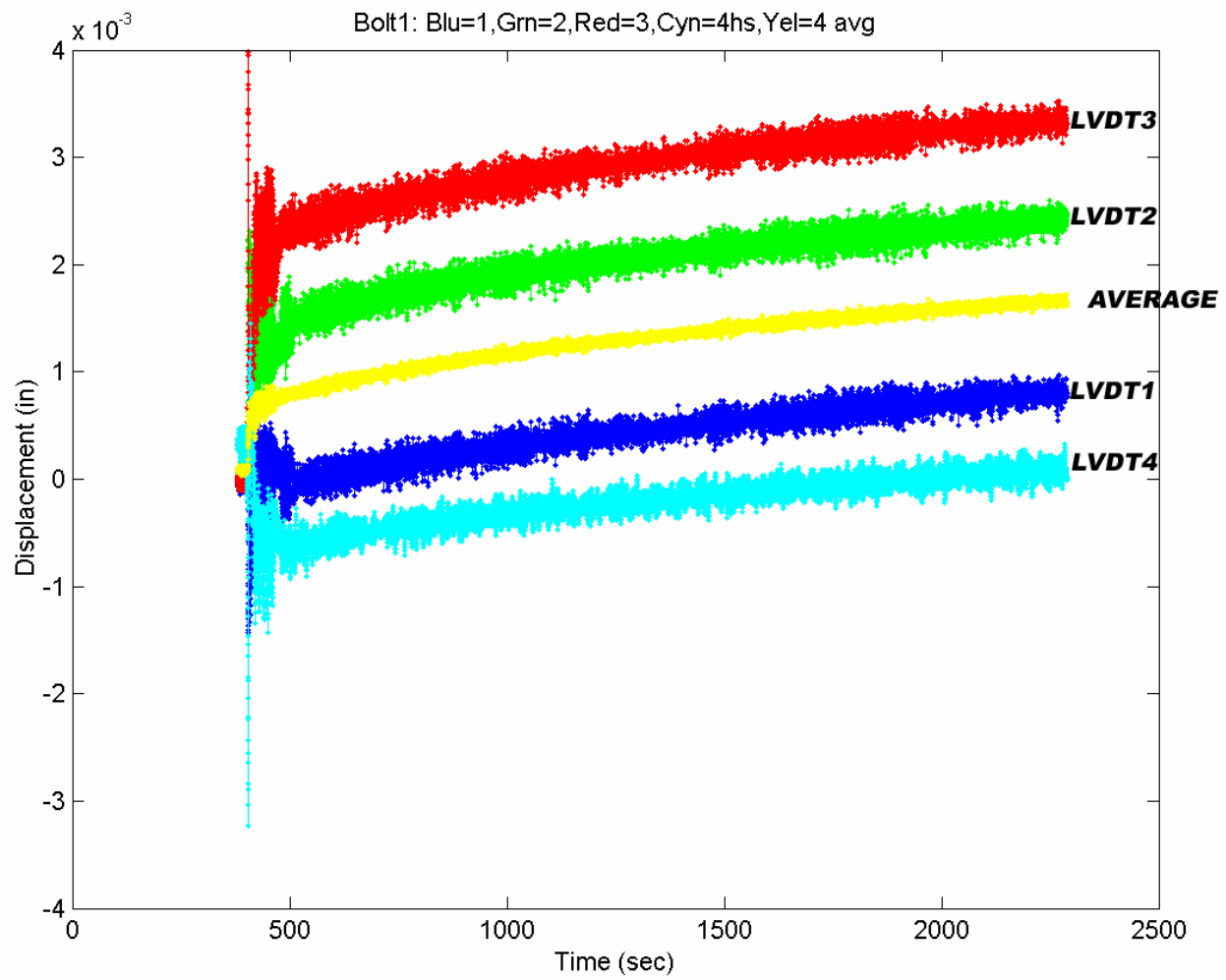


Figure 39. Individual displacement sensors and average displacement data for Anchor1.

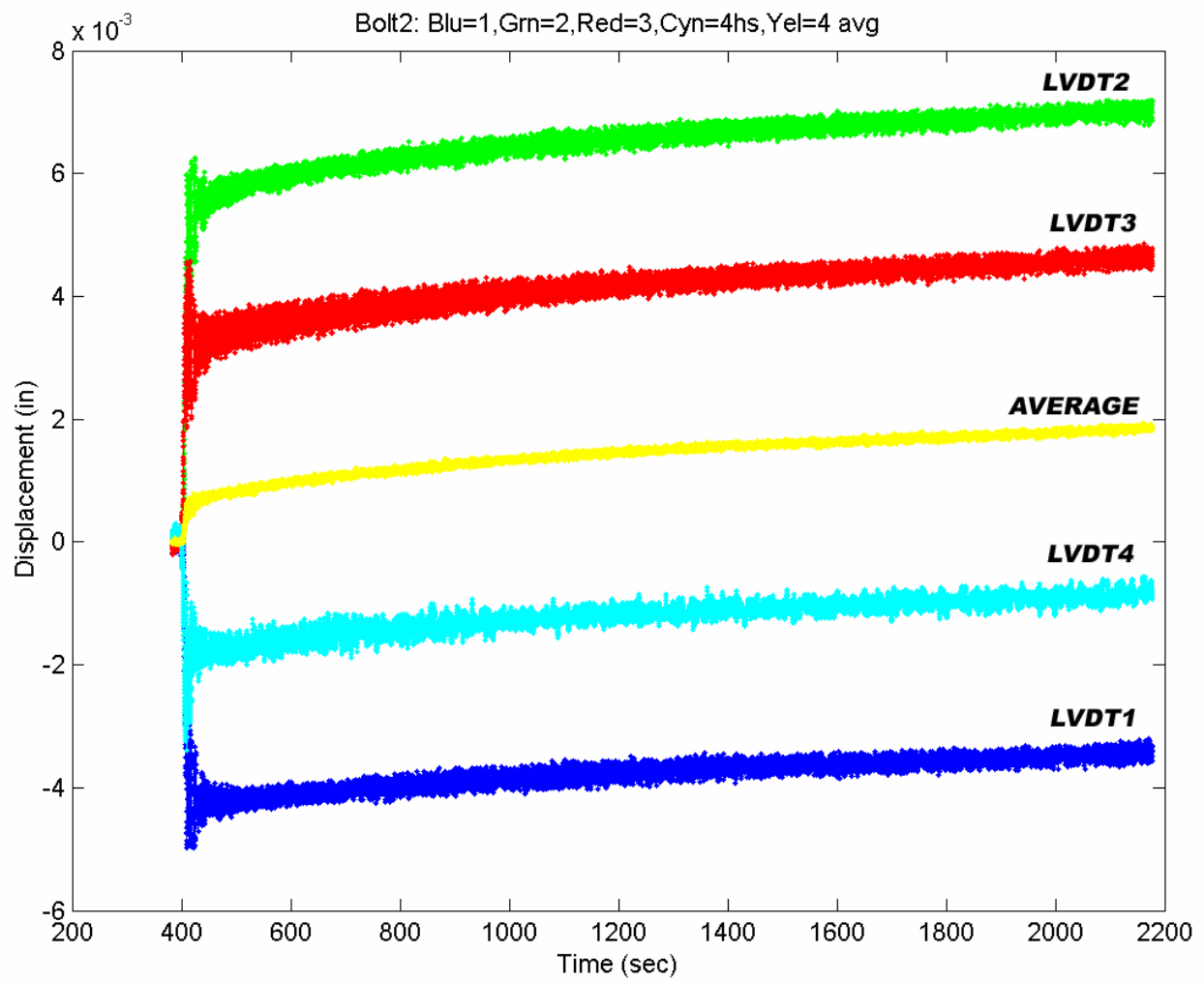


Figure 40. Individual displacement sensors and average displacement data for Anchor2.

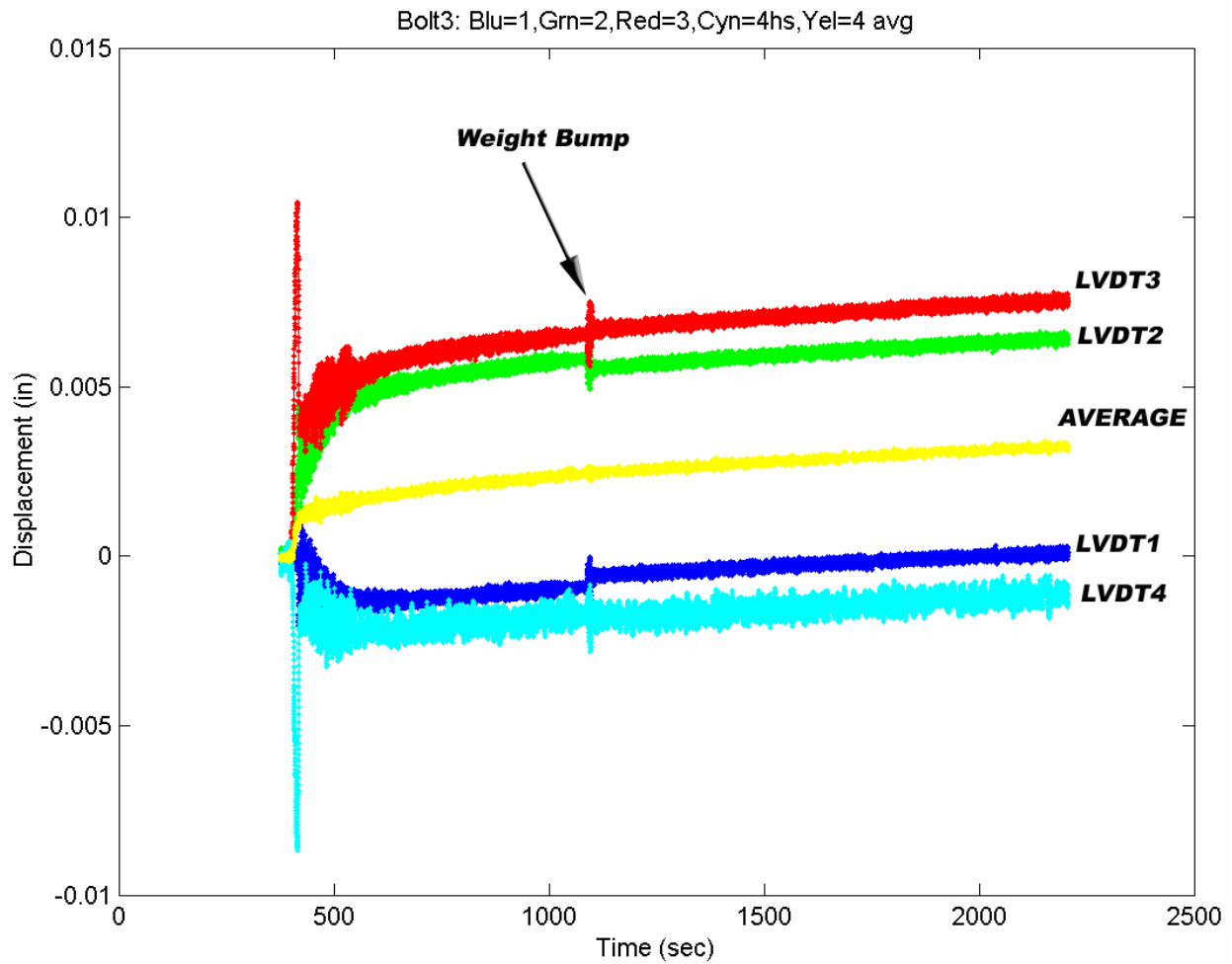


Figure 41. Individual displacement sensors and average displacement data for Anchor3.

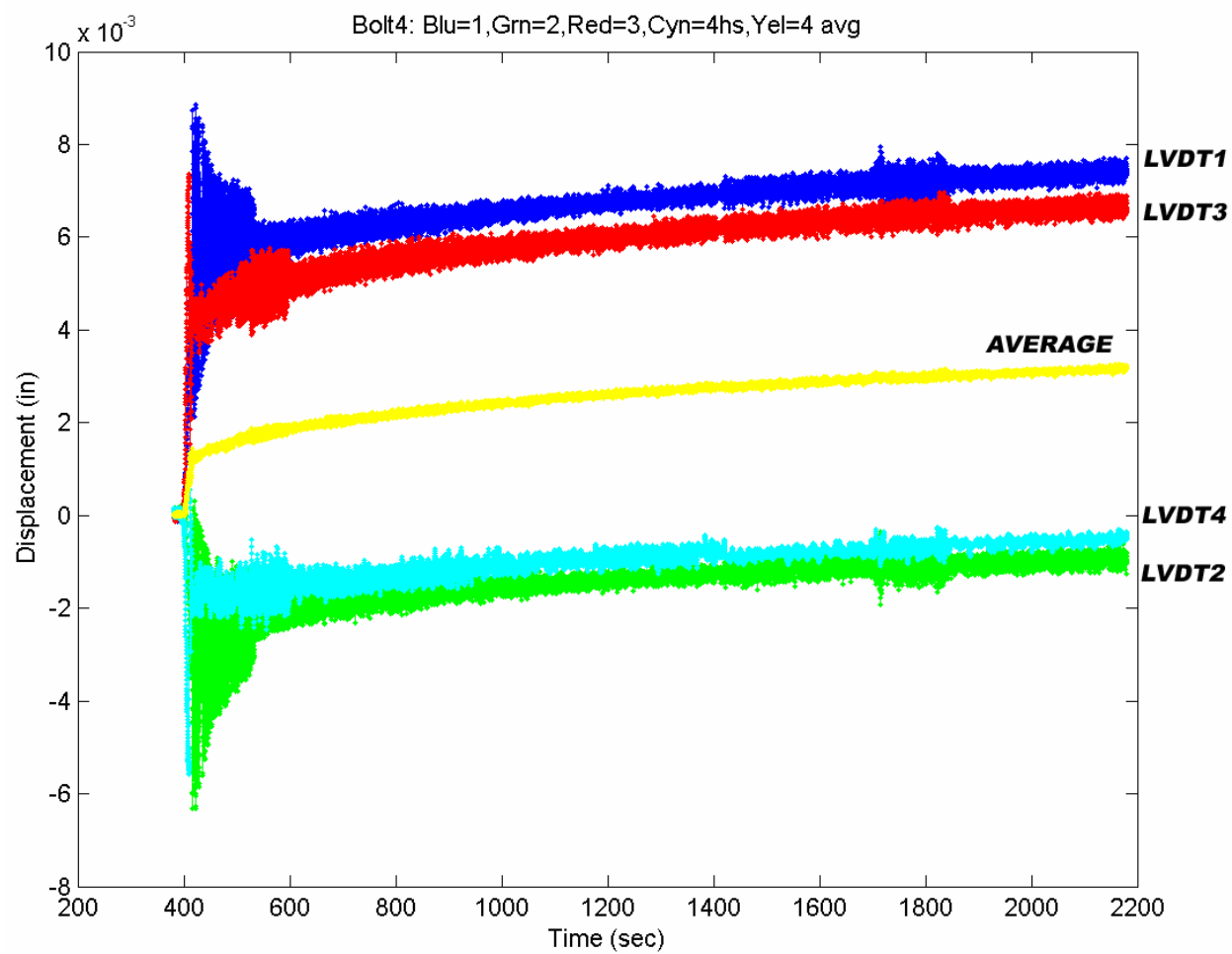


Figure 42. Individual displacement sensors and average displacement data for Anchor4.

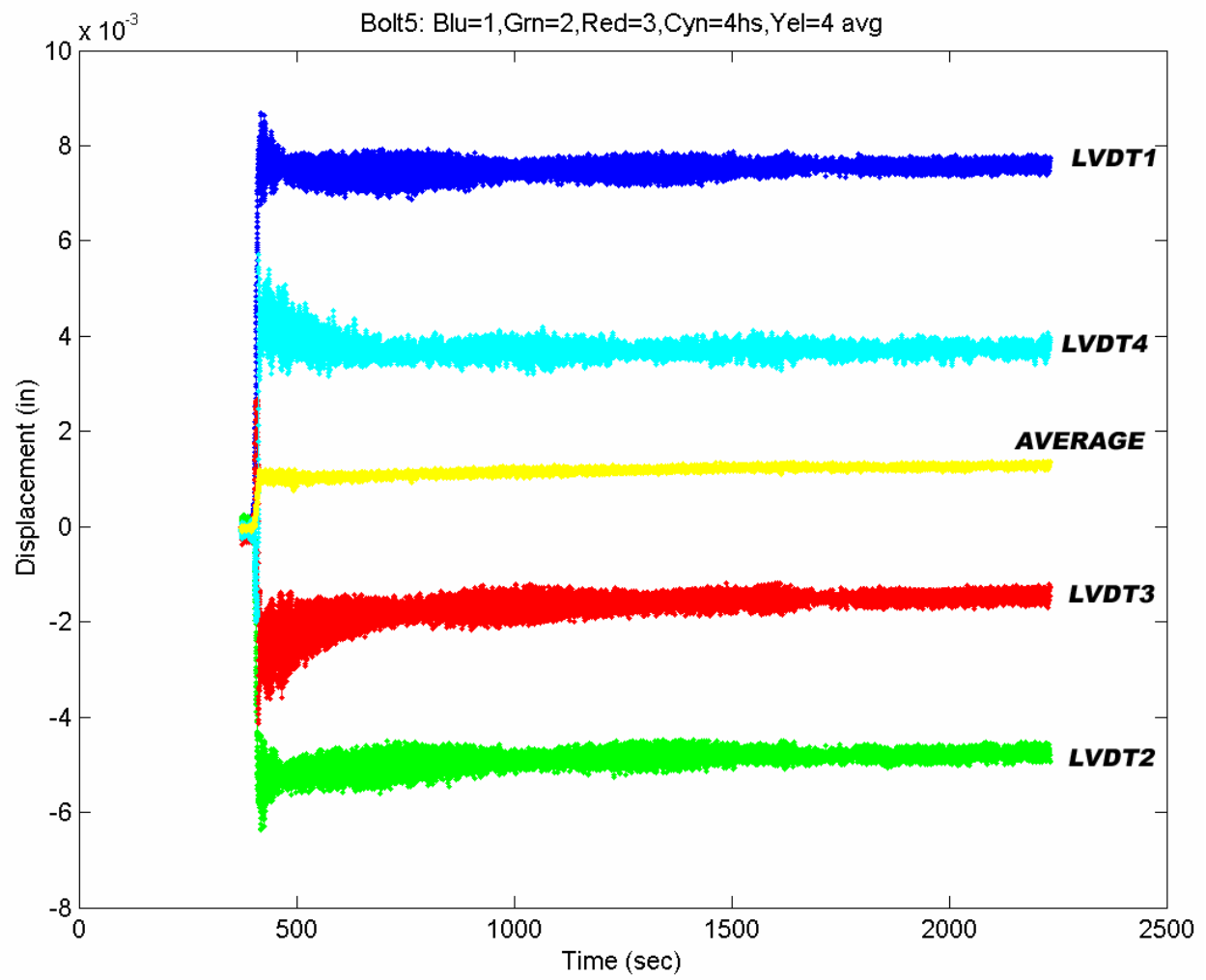


Figure 43. Individual displacement sensors and average displacement data for Anchor5.

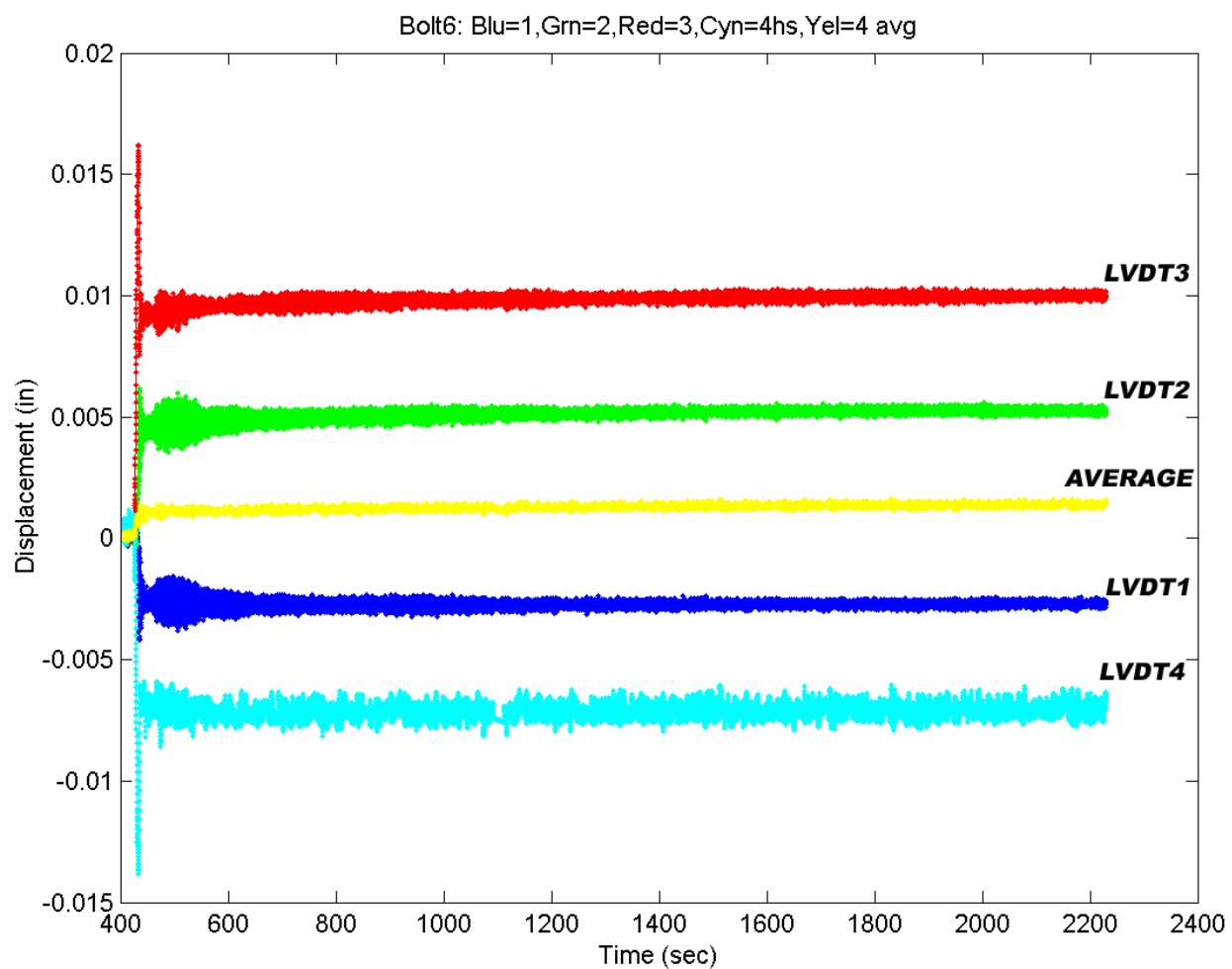


Figure 44. Individual displacement sensors and average displacement data for Anchor6.

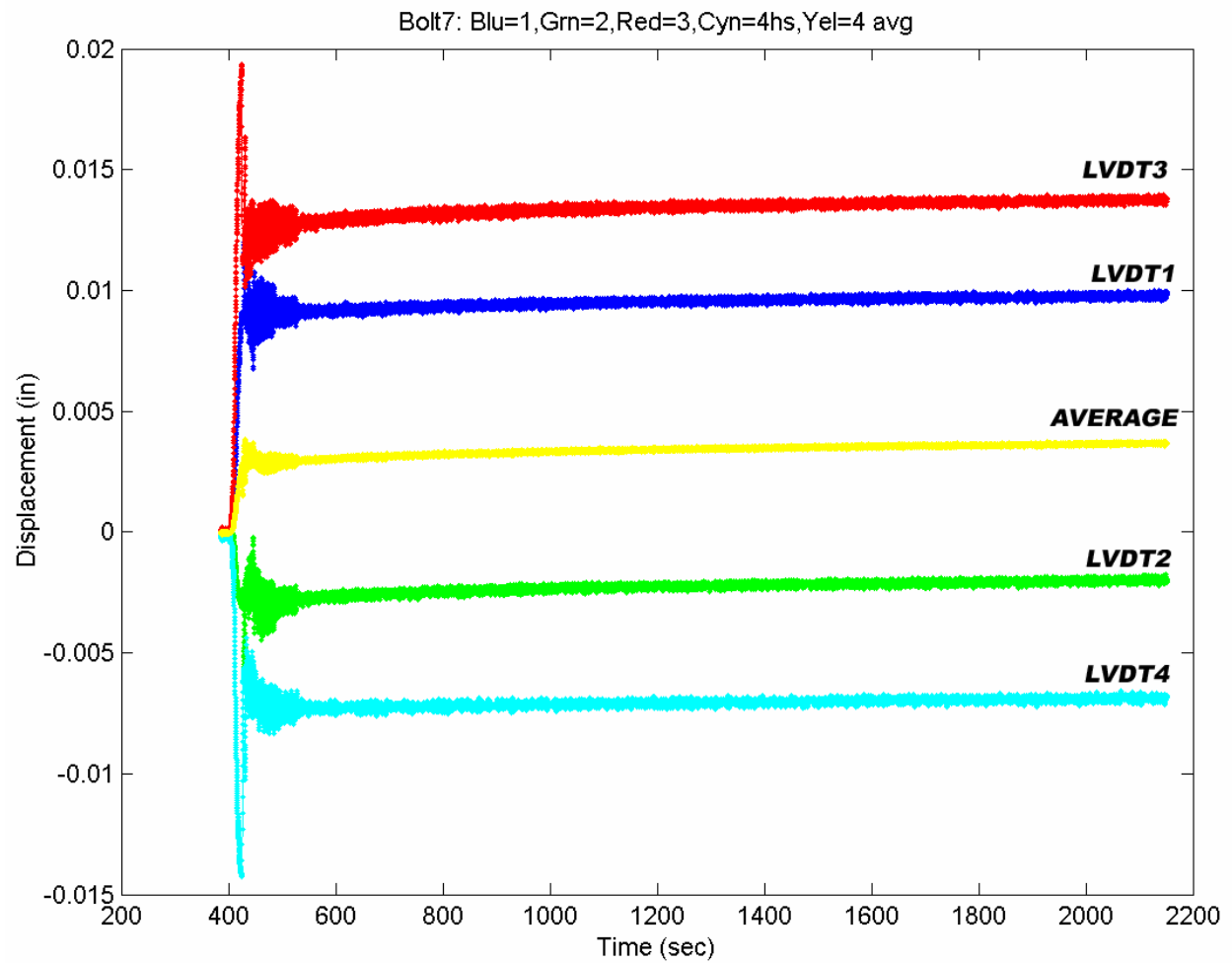


Figure 45. Individual displacement sensors and average displacement data for Anchor7.

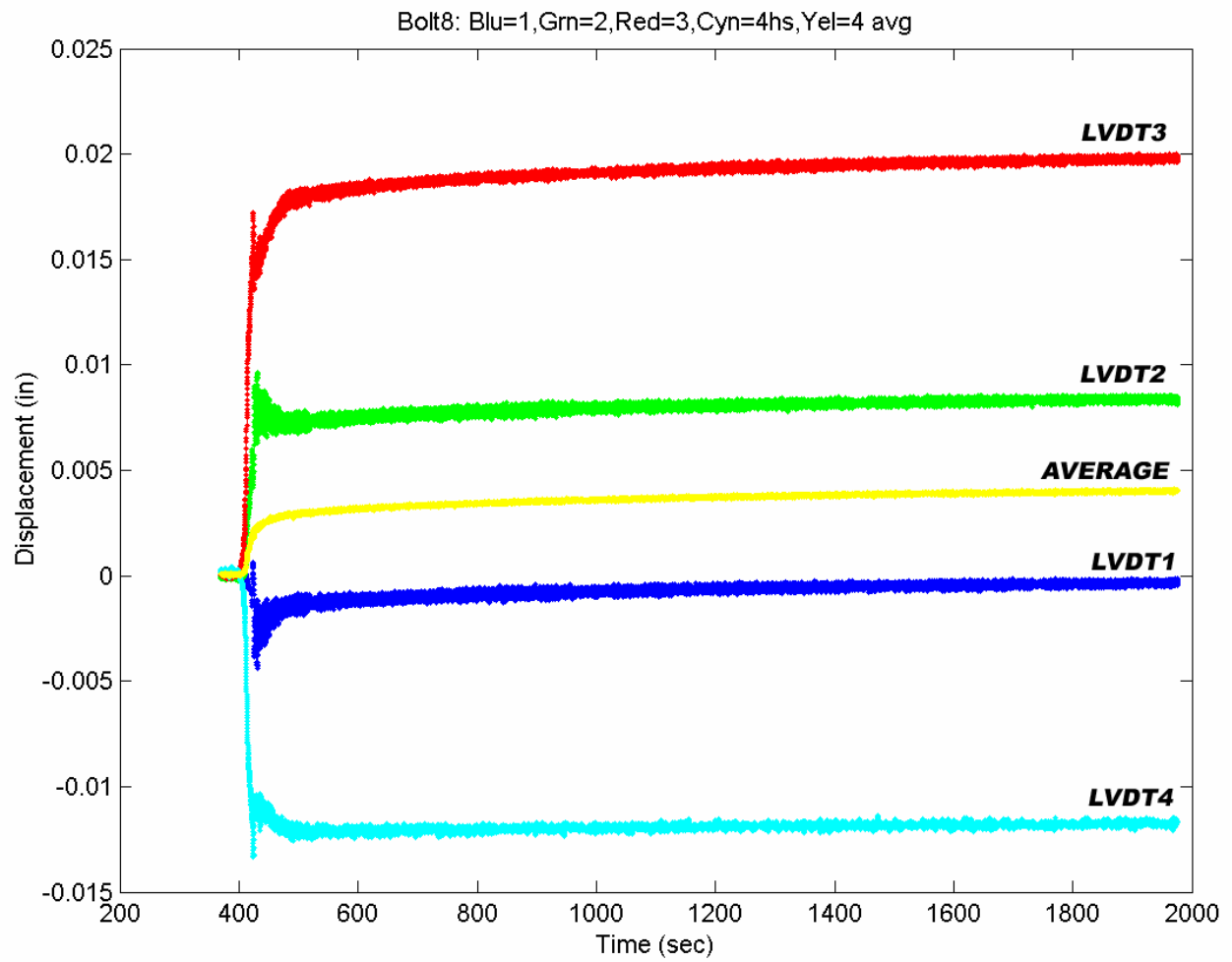


Figure 46. Individual displacement sensors and average displacement data for Anchor8.

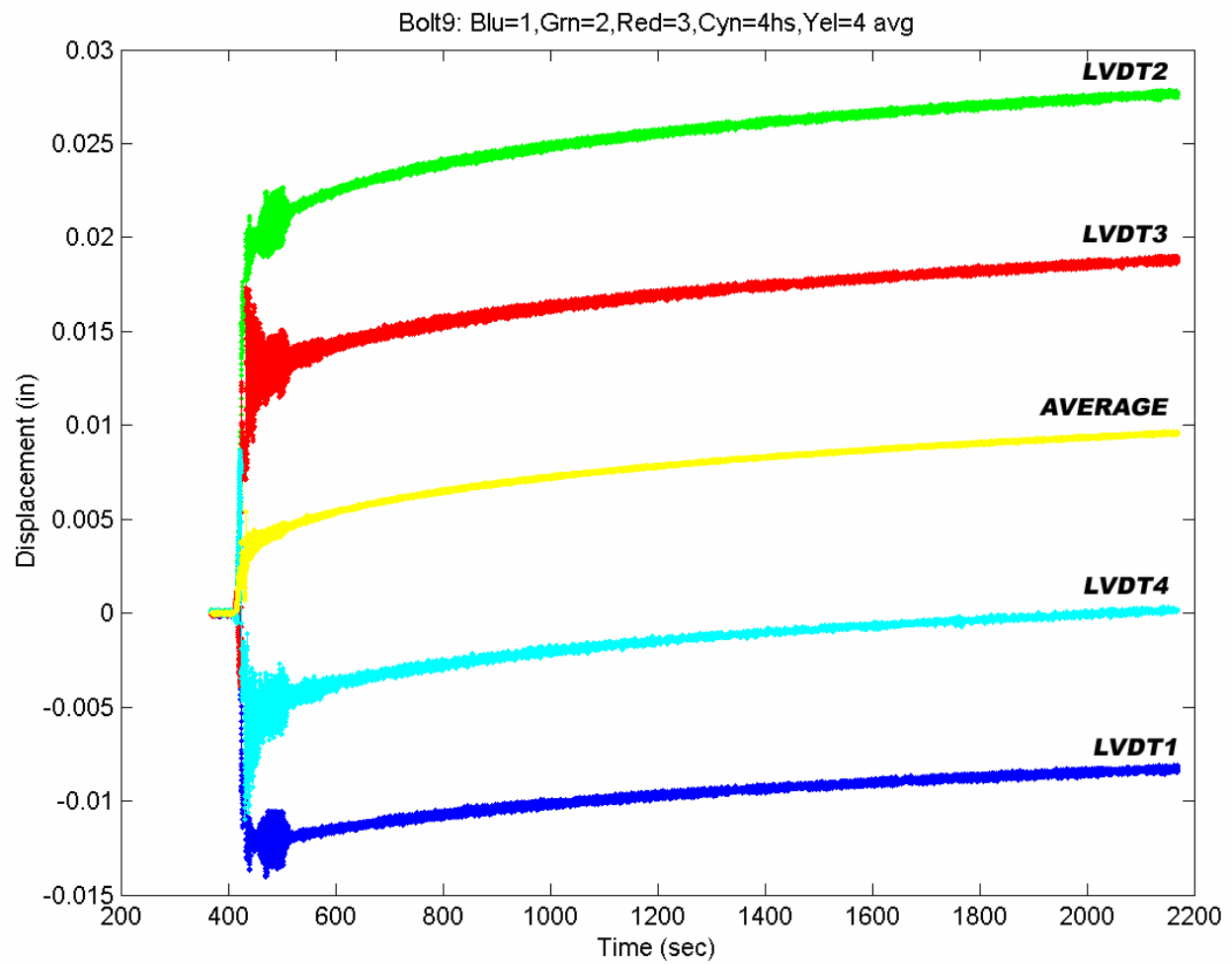


Figure 47. Individual displacement sensors and average displacement data for Anchor9.

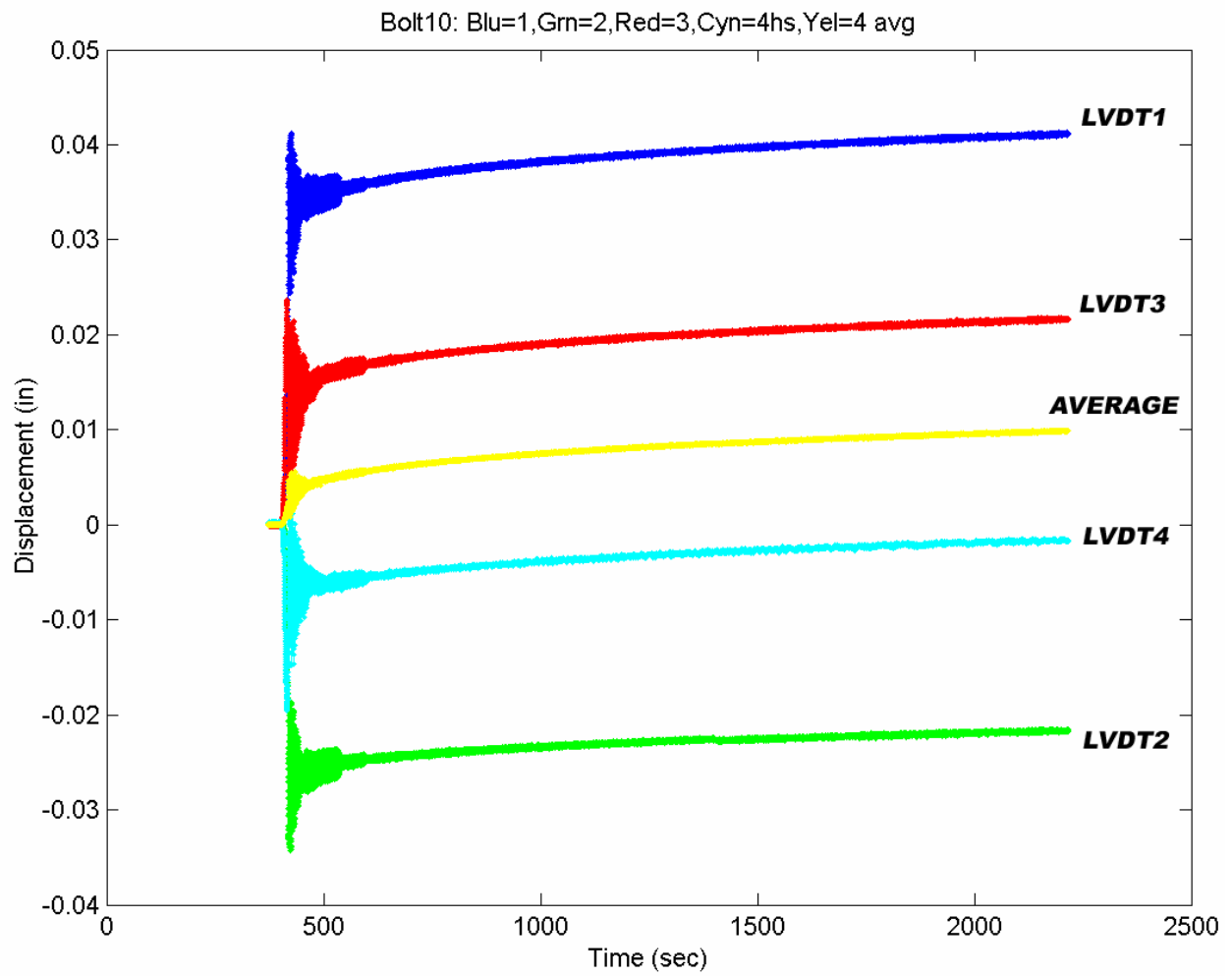


Figure 48. Individual displacement sensors and average displacement data for Anchor10.

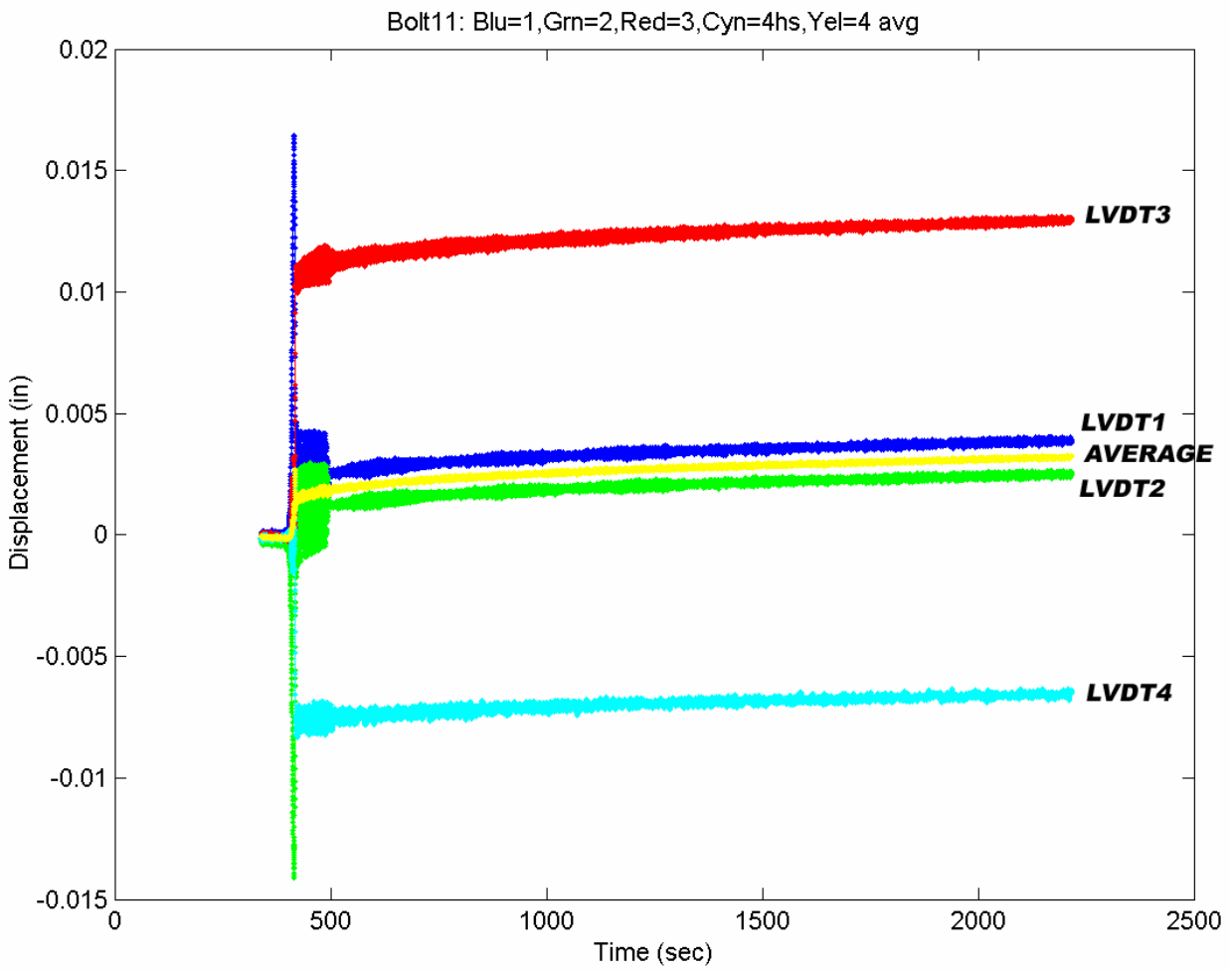


Figure 49. Individual displacement sensors and average displacement data for Anchor11.

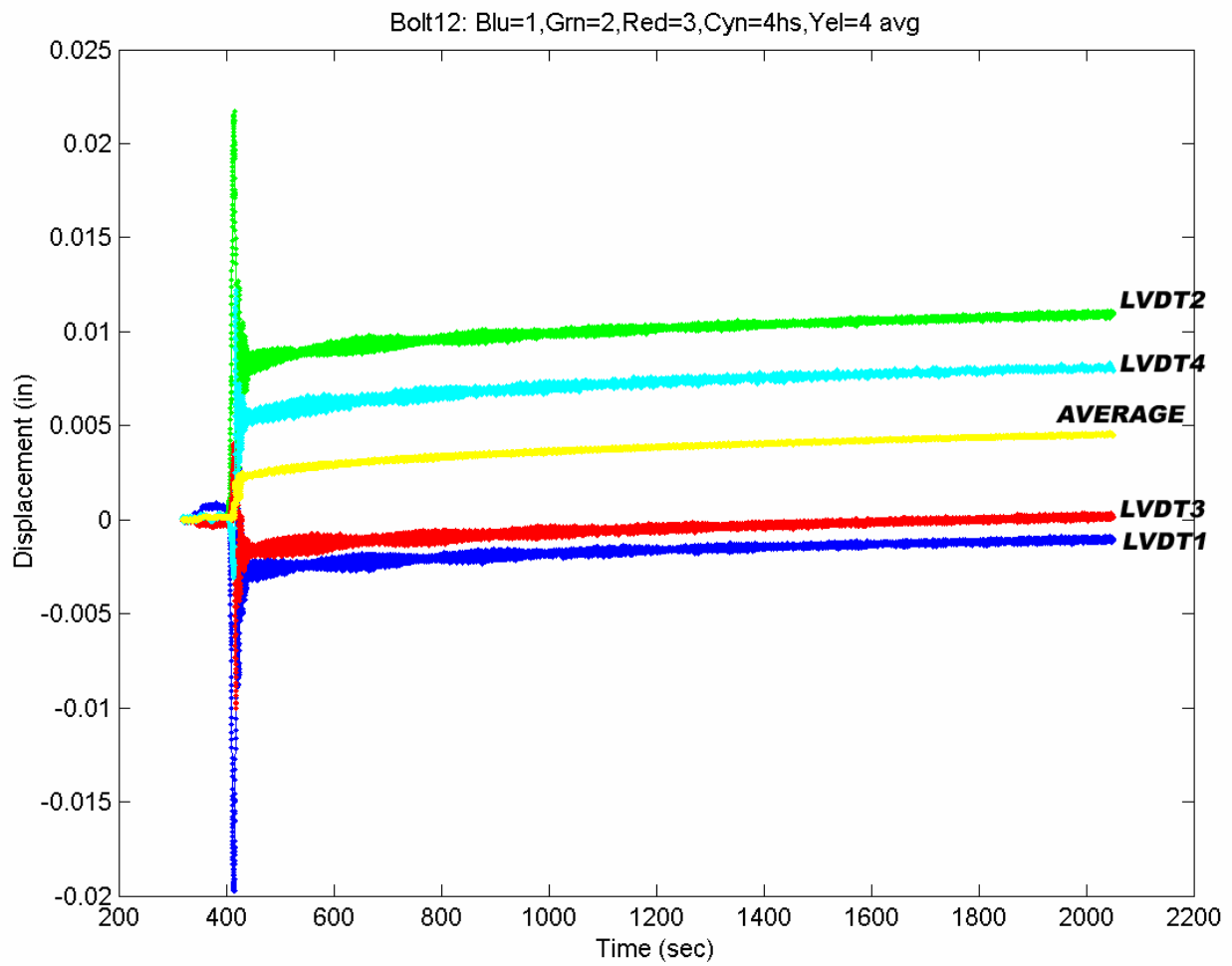


Figure 50. Individual displacement sensors and average displacement data for Anchor12.

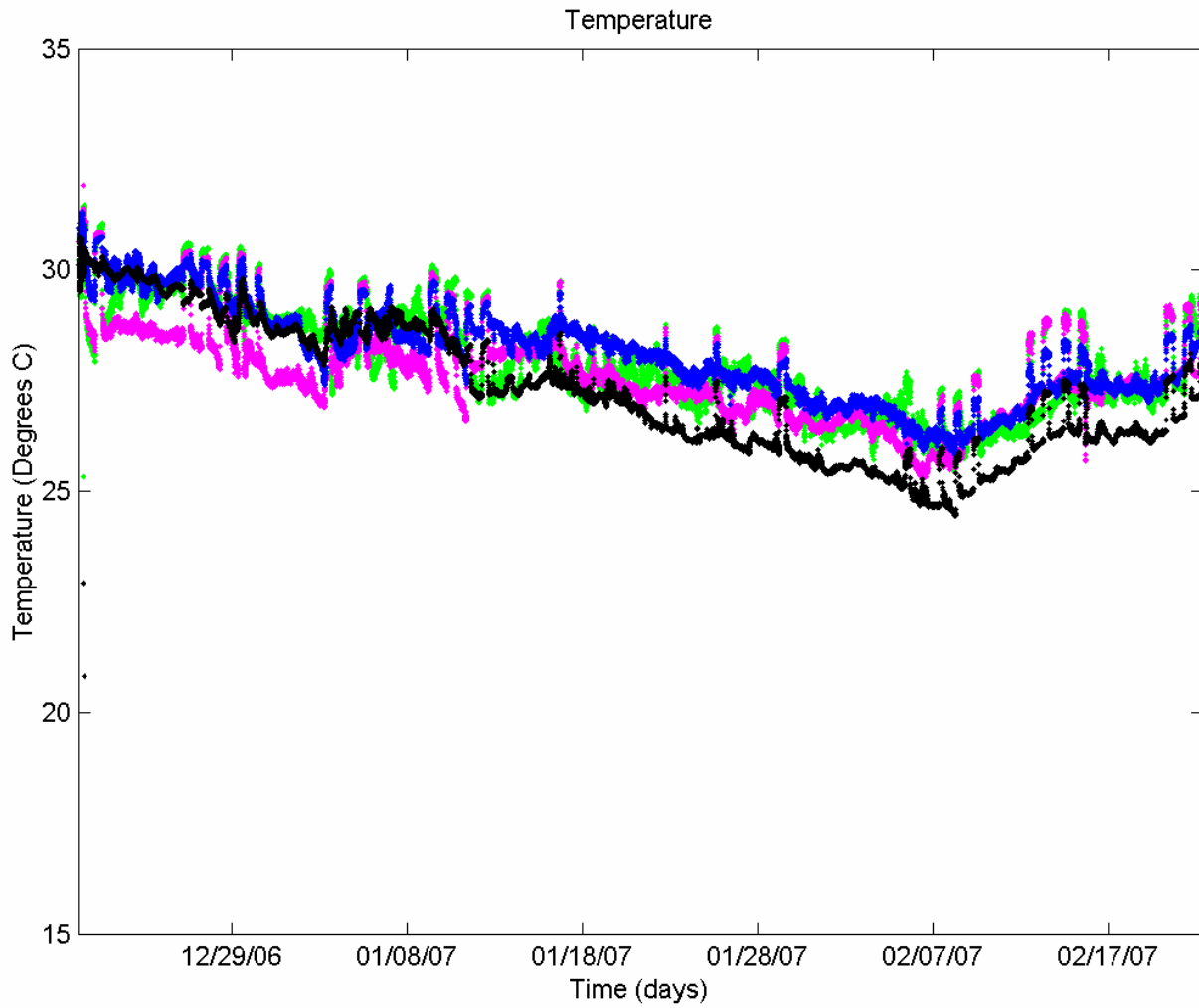


Figure 51. Temperature changes in the laboratory tunnel during testing.

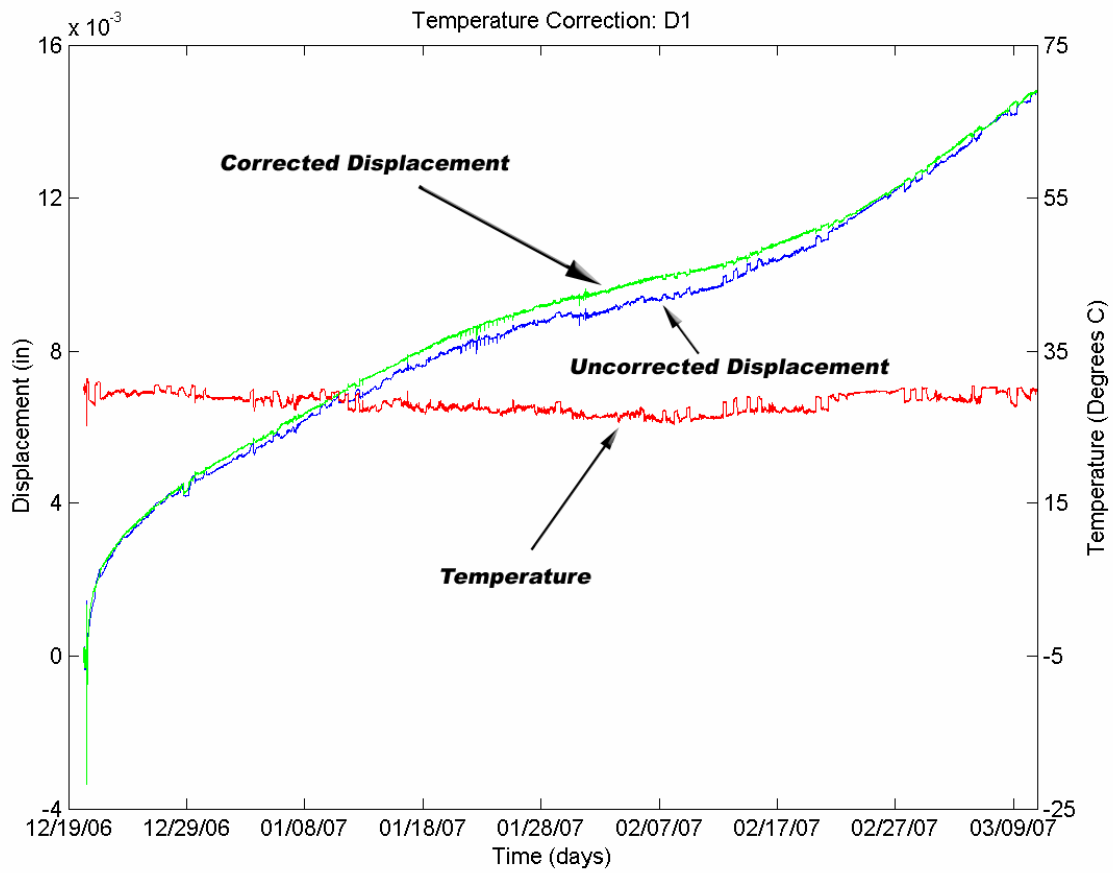


Figure 52. Temperature correction for Anchor 1 showing data before and after temperature correction; uncorrected displacement (blue), scaled temperature (red), corrected temperature (green).

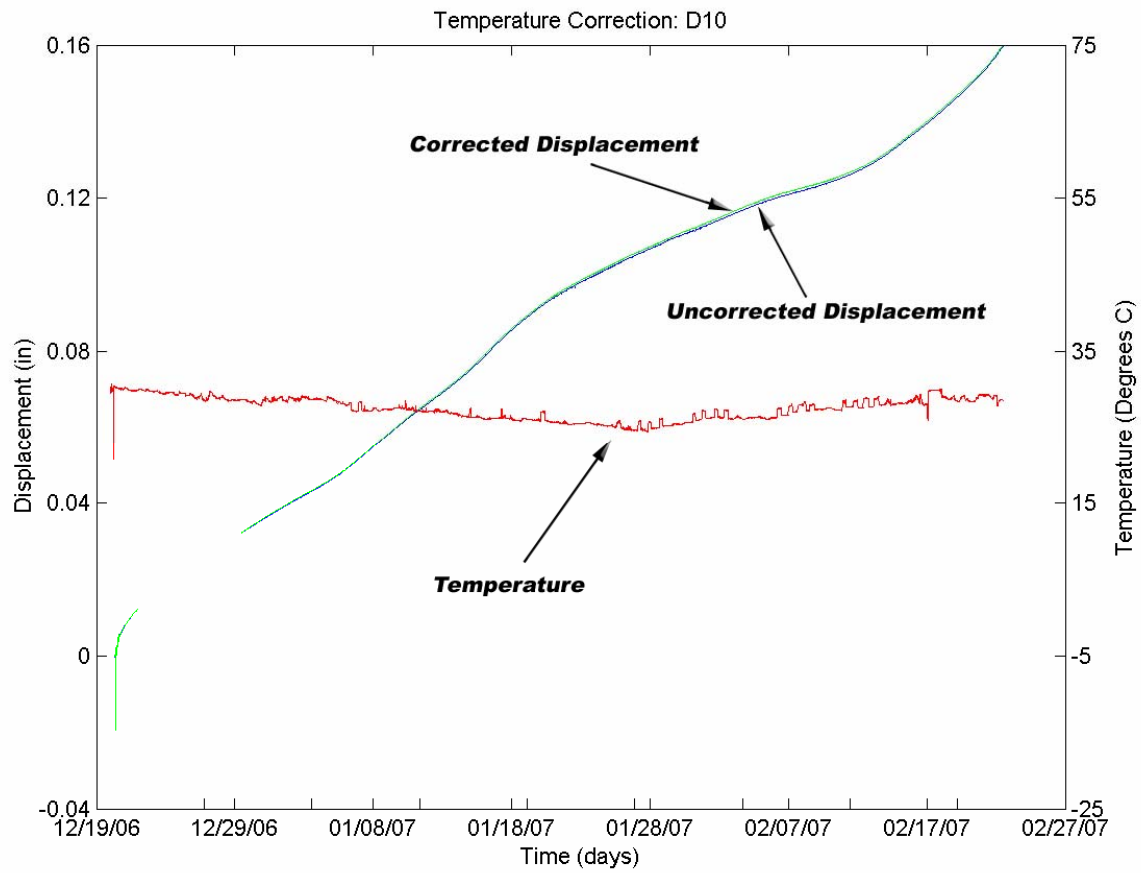


Figure 53. Temperature correction for sensor 10 showing data before and after temperature correction; uncorrected displacement (blue), scaled temperature (red), corrected temperature (green).

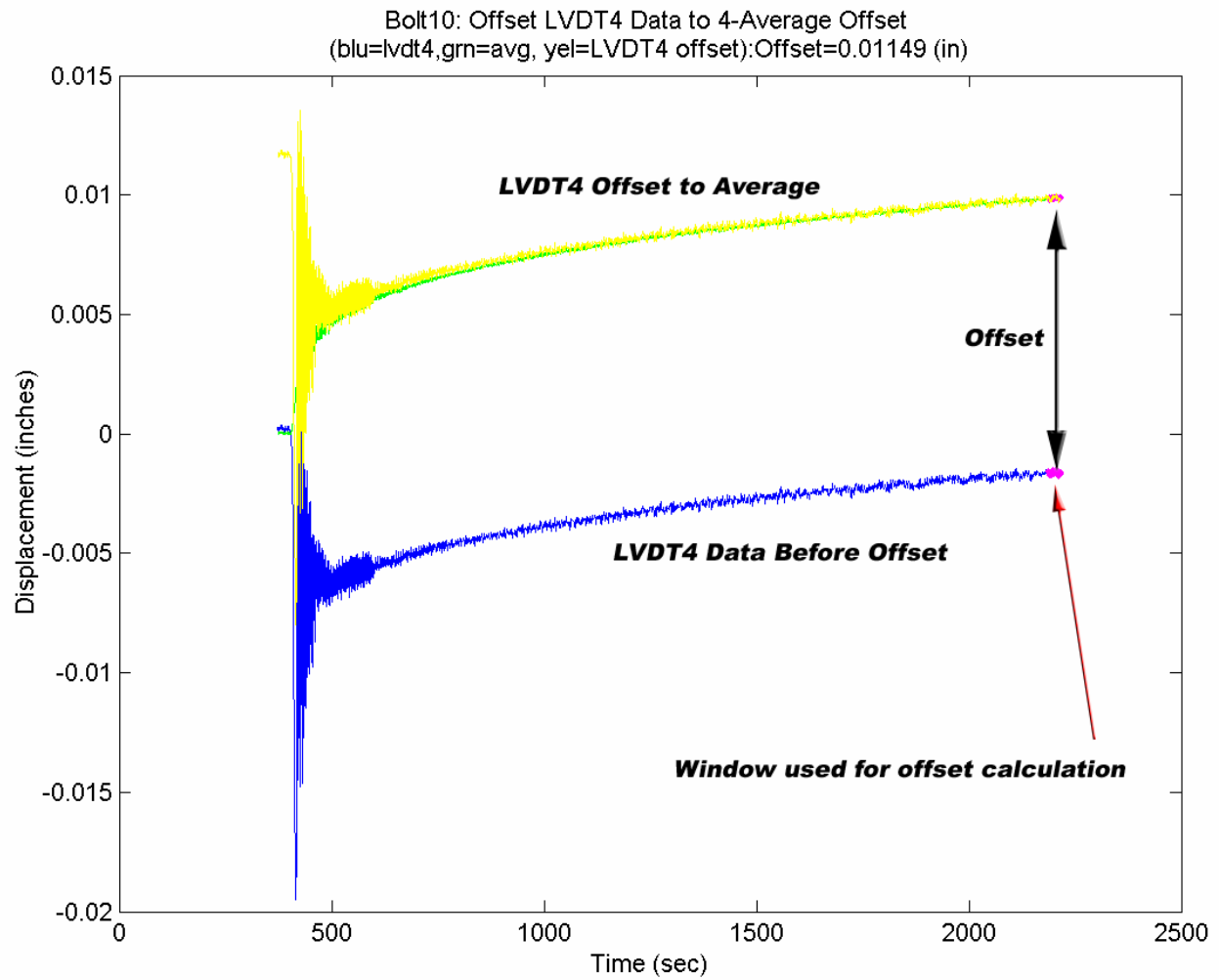


Figure 54. Offset of LVDT4 data to match average displacement data.

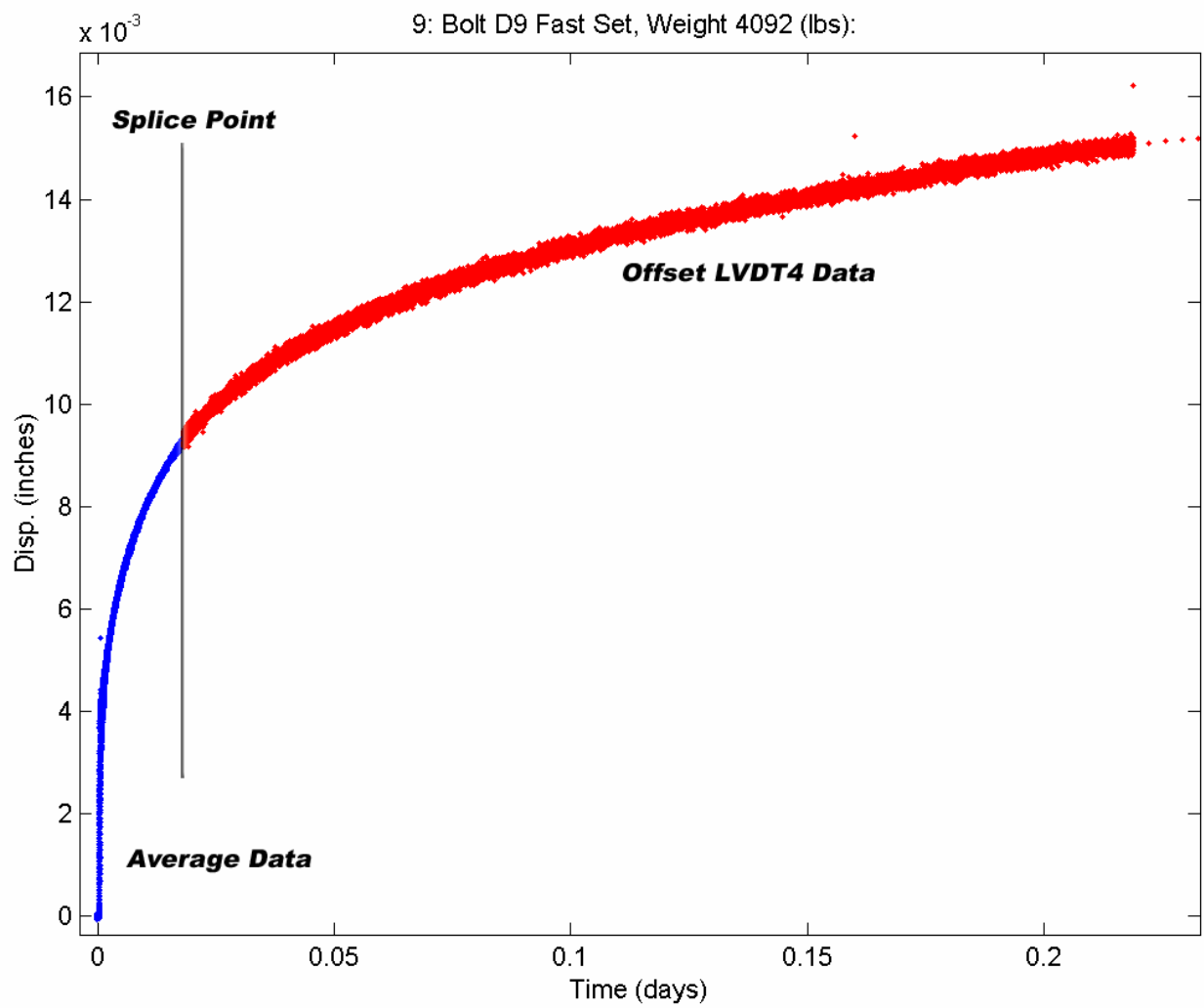


Figure 55. Average data (blue) spliced with long-term data (red) after offset.

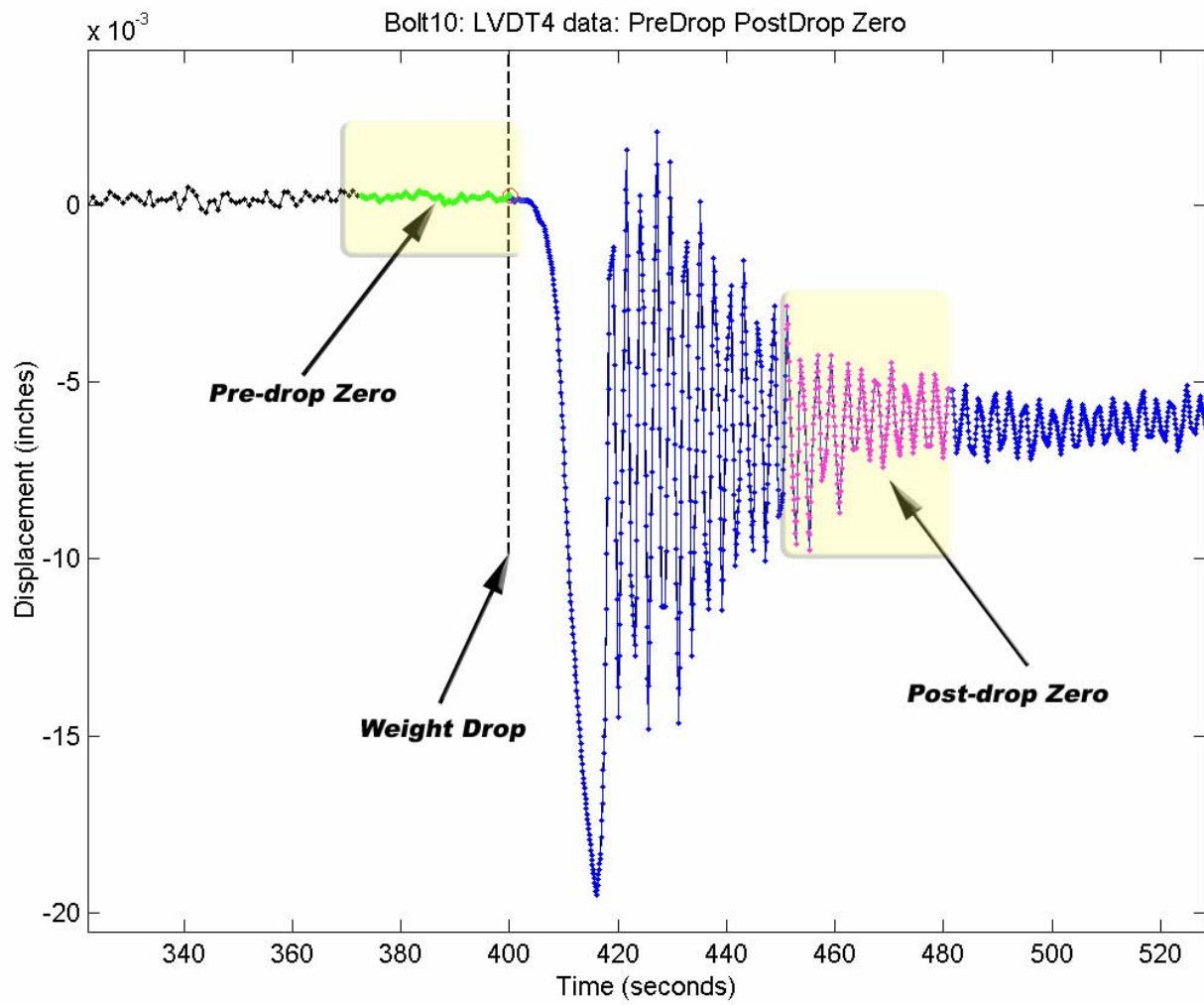


Figure 56. Long-term LVDT4 pre-drop and post-drop displacement values.

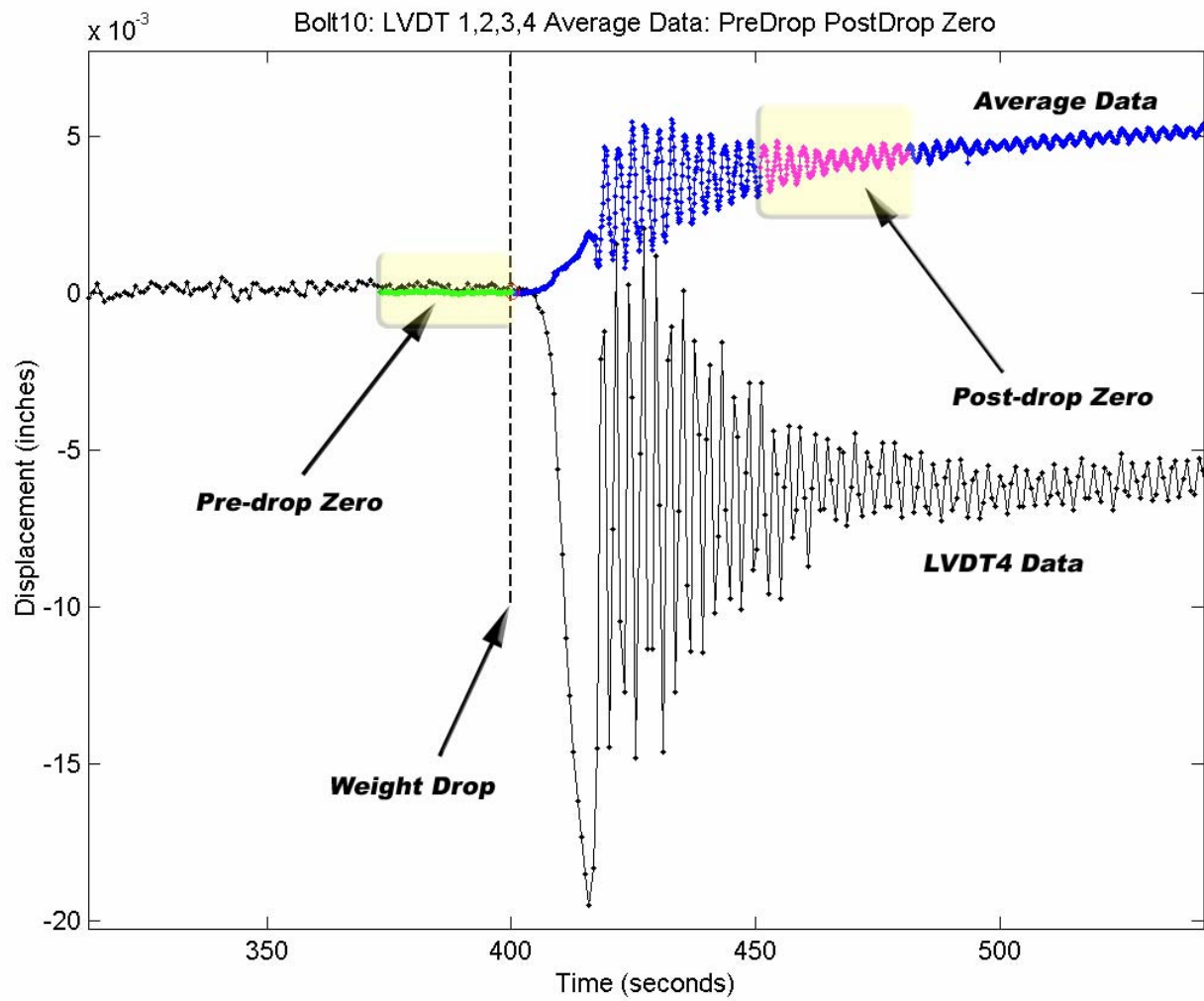


Figure 57. Average data pre-drop and post-drop displacement values.

Table 10 Pre-drop and post-drop zero values; Offset to move LVDT4 long-term output to average displacement

Anchor	LVDT4			Average			LVDT4 Offset to Average (in)
	Pre drop zero (in)	Post drop zero (in)	Change in Displacement Pre to Post Drop (in)	Pre drop zero (in)	Post drop zero (in)	Change in Displacement Pre to Post Drop (in)	
1	0.0003	-0.0006	-0.0009	0.0001	0.0007	0.0007	0.00157
2	0.0001	-0.0017	-0.0019	0.0000	0.0008	0.0008	0.00267
3	-0.0001	-0.0018	-0.0017	0.0000	0.0014	0.0014	0.00436
4	0.0000	-0.0017	-0.0017	0.0000	0.0015	0.0015	0.00367
5	-0.0001	0.0041	0.0042	0.0000	0.0010	0.0011	-0.00247
6	0.0004	-0.0069	-0.0073	0.0001	0.0011	0.0010	0.00845
7	-0.0001	-0.0072	-0.0070	0.0000	0.0029	0.0029	0.01055
8	0.0002	-0.0117	-0.0119	0.0001	0.0028	0.0027	0.01577
9	0.0000	-0.0052	-0.0053	0.0000	0.0040	0.0040	0.00942
10	0.0002	-0.0062	-0.0064	0.0000	0.0042	0.0042	0.01149
11	-0.0001	-0.0076	-0.0075	-0.0001	0.0017	0.0019	0.00976
12	0.0001	0.0056	0.0055	0.0001	0.0025	0.0024	-0.00355

APPENDIX B

Load-Displacement Test Results

Table 11 Creep #1 (1052 pounds)

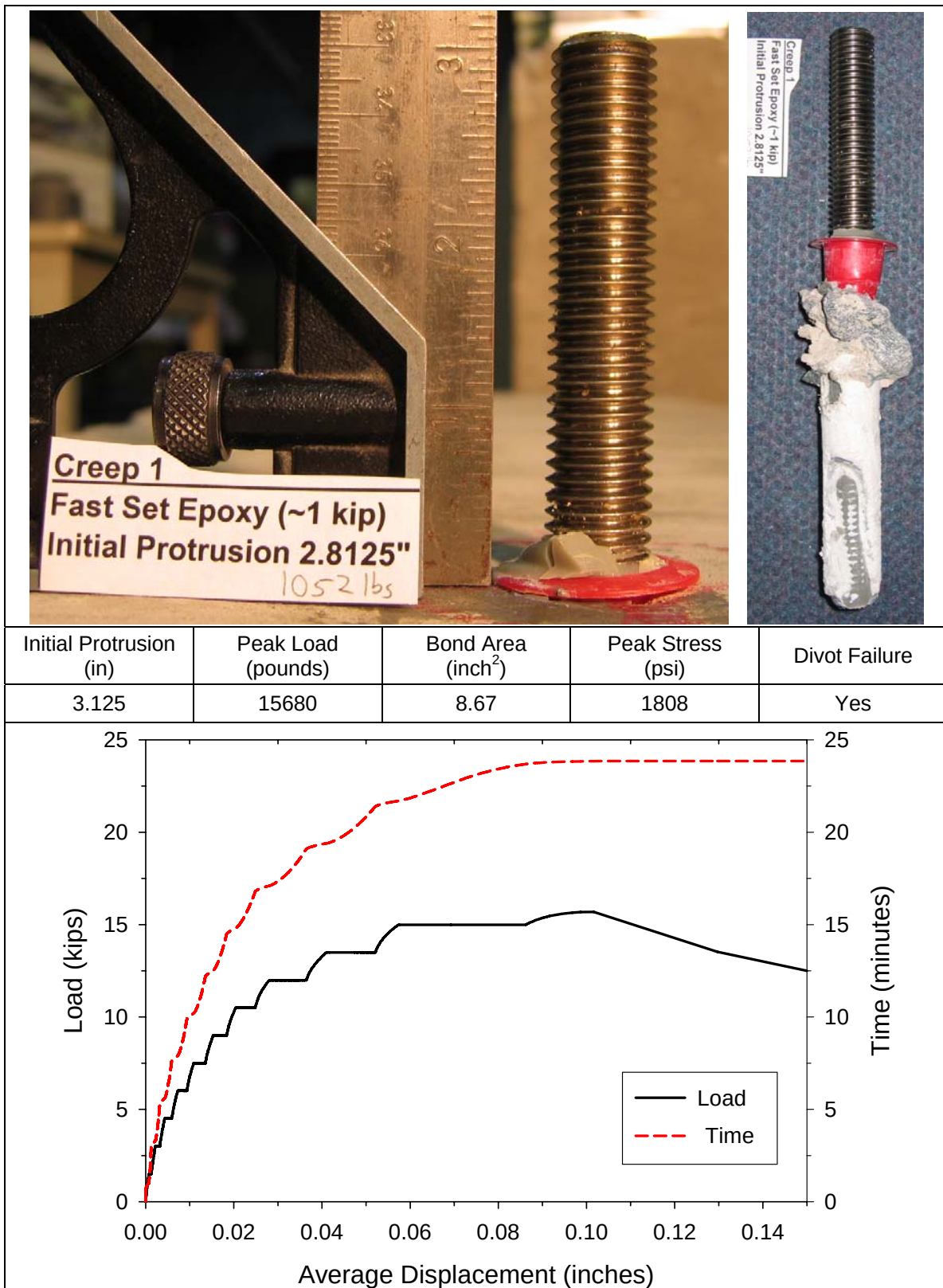


Table 12 Creep #2 (1057 pounds)



Initial Protrusion (in)	Peak Load (pounds)	Bond Area (inch ²)	Peak Stress (psi)	Divot Failure
2.813	14985	9.13	1641	Yes

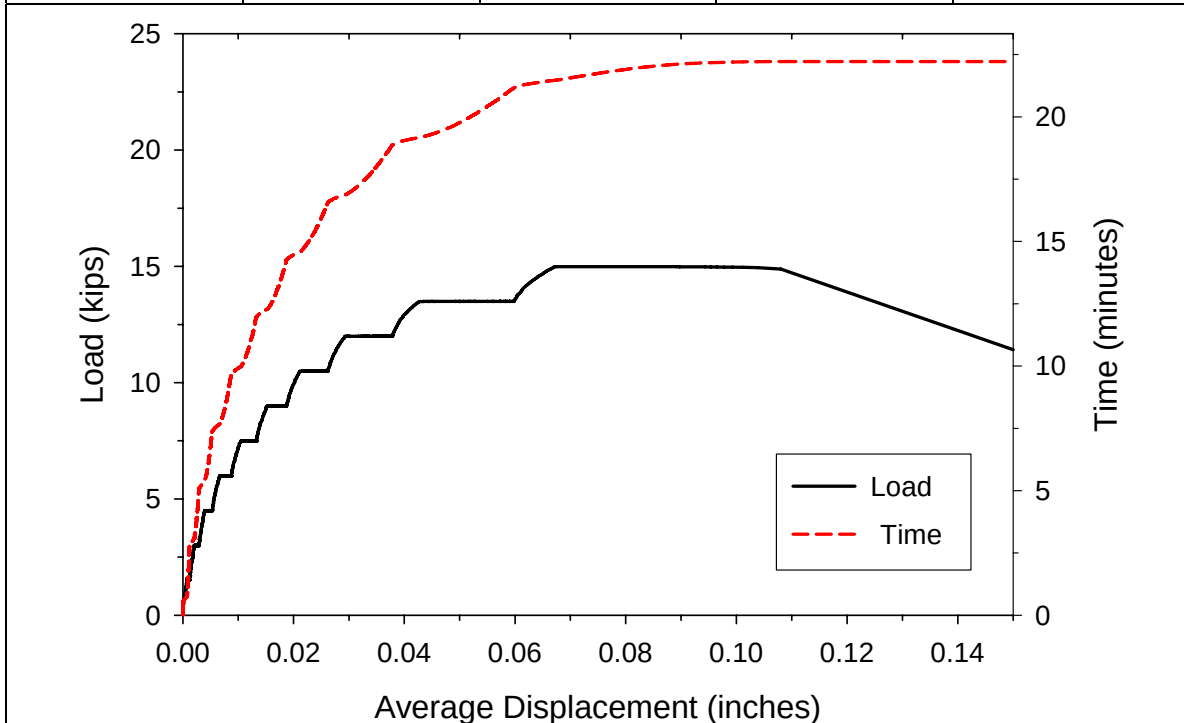
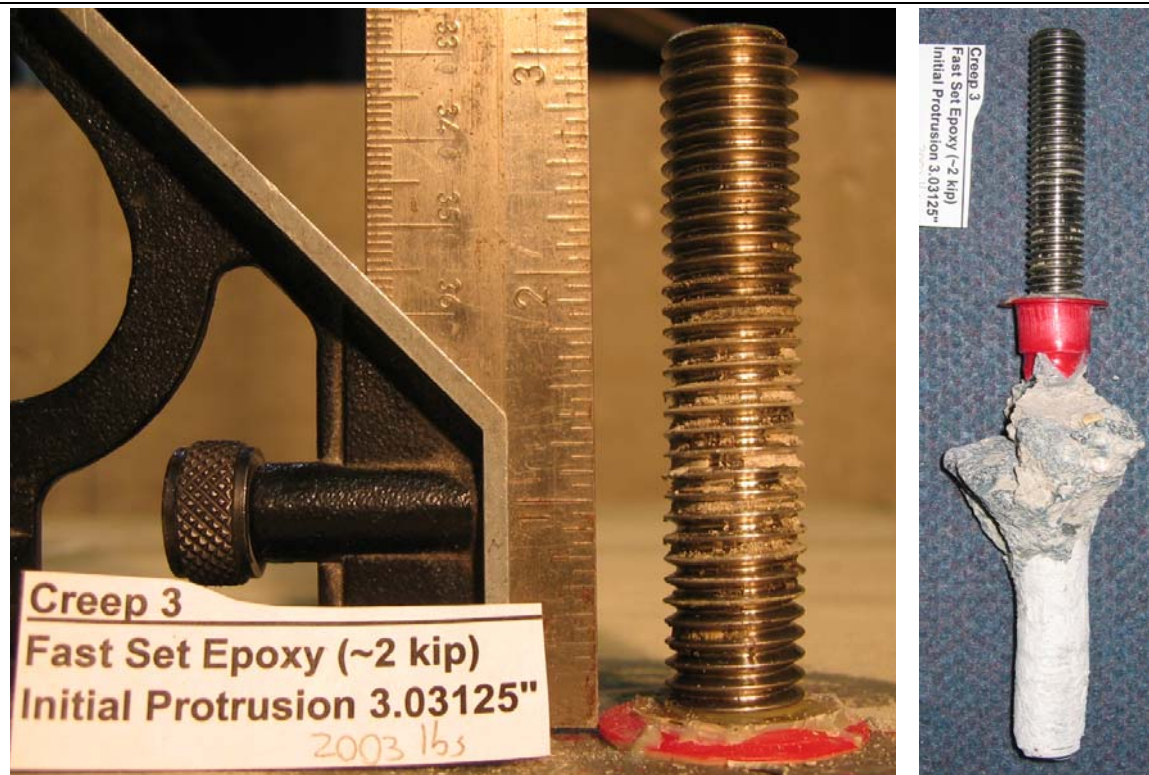


Table 13 Creep #3 (2003 pounds)



Initial Protrusion (in)	Peak Load (pounds)	Bond Area (inch ²)	Peak Stress (psi)	Divot Failure
3.063	17971	9.84	1827	Yes

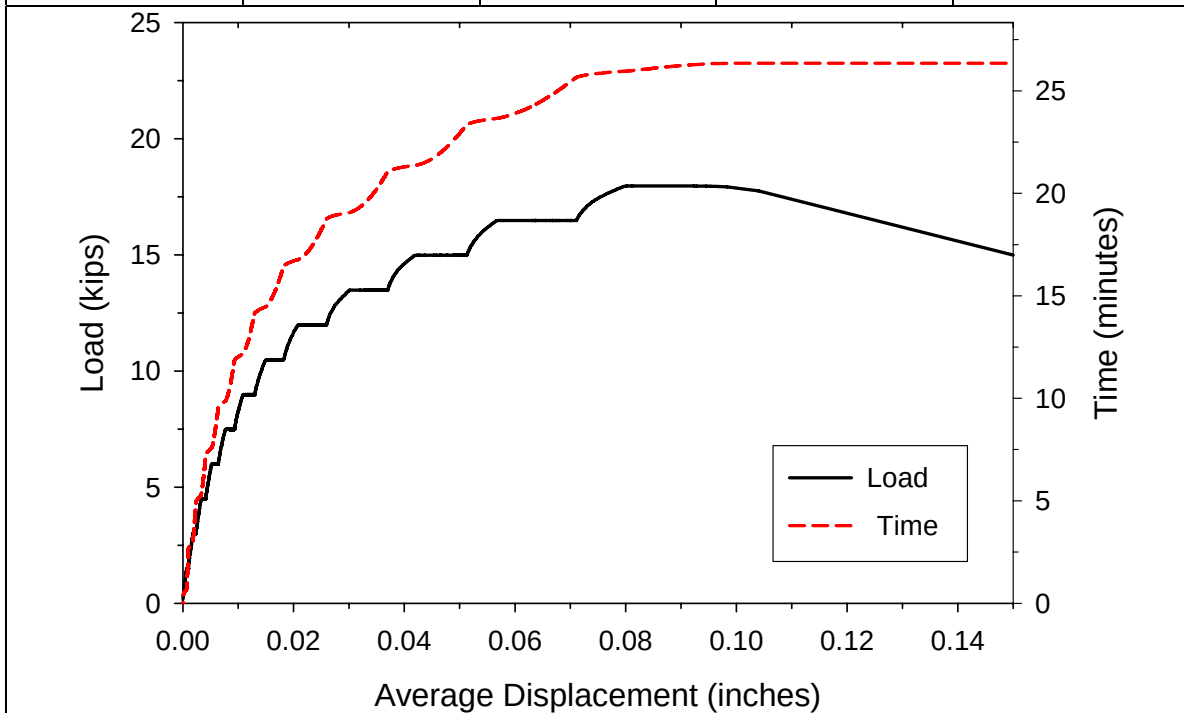
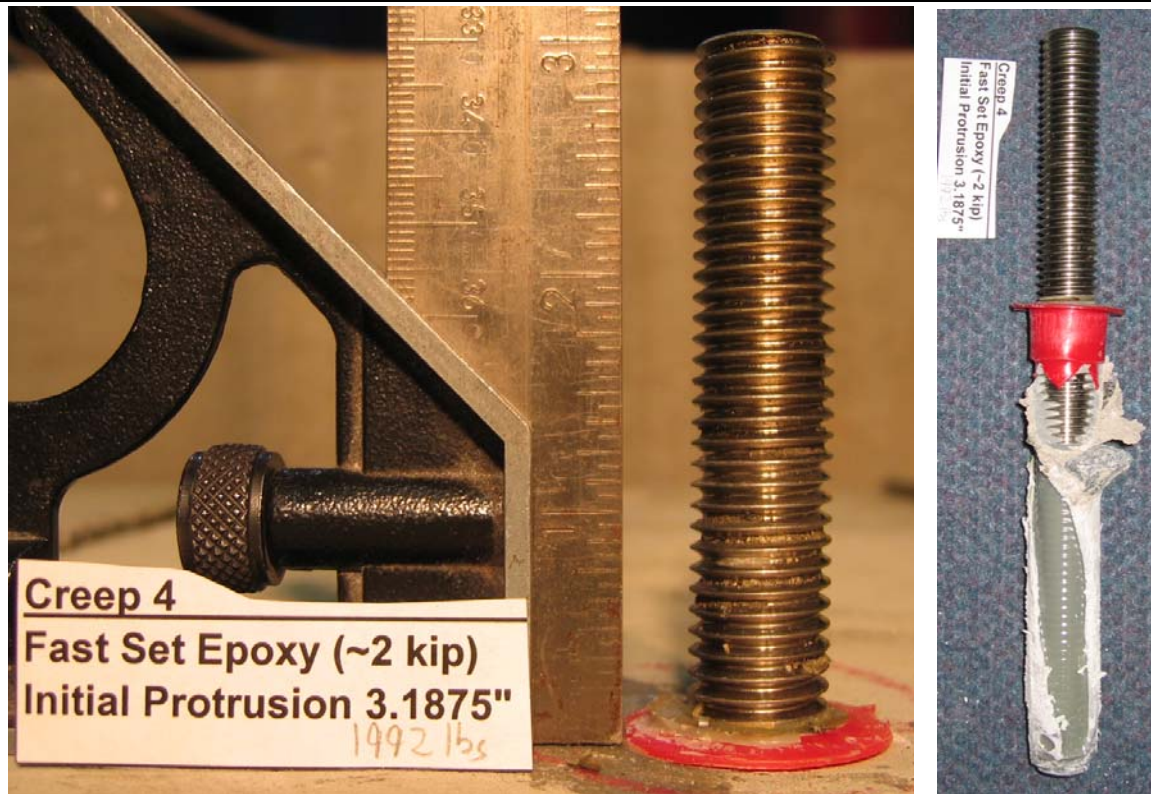


Table 14 Creep #4 (1992 pounds)



Initial Protrusion (in)	Peak Load (pounds)	Bond Area (inch ²)	Peak Stress (psi)	Divot Failure
3.031	14983	7.69	1949	Yes

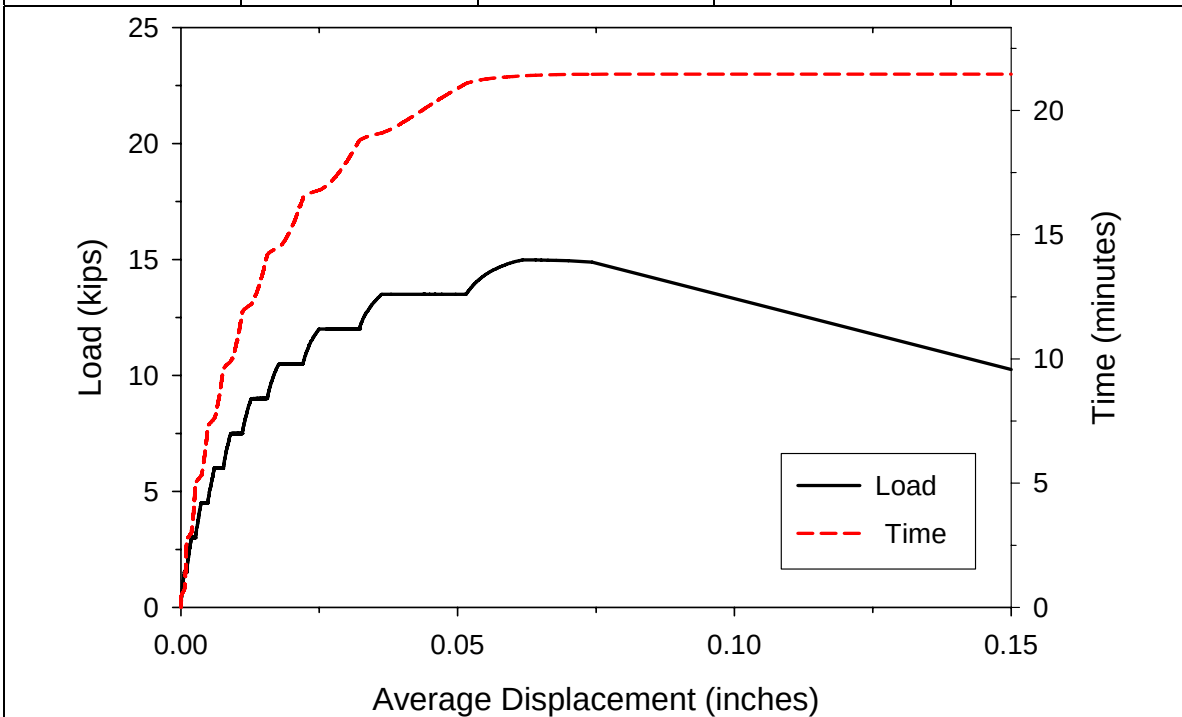


Table 15 Creep #5 (1989 pounds)

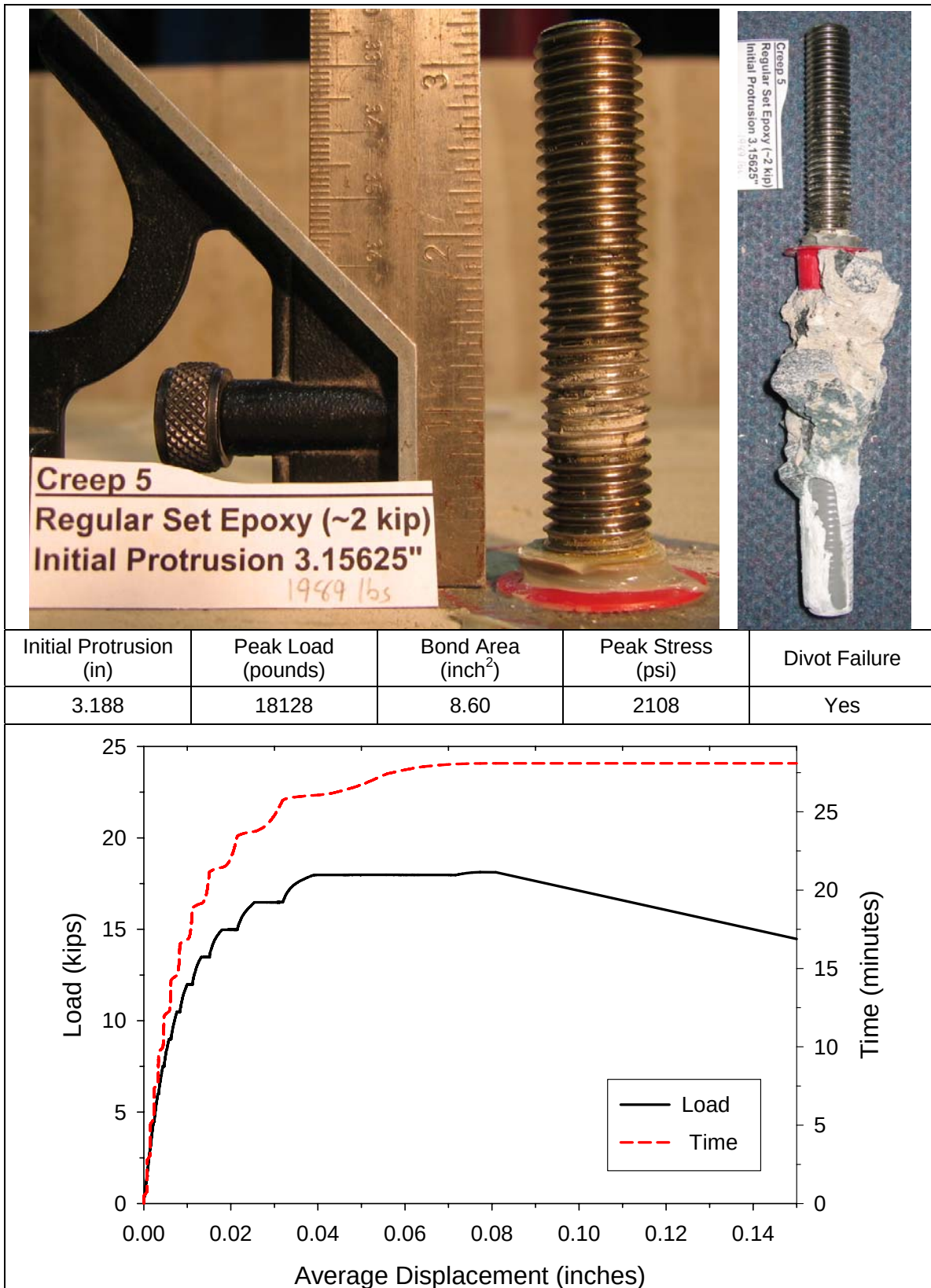


Table 16 Creep #6 (1991 pounds)

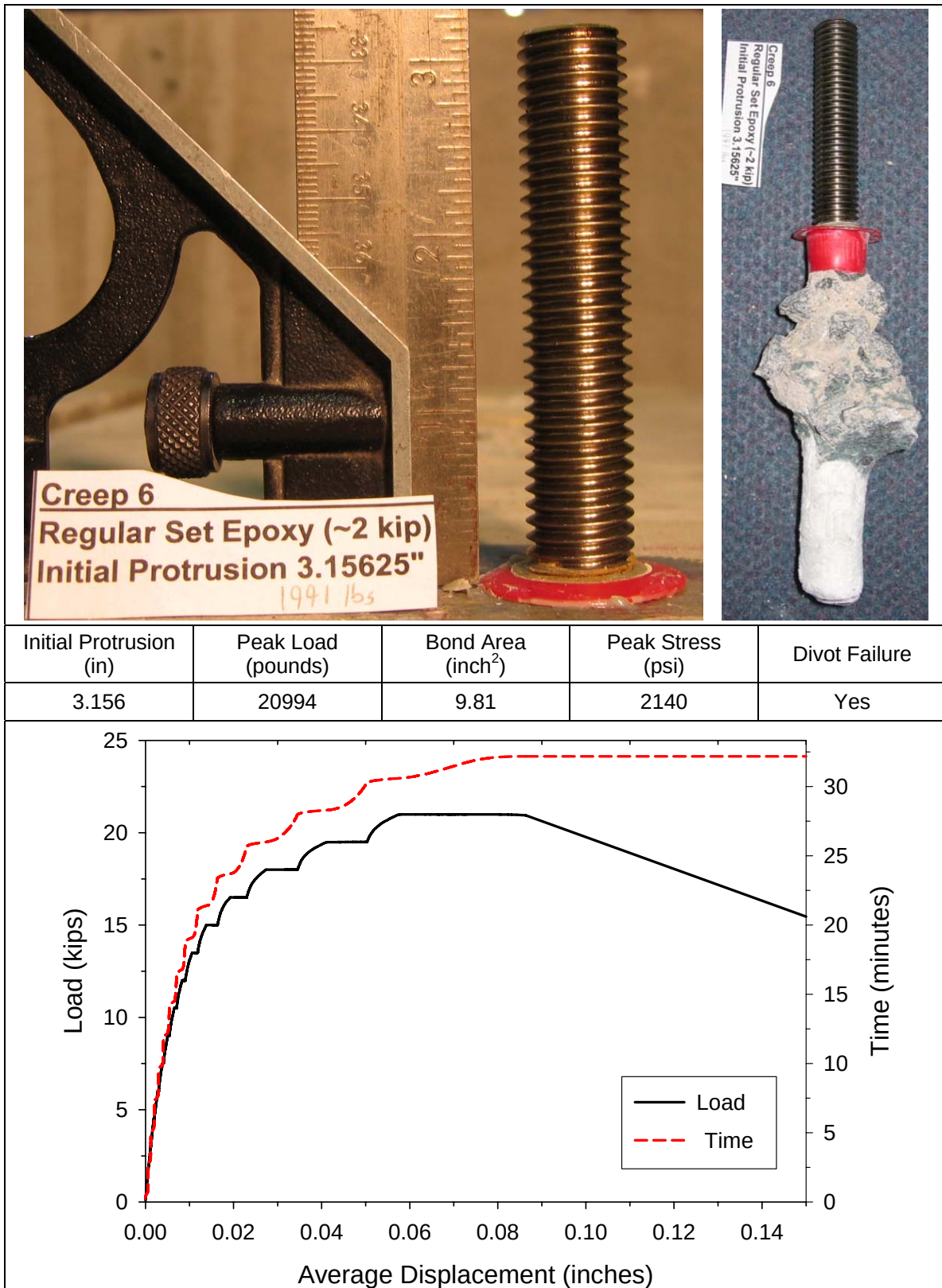


Table 17 Creep #7 (4075 pounds)

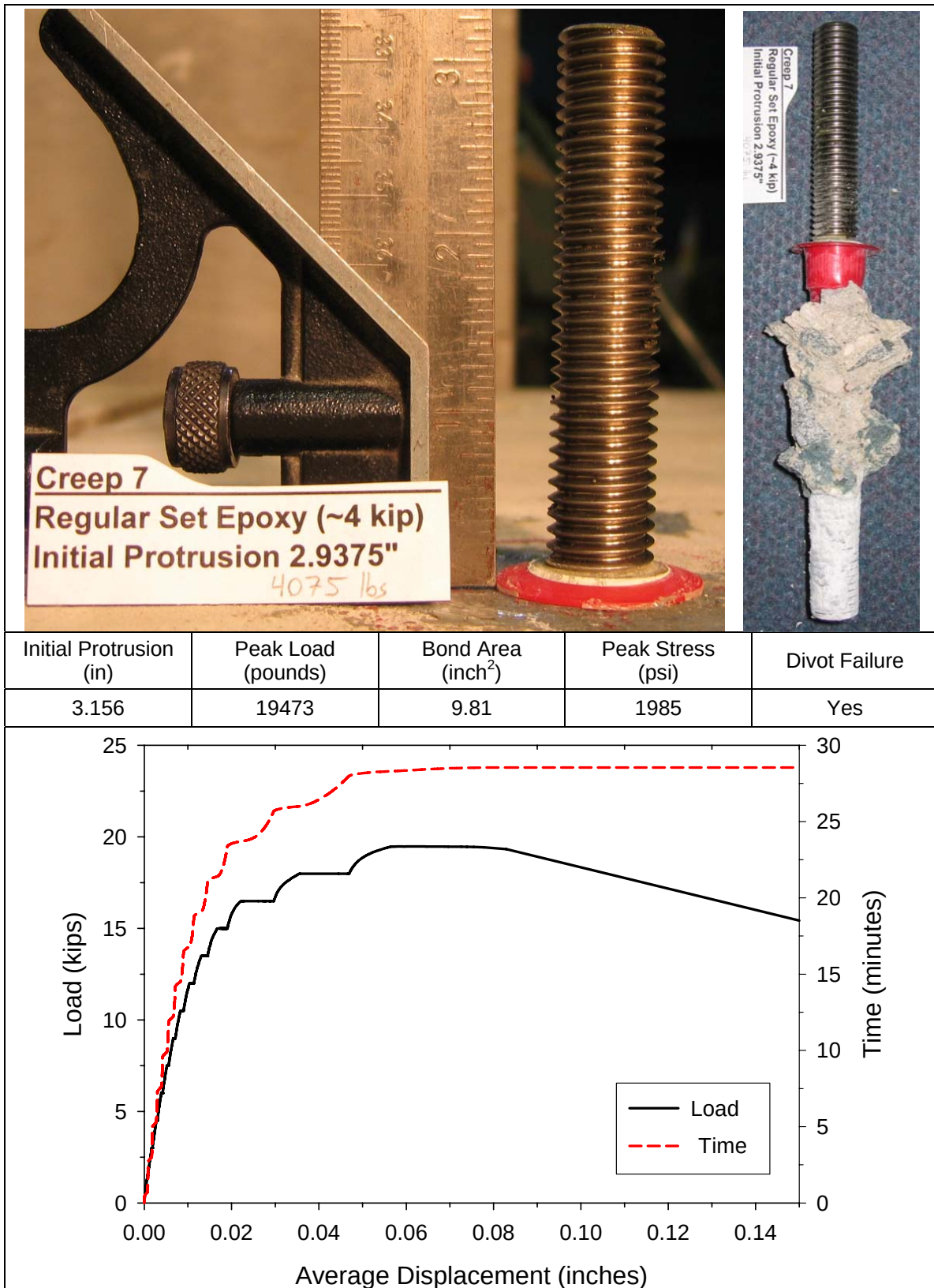
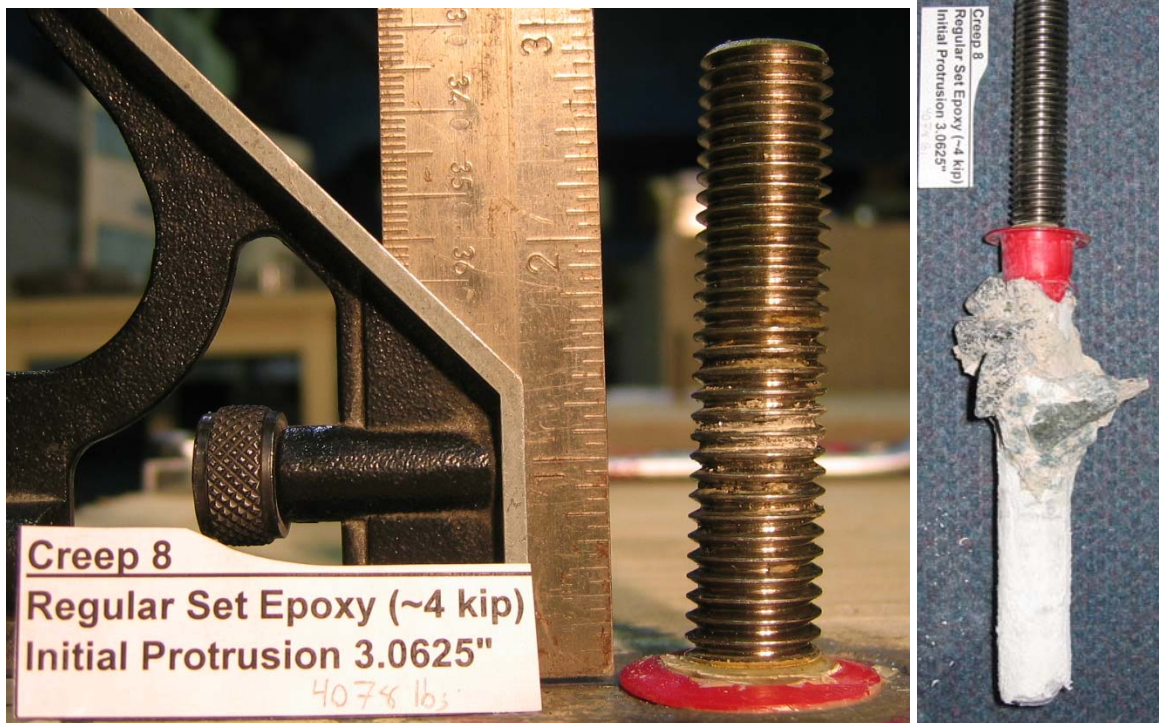


Table 18 Creep #8 (4078 pound)



Initial Protrusion (in)	Peak Load (pounds)	Bond Area (inch ²)	Peak Stress (psi)	Divot Failure
2.938	19478	10.32	1886	Yes

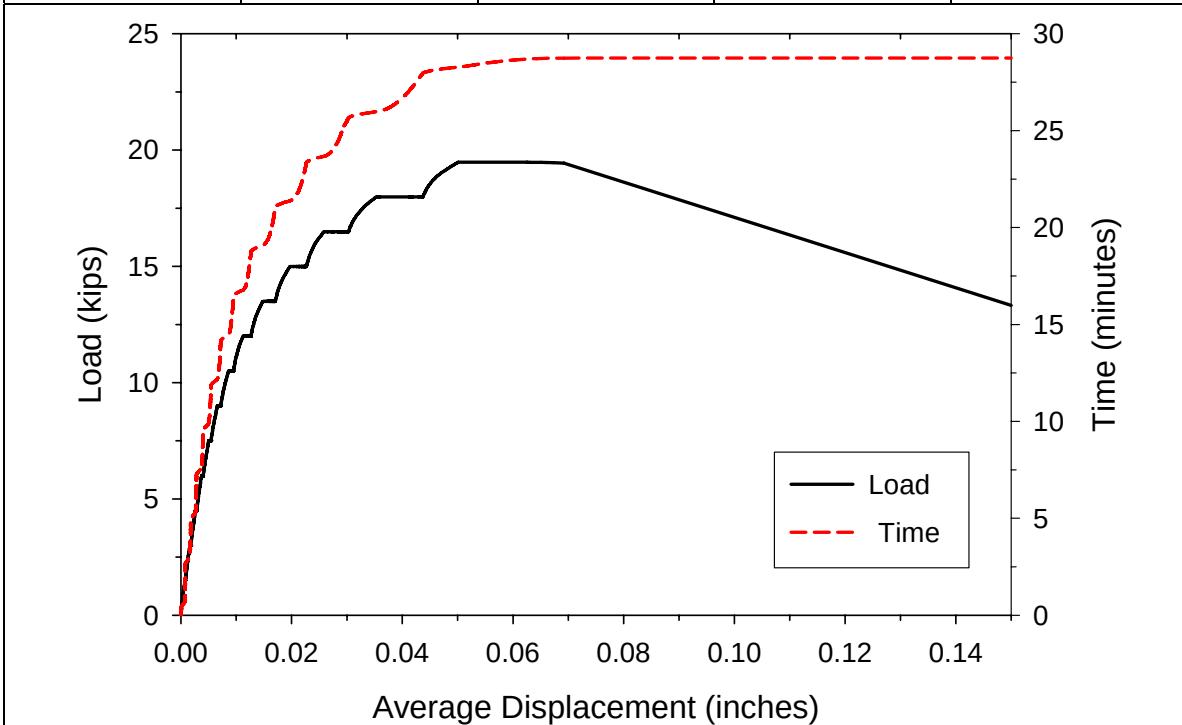


Table 19 Creep #9 (4092 pounds)

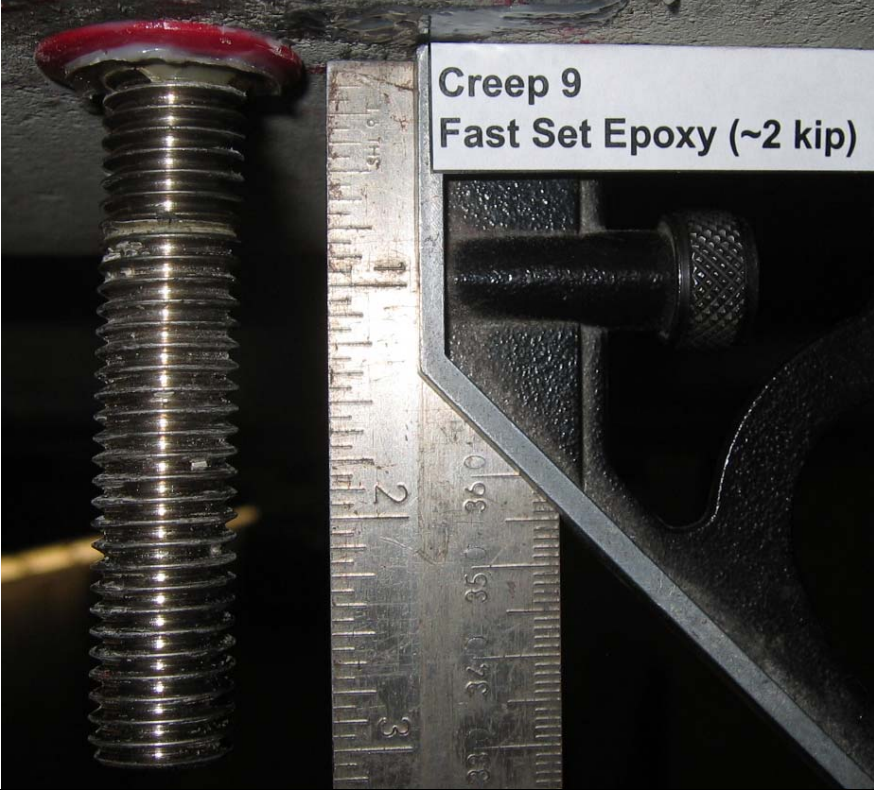
 				
Creep 9 Fast Set Epoxy (~2 kip) Creep 9 Fast Set Epoxy (~4 kip) Initial Protrusion 3.000"				
Initial Protrusion (in)	Peak Load (pounds)	Bond Area (inch ²)	Peak Stress (psi)	Divot Failure
3.063	-	8.16	-	-
This anchor pulled out after 63.5 days, therefore a static pull test could not be conducted.				

Table 20 Creep #10 (4013 pounds)

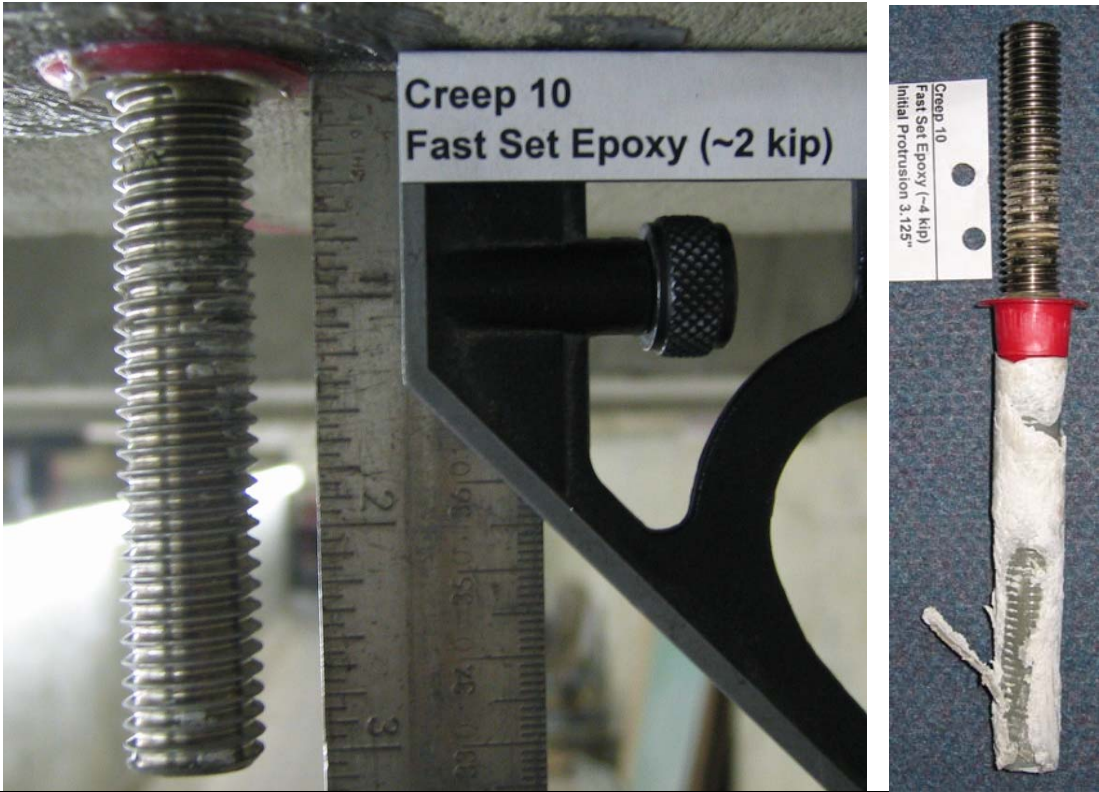
				
Initial Protrusion (in)	Peak Load (pounds)	Bond Area (inch ²)	Peak Stress (psi)	Divot Failure
3.000	-	8.46	-	-
This anchor pulled out after 75.8 days, therefore a static pull test could not be conducted.				

Table 21 Creep #11 (3058 pounds)

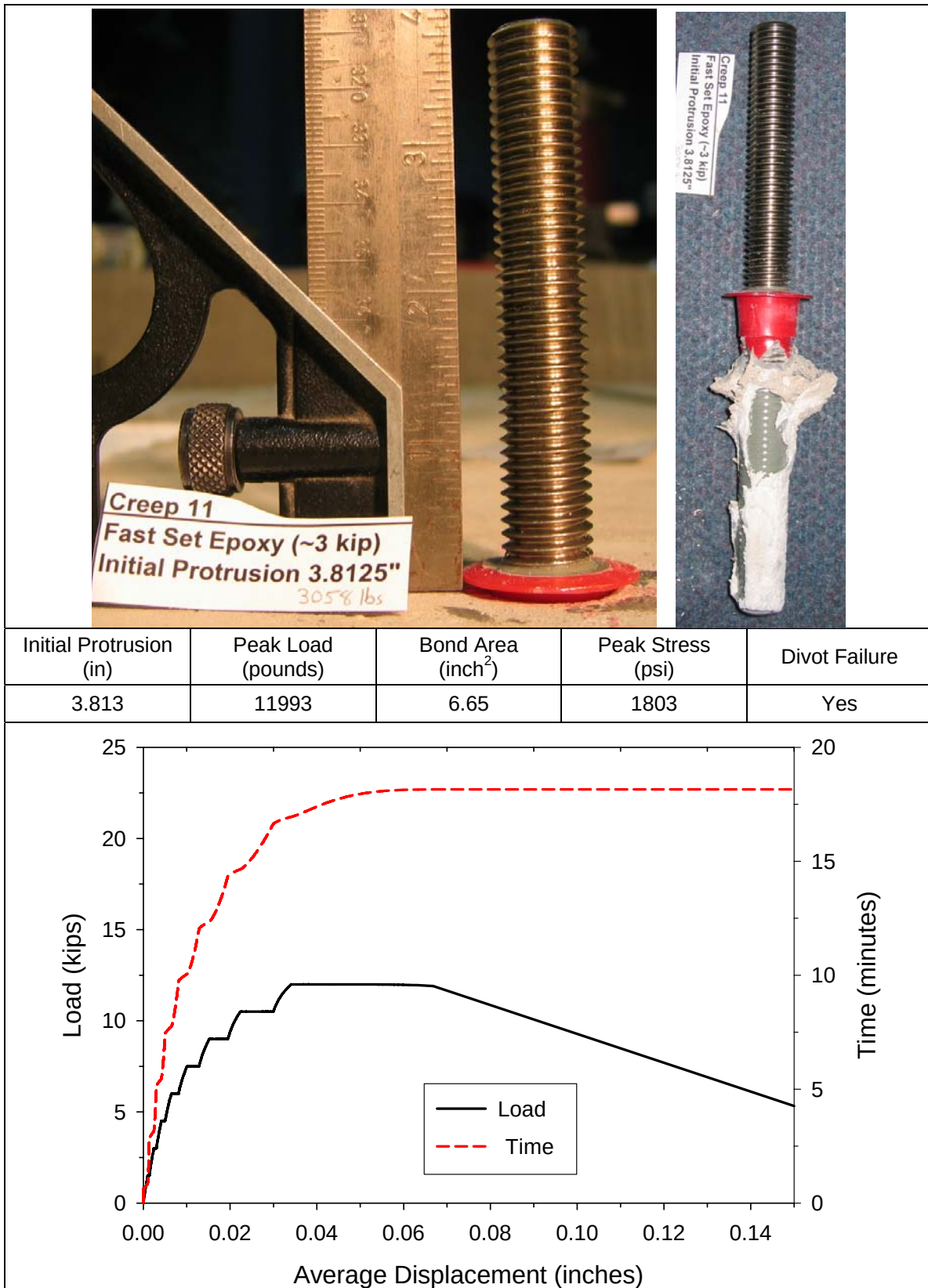


Table 22 Creep #12 (3143 pounds)



Initial Protrusion (in)	Peak Load (pounds)	Bond Area (inch ²)	Peak Stress (psi)	Divot Failure
3.125	17995	9.54	1882	Yes

